

Super Grids in Africa - Could they release the economic potential of concentrating solar power?

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Abstract

The way its future power systems are designed will have significant impact on sub-Saharan Africa's (SSA) aspirations to move from low electricity consumption rates to enhance life quality and further increase economic opportunity. At present, Africa is experiencing higher economic growth rates than other continents (including Asia). And so is its need for electric power. However, all too often the options that are chosen are the ones with lowest risk and that require little coordination. In part, this is because region-wide planning, coordination and institutions are in their infancy. “Low risk” power plants typically include oil generators that can be sited close to loads, other fossil fuel power plants, and hydro plants that can easily be connected to the continent’s grid. However, hydropower production has been limited due to changes in weather and climate and socio-economic impacts (Collier, 2011). Additionally, its potential has also not been reached as large sites are far from adequate grids. A restructuring of the energy system that considers both the potential for increased geographical integration while moving gradually towards more sustainable electricity generation may hold significant promise.

This work considers the potential of another renewable technology namely concentrating solar power (CSP) and connecting supply and demand centers via high voltage direct current (HVDC) power lines. Specifically, the focus is on utility-scale solar power generation to supply the needs of growing urban centers of demand. It develops a Geographic Information System-based (GIS) model with a spatial resolution of 30 arc-seconds to calculate the cost evolution of the electricity produced by different technologies of CSP plants and the costs of grid development to selected centers of demand. The results show that major SSA metropolis can benefit from distant CSP economically attractive to compete with inlaid coal-based generation. In 2010, total imports of coal exceeded 1.4 million short tons with consequent economic and environmental costs. Solar towers plants endowed with thermal storage may become a leading technology for smoothing purposes with zero fuel costs. Furthermore, Africa’s vast solar resources are far from urban centers of demand and a transmission system capable to integrate high levels of renewable energy while improving reliability of supply is required. The results of this study point to the importance of SSA centers to rely on a Super Grid approach to take advantage from CSP least-cost potential and to discontinue expensive traditional sources. Overall, solar corridors can integrate with geographically-wide wind and hydro potentials to create clean energy corridors and encourage a transition towards more sustainable energy systems.

Key words: Concentrating solar power, HVDC, Sub-Saharan Africa

Preamble

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Notations and Conventions

Country names and the composition of geographical areas follow those presented in “Standard country or area codes for statistical use”, available at: <http://unstats.un.org/unsd/methods/m49/m49.htm>. Names of cities or urban agglomerations are presented following the names used by National Statistical Offices or the United Nations Demographic Yearbook. When necessary, the administrative subdivision to which a city belongs is added to the city name to identify it unambiguously.

Data collection

This study is based on current and public accessible documents. When required, data was collected through requests and personal communication with relevant specialists.

The following procedure was used to decide which data use when competing data was found:

- Where possible, international databases were used, such as EIA, OCDE Factbook, the World Bank and the African Development Bank Group. For electricity access and energy access targets, country data reported in IRENA was used. For electricity consumption and electricity shares, data reported in the EIA was used as it is the most transparent and complete in terms of accessible country time-series data. For population and development figures, the Department of Economic and Social Affairs (Population Division) data from the United Nations was used.
- When country data was not available from international or national statistical sources, data was obtained from government websites, regional intergovernmental bodies, policy documents and other reports. Most of the African sub-regions have carried out energy forecasts with projections normally based on studies conducted at national level. Regardless of forecasting methods that may vary considerably among countries, the regional plans and related documents entail a wealth of information that was utilized in this study.

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1

Introduction

Nowhere in the world is the gap between available energy resources and access to electricity greater than in SSA. Except for conventional hydropower, the share of renewable energy in the SSA electricity mix is still negligible, although very large potentials of solar and wind energy are available in the region. A critical issue in the development of solar, wind, and other renewable energy sources is the cost. In a policy vacuum, costs of most forms of renewable electricity continue to be relatively expensive when compared with the electricity produced from coal and gas. Although renewables are attractive for many reasons, their use must be seen in the light of competing with fossil-fuel options.

Many net energy importing countries in SSA claim to be too poor to afford renewable sources other than hydro and say that their citizens cannot afford electricity generated by wind or solar (Trieb, et al., 2010). Given the common assumption that renewable energy costs more, this may seem reasonable, but the reality of the oil market indicates that the economic argument may be weak. Oil prices peaked at almost USD 150 a barrel in July 2008 (EIA, 2013), and fell sharply as the global financial crisis hit. Subsequent oil and gas prices have been volatile, with a steady increase driven by the tensions between the West and the Middle East. Exemplary, oil prices were driven higher because of the revolt in Libya in 2011. As is evident, the oil market is erratic, with prices linked to events in the most unstable regions of the world. Coal prices, which historically have been the most stable of the traditional fossil fuels, are now also subject to strong fluctuations because of regulatory restrictions and reserve depletion (EIA, 2013). Many experts believe that these price fluctuations and the factors that drive them mean that the era of stable-priced and abundant fossil energy is coming to an end (Murray, et al., 2012).

Climate change poses a serious threat to Africa's economic growth. While Africa only accounts for less than 4 percent of global greenhouse gases (GHG) emissions, the region is prone to suffer the most from the effects of climate change because of its high climate sensitivity and relative low adaptive capacity (AfDB, 2013). Climate change and changes in fossil-fuel production are generally seen as separate phenomena; however, they are closely related (Murray, et al., 2012). The strategy needed to tackle the economic impacts of resource scarcity and climate change follows the same scheme: to move away from a dependence on fossil-fuel energy sources. Analysts demonstrated that sustainable development and poverty eradication can go jointly with mitigating climate risks when alternatives to dependence of fossil fuels are taken into account (Rogelj, et al., 2013). This brings the vast undeveloped renewable energy resources in Africa essential in discussions about enhancing economic activity and improving quality of life. How can the high populated SSA urban centers of demand, supply their growing economies with secure and affordable electricity? How can developing countries help to reach the world's ambitious climate action goals? Further, due to locational differences between resource availability and where electricity is needed, the role of assessing future transmission systems has become crucial.

SSA needs an electricity supply structure to be secure, inexpensive and compatible with society and the environment. Electricity blackouts occur daily and force people and companies to often rely on expensive diesel generators to meet their electricity demand. Furthermore, African manufacturing enterprises experience power outages on average 56 days per year (Bank, 2013), and this dependence on emergency systems costs to some African economies between 1 percent and 5 percent of their GDP each year (IRENA, 2011). Despite expanding electrification is a significant challenge, the costs associated with the extension of the transmission network are minor in comparison with the cost of inaction when lagging behind basic energy needs (IRENA, 2013).

Previous studies

Multiple studies have established the technical feasibility of developing renewable energy projects in the Sahara Desert for export to Europe. Commissioned by the German Federal Ministry of the Environment, Nature Conservation and Nuclear Safety (BMU), the Trans-CSP study analyzed the potential of renewable electricity in Europe and the capability to provide firm power capacity on demand. The concept comprises an interconnection of the electricity grids of Europe, the Middle East and North Africa (EUMENA) and evaluates the potential benefits of solar power imports from the South. This was first proposed by researchers from the German Aerospace Centre and conceptualized by the Club of Rome as the Desertec concept. In the year 2008 the European Union launched the Mediterranean Solar Plan (MSP) that complements the expansion of national renewable sources of electricity in MENA for its own economic development with solar electricity exports to Europe. High voltage direct current (HVDC) overhead lines and sea cables represent the key technologies for implementing these options for renewable energy import/export. From a different perspective, multiple scattered generation sources aggregated together and managed by intelligent technologies are known as Smart Grid. The Smart Grid approach was introduced in the African roadmap by (Sebitosi, et al., 2009), (Bazilian, et al., 2011) and (Welsch, et al., 2012). Other publications have focused their attention on the role of rural electrification in Africa and improving access to electricity (Turkson, et al., 1999) (Deichmann, et al., 2011). Additionally, a focus was done on grid expansion for countries with low pre-existing electricity coverage (Parshall, et al., 2009).

Moreover, a number of scenarios and strategy studies for the African power sector have been published in recent years (Foster, et al., 2010) (EREC, 2010) (IEA, 2011) (IRENA, 2011). Likewise, several economic-wide scenarios specifically for the SSA sector from current dates to 2030 were also carried out to inform policy-makers and investors and guide them in the power system design (Bazilian, et al., 2011).

Objectives

Despite the effort made by previous studies, the way SSA countries may enhance future power system still uncertain. Some of the immediate challenges that suffers SSA are i) vast gap in generation capacity ii) limited coverage of energy infrastructure in terms of geographic area and end users iii) obsolescence of current generation and transmission technologies and poor state of the overall energy infrastructure iv) and high costs of electricity prices often kept down by subsidies from already constraint public budgets. SSA needs to shift from carbon-intensive to carbon-efficient and sustainable options to enhance future power sector. However, the vast renewable sources are far from urban centers of demand. Therefore, a system capable to transmit large amounts of renewable energy through long distances to centers of high demand may be required. In an effort to address the future African energy roadmap, this study investigates the possibility of Africa to take advantage of its renewable solar potential and to rely on a cross-system electricity mechanism that transmits sustainable utility-scale power generated in remote areas to Africa's growing centers of demand.

Methodology

First, we performed a collection of geographical and meteorological data on the continent for the detection of i) the most appropriate land resources to install concentrating solar plants and ii) its subsequent transmission corridors to urban centers of demand. Consequently, we implemented the siting criteria regarding CSP plant location and deployment of transmission corridors. This results in the development of a mask of non-suitable areas to be excluded when implementing CSP plants and transmission corridors.

Second, we selected the most important sub-Saharan million-plus urban centers of demand. This results in the detection of the potential transmission corridors to connect the centers of demand with the suitable fields to install concentrating solar plants.

Third, we developed a model to identify the cost evolution of the electricity produced by means of different CSP technologies and transmitted through HVDC overhead lines for electricity imports. We obtained a detailed cost curve for CSP generation and transmission in relation with selected centers of demand. Thereafter, we examined the possibility of selected SSA demand nodes to benefit from CSP from an economic and geographical approach. Additionally, we examined into which extent imports of CSP for different regions are an affordable replacement for the electricity produced from traditional fossil fuel sources.

Fourth, we identified the required structure capable to transmit economically attractive renewable electricity from distant solar fields to million-plus urban centers. This analysis is optimized for one resource while enormous renewable potential can be harnessed and integrated into current power systems. With the aim to raise awareness about the tremendous renewable potential and the benefits accrue to cleaner and more secure participatory networks; we identified the possibility of grid integration between the identified solar transmission corridors with SSA renewable potential.

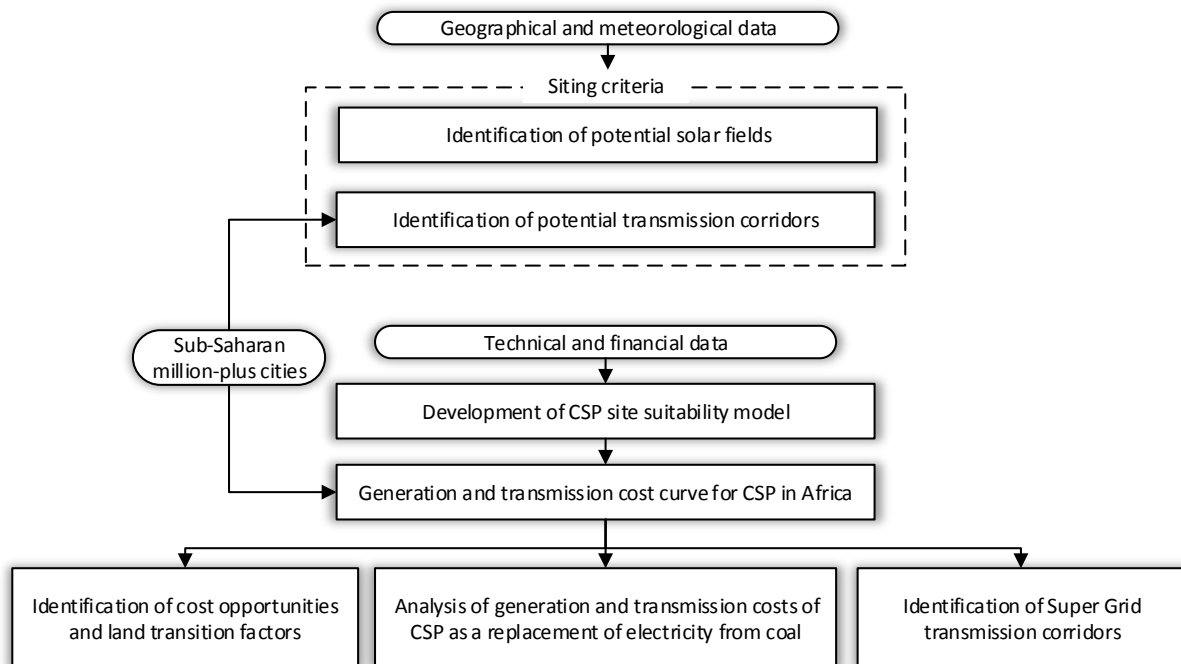


Figure 1 Flowchart of the methodology.

2

Energy problem in sub-Saharan Africa

2.1 Energy demand and consumption patterns

There is a close relationship between economic growth and energy consumption (UNECA, 2011). Emergent global energy consumption will be mostly from developing countries where demand is driven by long-standing economic growth (UNECA, 2011). Increasing per capita income levels, increasing population growth and increasing urbanization rates will drive energy demand and consumption patterns in Africa in the coming decades.

At the start of the 21st century, much of SSA experienced a remarkable economic and population development. Six of the top ten fastest-growing economies worldwide between 2000 and 2010 were in this region, increasing at an average rate of 4 percent per year (IRENA, 2013). If current rates continue, Africa's GDP will increase three-fold by 2030 and seven-fold by 2050 (IRENA, 2013). In addition to GDP growth, population growth and the change in the lifestyle are the other key factors to be considered when analyzing future energy demand. In 2050, total population in SSA is expected to be twice as much as current dates; while urban population is expected to be three times as much as current dates, representing the 60 percent of total population (see Table 1).

Table 1 SSA population growth 2010-2050. The percentage population growth from 2010-2050 is approximately 200 percent, doubling the population in current dates.

Sub-Saharan Africa	2010*	2030*	2050*	%Growth 2010-2030*	%Growth 2030-2050*	%Growth 2010-2050*
Total population	866,948	1,308,461	1,760,724	150	135	203
Urban population	323,525	630,351	1,064,736	195	169	330
Urban % of Total	37.3	48.2	60.5			

Source: (United Nations, 2010). *Projections

In a closer look, we observe how some urban agglomerations will face an unprecedented growth in the incoming decade (see Table 2). Between 2005 and 2010, Lagos (Nigeria, by 1.8 million), Kinshasa (Democratic Republic of the Congo, by 1.6 million) and Luanda (Angola, by 1.2 million) grew by more than one million inhabitants. Nairobi (Kenya) and Abuja (Nigeria) were the next fastest growing agglomerations with additions of 709,000 and 680,000 respectively. Total population of Africa's million-plus cities increased 17.3 million between 2005 and 2010. Since the total urban population increased 63.8 million over the same period (Foster, et al., 2010), it appears that the largest cities are absorbing a significant share (27.1 %) of Africa's urban transition.

Table 2 Africa's ten fastest growing large cities (2005-2010) (2010-2020) Source: (United Nations, 2010). * Projections.

City	2005-2010* Absolute Growth (000s)	City	2010-2020* Absolute Growth (000s)
Lagos	1,811	Kinshasa	4,034
Kinshasa	1,648	Lagos	3,584
Luanda	1,239	Luanda	2,308
Nairobi	709	Dar es Salaam	1,754
Abuja	680	Nairobi	1,669
Dar es Salaam	669	Ouagadougou	1,548
Ouagadougou	581	Cairo	1,539
Abidjan	561	Abidjan	1,375
Dakar	429	Kano	1,100
Alexandria	414	Addis Ababa	1,051

In terms of proportional growth, SSA million-plus cities are expected to expand by an average 32 percent between 2010 and 2020 (Foster, et al., 2010). Abuja (Nigeria), Luanda (Angola), Bamako (Mali), Lubumbashi (Democratic Republic of the Congo) and Nairobi (Kenya) are projected to grow at rates around 48 percent over the current decade, while in Dar es Salaam (Tanzania), Mbuji-Mayi (Democratic Republic of the Congo), Kampala (Uganda) and Niamey (Niger) the range is projected around 54 percent.

In terms of distribution by settlement type (rural hinterlands 50%, remote rural areas 16%, peri-urban cities 10%, intermediate cities 10%, megacities 14%), 34 percent of the population lives in cities and megacities, accounting for a remarkable percentage of the total share (Foster, et al., 2010). An energy system capable to handle the unprecedented growth of sub-Saharan centers is essential for a successful renewable power strategy. Therefore, in following years, the emphasis will lie in a large-scale grid-based energy system capable to supply significant daytime peak demands (IRENA, 2011).

2.1.1 Investment to improve electricity access

Along with India, SSA is the region in the world with the lowest consumption of electricity per capita and the lowest rate of electricity access (Rosnes, et al., 2012). In 2011, the population in SSA excluding South Africa accounted for as much as 12 percent of total world population; however, the total average per capita electricity consumption only reached 153 kWh/year¹ (IRENA, 2011). This figure is very far from the world average electricity consumption of 2,974 kWh/year (see Figure 2) or the electricity consumption in Organisation for Economic Co-operation and Development (OECD) countries, representing just a 5 percent of world's average share.

¹ South Africa is excluded from SSA to give a more focused perspective on energy access, as access to electricity in South Africa is roughly 76 percent (IEA, 2012).

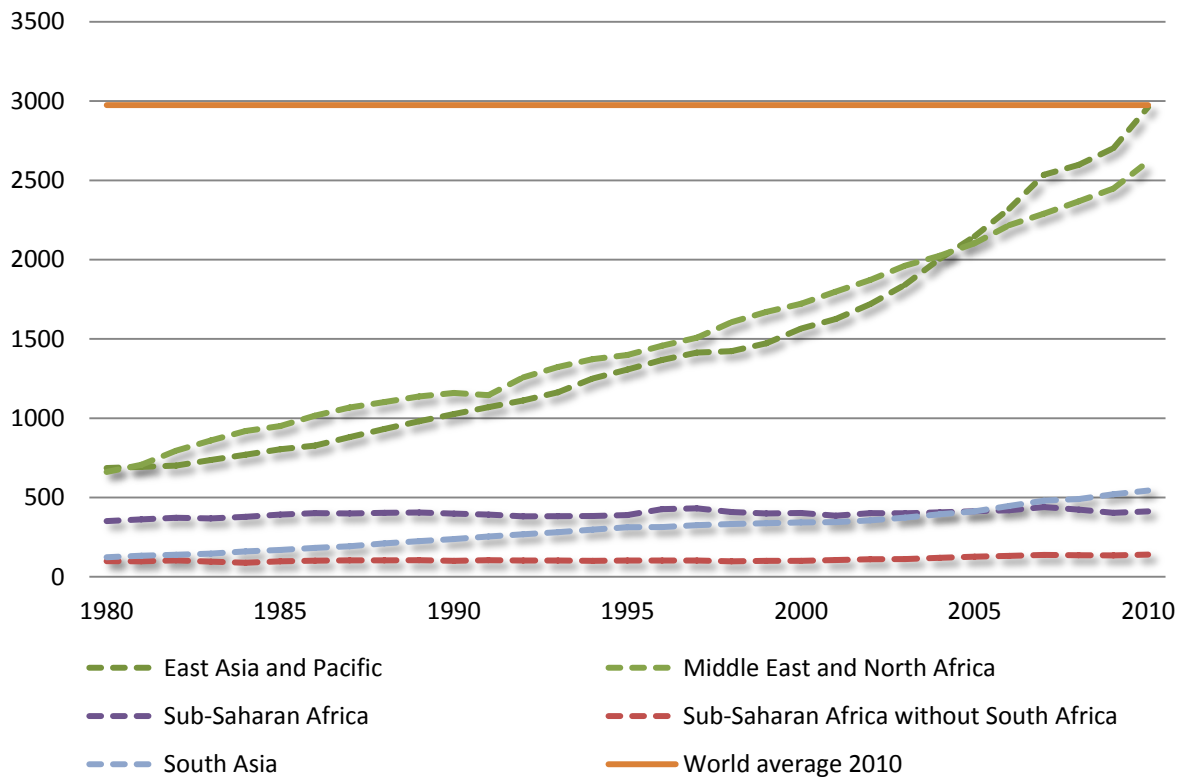


Figure 2 Electric power consumption (kWh/capita) by region, 1980-2010. Electric power consumption measures the production of power plants and combined heat and power plants less transmission, distribution, and transformation losses and own use by heat and power plants. Source: www.worldbank.org

In Africa, not only electricity consumption is low, but access to modern energy differs significantly across the countries. While Northern Africa is virtually electrified, the average electrification rate for SSA is just 35 percent (IRENA, 2011). In 2010, about 590 million African people (57% of the population) had no access to electricity. If current energy access trends continue, this figure is expected to increase significantly to about 655 million people (42% of the population) by 2030 (IRENA, 2013), depriving the majority of the population from the opportunity to improve their quality of life.

To provide Africa with universal electricity access, around USD 20 billion per year are required by 2030 (IEA, 2011). For a sense of scale, currently total African power sector investment, including operation and maintenance, is roughly USD 50 billion per year (IRENA, 2011). This figure can significantly decrease with effective efficiency measures to increase the reliability of energy production and transmission infrastructure. In a closer look to SSA, the investment cost of providing energy access over a 10-year period is between USD 160 and USD 215 billion (Rosnes, et al., 2012). This amount accounts for 100-120 GW of new and refurbished electricity generation capacity plus associated distribution and transmission investments. Additionally, a second study quantifies that the investment cost of providing energy access in SSA is USD 25 billion per year (Bazilian, et al., 2010). Although these figures may seem a significant amount, the cost of inaction may be higher than the cost of increasing electricity supply (IRENA, 2013). To prevent Africa to lag behind the basic energy needs, an increase of generation capacity is required. In a country where renewable sources are large enough, it may seem reasonable that the new generation capacity is provided by cost effective renewable sources. Cost effective renewable sources may not only help to decrease the dependency of many countries in fossil fuel technologies but also help to diversify the energy portfolio and increase security of supply.

2.2 High costs of electricity generation

To meet suppressed demand, keep pace with projected economic growth and offer additional capacity to support electrification expansion, current installed capacity in Africa will need to grow by more than 10 percent annually (Eberhard, et al., 2011). However, an obstacle to increase the generation capacity is the relatively high cost of electricity production². These high costs of electricity production have forced governments to subsidize consumption in several African countries (Ncube, 2013). To illustrate, in 2010, the average effective electricity tariff in Africa was USD 0.14 kWh, while the average production costs were USD 0.18 kWh (Ncube, 2013). This results in an average electricity tariff 23 percent lower than average production costs. Using cost-effective electricity generation technologies may significantly reduce the burden on governments in subsidizing measures. Subsidies in electricity consumption have significant disparities among countries, leading to high differences among effective tariffs. E.g., while electricity tariffs in South Africa and Zambia are among the lowest in the world (USD 0.0364 kWh), prices in Gabon and Djibouti are among the highest (USD 0.1723 kWh)³ (UPDEA, 2009) (Ncube, 2013). Nevertheless, despite substantial variability among countries, average tariffs in Africa are much higher than in other developing regions. Thereby, compared to the USD 0.14 kWh in Africa, the average tariff in East Asia was USD 0.07 kWh, while in South Asia was USD 0.04 kWh (Ncube, 2013).

Fossil-fuel based power generation is the largest source of electricity generation in Africa and is either used in large power plants or emergency generators. Daily power outages in SSA cost more than 5 percent of GDP in Malawi, Uganda and South Africa, and 1-5 percent in Senegal, Kenya and Tanzania (Eberhard, et al., 2011). Emergency solutions such as diesel generators are used to overcome the daily power outages and constitute a high share of total installed capacity in many countries (see Table 3). In West Africa, the installed capacity of diesel generators achieves the 17 percent of total installed capacity, and in the Democratic Republic of the Congo, Equatorial Guinea and Mauritania, the share is higher than 50 percent (IRENA, 2011). Costs associated with emergency generation are in the range of USD 0.2 kWh (Ncube, 2013) to USD 0.4 kWh (IRENA, 2011), which are much higher than the costs of renewable energy solutions⁴. Furthermore, the costs are affected by the volatility of fuel prices.

² Hydropower is not subject to this assumption as produces electricity cheaply. The obstacle for hydropower is the difficult access to capital finance.

³ Comparative of electricity tariff for commercial use (12kW, E=1,800 kWh/month) for South Africa and Gabon.

⁴ Average electricity generation costs without accounting for external costs, are USD 0.065 kWh to USD 0.088 kWh for coal-fired plants, and USD 0.06 kWh to USD 0.13 kWh for natural gas-fired plants (EIA, 2013). When considering external costs, the electricity generation cost for coal increases up to USD 0.1 kWh to USD 0.12 kWh (Greenpeace, 2013). For renewable sources, typical average electricity generation costs for large-scale hydro are USD 0.03 kWh to USD 0.08 kWh, USD 0.05 kWh to USD 0.1 kWh for geothermal and USD 0.05 kWh to USD 0.14 kWh for wind power (IRENA, 2013). While typical average electricity generation costs for large ground-mounted PV systems are from USD 0.16 kWh to 0.27 kWh in southern and northern Europe (IEA-ETSAP, 2013), and costs for CSP are in the range of USD 0.2 kWh to 0.33 kWh for parabolic trough plants and from USD 0.17 kWh to 0.28 kWh for solar tower plants (IRENA, 2013).

Table 3 Diesel-fired emergency generation capacities of selected countries in sub-Sahara Africa. In Angola, the reliance on emergency solutions is almost up to 20% of country's generation capacity.

	Emergency generation capacity (MW)	Emergency Generation Capacity as % of Total	Costs of Emergency Generation as % GDP
Angola	150	18.1%	1.0%
Ghana	80	5.4%	1.9%
Kenya	100	8.3%	1.5%
Senegal	40	16.5%	1.4%
Tanzania	40	4.5%	1.0%

Source: (Eberhard, et al., 2011)

Africa has sufficient renewable unexploited resources to supply the required electricity demand for the power sector. Oppositely, socio-economic developments in recent years suggest that the period of cheap, stable-priced and abundant fossil fuel energy is coming to its end. Either is renewable energy or fossil-fuel based, reducing the cost of generating electricity may significantly improve the overall energy supply in Africa. The most cost-effective ways of generating electricity are:

- i. Through large power plants that benefit from economies of scale. In Africa, the power generation industry is often dominated by costly small-scale power systems that connect to the national grid; these systems are broadly used in remote areas and lead to increase general transmission and distribution costs mainly due to high electrical losses⁵.
- ii. Exploitation of cost-effective renewable energy resources. Exploitation of renewable energy resources especially of those whose costs are rapidly declining and those who can supply electricity at times when it is demanded.
- iii. Restructuration of the power generation industry to attract private-sector participation. Mobilizing private sector investment will strengthen supply conditions in the industry. However, there are several obstacles that constrain private investment. Thereby, to overcome these obstacles, it is essential that the electricity sector adopts some reforms. I.e., reforms in the tariff structure and remove subsidies to public power utilities⁶ to achieve competitive price-setting in the market (Ncube, 2013).
- iv. Increase investments in infrastructure. Total power capacity installed in Africa is 147 GW; this is equal to the total capacity that China installs every one to two years (Ncube, 2013). However, to provide full access to SSA population for 2030, 250 GW of capacity are needed to be installed starting in current dates (IRENA, 2011). Moreover, to decrease generation costs, the state of the power infrastructure has to improve to reduce transmission leakages. This would avoid the extra production needed to overcome the leaks.

⁵ System losses in Botswana and South Africa are the lowest in Africa, with 10 percent and 15 percent losses respectively. Highest system losses can be found in Nigeria, where utilities only capture the 25 percent of the revenues owed. Hidden costs of distribution losses in SSA are usually more than 0.5 percent of GDP, and in some countries may be as large as 1.2 percent of GDP (Antmann, 2009).

⁶ Industrial consumption of electricity and coal in South Africa is subsidized directly from the state as well as through cross-subsidies. This results in very inefficient patterns of energy use. E.g., in South Africa the double of energy is consumed to make a ton of steel compared to some of the most efficient countries. As a result, the country is among the highest emitters of CO₂ in the world (Greenpeace, 2013).

- v. Lastly, to meet the requirement of 250 GW of power capacity, transmission grid infrastructures need to increase in the most cost-effective way. Improving cross-regional power trade interconnections is essential to create economies of scale that lead to cost reduction in transmission infrastructure.

3

Energy sector in sub-Saharan Africa

3.1 Overview of the energy sector in SSA countries

The bulk of power generation in Africa is coal-based. South Africa coal deposits account for the 90 percent of the 53 billion coal reserves in Africa (Eskom, 2013). To overcome the high losses incurred along the transmission lines to distant centers of demand, generation capacity must overproduce to compensate these losses. When the generation capacity is coal-based, this results in additional stress on the environment in terms of climate change and, from a local perspective, in respiratory disorders particularly in young. Likewise, it results in additional stress to water resources both at surface and underground level (when coal mining and cooling of power plants are considered). Oil-fired power in Africa accounts for 17 percent of capacity and 12 percent of generation (IRENA, 2011). This relatively high share when compared with the world average of 5 percent is somewhat due to the costly decision made in the 1990s, when several African nations decided to build oil-based power plants when oil was at its minimum prices (IRENA, 2011). Recent decisions to build coal-fired power plants may revive the same situation lived with oil-fired plants when fuel prices were low. However, future fuel cost developments are uncertain and both oil-fired and coal-fired plants may be affected by several risks. E.g., risk related to the increment of fossil fuel prices and risk related to the increment of CO₂ price, since future regimes of CO₂ emissions in developing countries are highly uncertain (IRENA, 2011).

Hydropower generation accounts for 16 percent of total generation and is mainly concentrated in the lower Congo River and the upper Nile River. The Grand Inga project in the lower Congo has a potential of around 44 GW, with just 1.7 GW being installed, and the potential on the upper Nile is around 30 GW just in Ethiopia. Further hydropower potential can be found in the tributaries of Angola (18 GW), Mozambique (12 GW), Niger (10 GW in Nigeria), and in the Congo (6 GW in Gabon) (IRENA, 2011).

Installed wind capacity in Africa was roughly 1 GW in 2010 (GWEC, 2010). Egypt, with the largest installed capacity (550 MW), together with Morocco (286 MW), Tunisia (114 MW) and Cape Verde (26 MW), account for roughly 98 percent of Africa's total installed capacity with ambitious plans to continue growing (IRENA, 2011). The 87 percent of the high quality resources are located in the Northern, Eastern and Southern coastal zones. These regions embrace world-class wind resources and, if no large hydro generation resources are nearby, wind may be the cheapest renewable technology for those regions.

Solar photovoltaic (PV) accounts for more than 95 percent of the solar power generation, while concentrating solar power (CSP) accounts for less than 5 percent. Africa has only one utility-scale PV plant (7.5 MW in Cape Verde) and the total installed PV capacity is in the order of 160 MW

(Greenpeace, 2011). PV and CSP compete to some extent for utility-scale applications. Nevertheless, given important operational differences between one and the other, it is likely that they will coexist as they have the potential to complement each other.

Geothermal energy contributes to Africa's electricity generation with around 210 MW of installed capacity (IRENA, 2011). However, Africa's geothermal resources are not evenly distributed. East Africa, especially the East African Rift system, has the best resources. The generation potential along the 4,000 km of the East African Rift system is estimated in 7 GW–15 GW (AUGRP, 2010), with Kenya and Ethiopia standing with the highest geothermal resources (IRENA, 2011).

Lastly, the most important source of bioenergy in Africa is sugar cane bagasse with 94 percent of the 860 MW of installed generation capacity in 2011 (IRENA, 2011). The technical potential of bioenergy for power generation is estimated in 2,631 TWh, with the 60 percent located in Central Africa.

However, to surpass increasing energy needs, Africa needs further generation and transmission capacity. Analysts from the World Bank Africa's Infrastructure estimated that 73,193 MW of new generation capacity and 22,120 MW of new cross-border interconnection capacity are required to overcome these needs (Foster, et al., 2010). To help achieving these capacity goals, several projects relying on fossil-fuel based and renewable generation are expected to be on the ground in short. E.g., South Africa is expected to add two new coal-fired power plants (Medupi and Kusile) with around 4,800 MW each, which will supply coal-based electricity to the national grid. These plants are among the largest coal plants worldwide and plan to be commissioned in phases between 2014 and 2020 (KPMG, 2011). Natural-gas fired power plants also will be used to fill the generation gap. Countries in Western Africa (along the Western African Gas Pipeline, WAGP) together with North and South Sudan, Angola and Mozambique, have significant gas reserves. E.g., Lake Kivu, on the border of Democratic Republic of the Congo and Rwanda, has methane reserves to support a gas turbine's installed capacity of 1,000 MW over the next 50 years (KPMG, 2011).

Renewable installed capacity will be mainly hydro-based. New hydropower capacity will be mostly located in the Grand Inga region with enormous hydropower potential. The Inga III project in the Democratic Republic of the Congo will add 4,300 MW–5,000 MW by 2017–2018, becoming the greatest hydropower project in Africa and fifth in the world (KPMG, 2011) (IRENA, 2011). High voltage (HVDC) lines will transmit the power generated from the Inga III to Zambia, Zimbabwe, the Republic of South Africa and the Republic of Congo (IRENA, 2011). Northern and Eastern coastal zones have vast wind potential that can supply the demand centers in Southern Africa, although transmission lines with distances up to 2,000–3,000 Km would be required to connect to the centers of demand (IRENA, 2011). In the region of Lake Turkana, Kenya, new geothermal plants as well as wind and solar projects are planned to be installed. E.g., the Lake Turkana wind power project will provide 300 MW to Kenya's national electricity grid. In Senegal, 125 MW of wind capacity is in an advanced stage. Also, South Africa promotes the development of renewable energy projects by Independent Power Producers (IPP) through the Renewable Energy Program. In the first phase, the RE –IPP program has allocated 3,750 MW to different renewable energy technologies (KPMG, 2011).

Adding all the projected renewable potential will only cover roughly 20 percent of the generation capacity required by the World Bank. However, many countries still have difficulties to supply sufficient electricity to the growing demand for households, commercial and industrial customers. Additional generation capacity is needed. Climate change regulations for developing countries may become eventually more restrictive, resulting in additional costs especially for coal and liquid fuel generated power. This brings an opportunity for renewable sources with null fuel costs -except for biomass and biogas- to provide reliable energy and help to reduce the dependency on fossil fuels.

3.1.1 Sub-Saharan Africa unexploited renewable resources

Africa's renewable energy resources are characterized by hydroelectric potential in Central and Eastern Africa, biomass in Central Africa, wind energy in Eastern and Northern Africa and a widespread solar potential depending on the generation technology. In SSA, hydropower has an enormous exploitable potential. Hydropower accounts for almost 20 percent of SSA's current electricity power generation, which represents only a small fraction of the commercially exploitable potential (IEA, 2013). Large hydropower is the least-cost renewable energy solution today, followed by onshore wind (if intermittency is not considered), biomass, geothermal and solar (IRENA, 2011). Although solar power has higher cost compared with other renewable technologies, solar power has by far the largest renewable resource potential in Africa when compared with the other renewable sources (see Table 4). Additionally, costs may reduce as a result of technology improvements and by selecting the highest-quality solar resources. PV potential is evenly distributed across Africa except for the central region where hydropower and biomass are the main actors. Instead, CSP potential is distributed along the northern, eastern and southern regions. When excluding North Africa, PV technical potential accounts for 5,477 TWh/year, closely followed by CSP with a technical potential of 3,784 TWh/year.

Table 4 Technical potential for power generation from renewable sources in Africa (TWh/year)⁷. Source: (IRENA, 2011).

	CSP	PV	Wind	Hydro	Biomass	Geothermal
Central Africa	299	616	120	1,057	1,572	
Eastern Africa	1,758	2,195	1,443	578	642	88
Northern Africa	935	1,090	1,014	78	257	
Southern Africa	1,500	1,628	852	100	96	
Western Africa	227	1,038	394	105	64	
Total Africa-Northern Africa	3,784	5,477	2,809	1,766	2,374	88
Total Africa	4,719	6,567	3,823	1,844	2,631	88

3.2 Power sector reform in progress

Power sector reform in developing countries initiated during the last two decades, was mainly influenced by the shortage of generation capacity and the inadequate funding for future investments. Matters like social and environmental welfare and public benefits, were not regarded as the main concern in the power sector (D'Sa, 2005). Many countries have followed a traditional approach of energy planning focused on supply side options. The traditional approach is driven by the idea of "supply and provide", which forecasts energy requirements and design the power systems to match accordingly (Graeber, et al., 2005). However, no demand-side measures are considered. To address the complex nature of modern electricity systems, an integrated planning that considers equitable access to society and integrates smart use of resources was well-aimed.

Integrated resource planning (IRP) is described as "an approach through which the estimated requirement for electricity services during the planning period is met with a least-cost combination of supply and end-use efficiency measures, while incorporating concerns such as equity, environmental protection, reliability and other country-specific goals" (D'Sa, 2005). When applied to the power sector, IRP approach no longer considers supplied-side but also demand-side options to meet energy

⁷ Numbers are subject to uncertainty, typically +/- 50%

needs while minimizing the costs accruing to producers and society (D'Sa, 2005). Moreover, IRP not only considers endogenic resources but may also extend beyond national borders. This is the case of small countries where energy resource potential is underdeveloped (e.g., hydropower) or where sharing cross-border infrastructure with neighboring countries may involve economies of scale (Bazilian, et al., 2011). To achieve the development goals of one particular region (e.g., sustainable development), and to overcome environmental protection and reliability issues, IRP is characterized by an upcoming higher penetration of distributed energy resources (DERs). With an increasing share of variable renewables connected to the grid, it becomes more important to consider the impact of renewable sources on grid stability and reliability. However, with the current transmission system and depending on the power generation system, local stability issues in Africa can arise when renewables just account for 10 percent of total generation (IRENA, 2011). The traditional approach of the energy system must evolve to increase grid stability in Africa's energy network. Diversification of renewable resources located along different climate zones, grid connection with neighboring regions, electricity storage (e.g., CSP and pumped hydro storage), backup capacity (e.g., gas turbines) and the integration of mini-grids, are essential in plans for grid expansion (IRENA, 2011) (Bazilian, et al., 2011). Among the priorities advocated by African policy makers, there is a plan to improve current grid interconnection with the purpose of improving grid stability and security of supply by facilitating the development of large electricity generation projects.

3.2.1 The promise of regional power trade

Although Africa is well endowed with renewable energy resources, only a small fraction of its potential has been developed. Moreover, some of Africa's most cost-effective energy resources are far from major centers of demand. One of the major initiatives implemented to increase system reliability and improve and ensure electricity supply, is the creation of regional power pools. Power pools aim to facilitate the provision of reliable and low cost electricity to consumers by diversifying the power mix and applying economies of scale⁸ (KPMG, 2011). The region's major exporters provide the electricity mainly from hydropower (Democratic Republic of the Congo, Mozambique, and Zambia), natural gas (Côte d'Ivoire and Nigeria), and coal (South Africa). Countries that rely on oil or diesel generators do not export electricity (Eberhard, et al., 2011).

Power pools started with the creation of the Southern African Power Pool (SAPP) in 1995, followed by the creation of the Western African Power Pool (WAPP) in 1999, the Central African (CAPP) in 2003, and the Eastern African Power Pools (EAPP) in 2006. After the four power pools were established, regional electricity regulators were constituted by the Southern African Development Community (SADC), the Economic Community of West African States (ECOWAS) and the Common Market for Eastern and Southern Africa region (COMESA) (Bazilian, et al., 2011).

Several studies developed by regional economic communities and national governments⁹ have identified the need for accelerated electrification rates. Those studies result in ambitious regional

⁸ Economies of scale not only have to consider an increase of the scale of renewable power plants but also the reduction in number of distribution system operators (DSO) among countries. The high number of separate DSOs in local municipalities results in loss of economies of scale. It is difficult for small municipalities to maintain the distribution networks and most need additional investment (KPMG, 2011).

⁹ The Forum of Energy Ministers of Africa's (FEMA) Position Paper on Energy and the MDGs (WHO and UNDP, 2009); The Economic and Monetary Community of Central Africa's (CEMAC) Energy Action plan with energy and electricity access goals (WHO and UNDP, 2009); The Treaty Establishing the Economic Community of Central African States (ECCAS); The economic Community of West African States' (ECOWAS) Energy Protocol (ECOWAS, 2003) and its White Paper for a Regional Policy (ECOWAS, 2006); The Common Market

energy strategies, policies and targets that are supported by national electrification policies. Consequently, the countries and regions of SSA are acutely aware of energy access issues, both in terms of quantity and quality. Therefore, the immediate need is to translate this understanding into the deployment of electricity services on the ground (Bazilian, et al., 2011). Figure 3 provides an overview of the electricity grid extension in 2009 and the grid extension foreseen by the regional power pools and utilities. As power pools can be operational only in regions with fairly developed grid interconnections, it is clear that the scale of opportunity of SSA for optimizing infrastructure design is still being large.

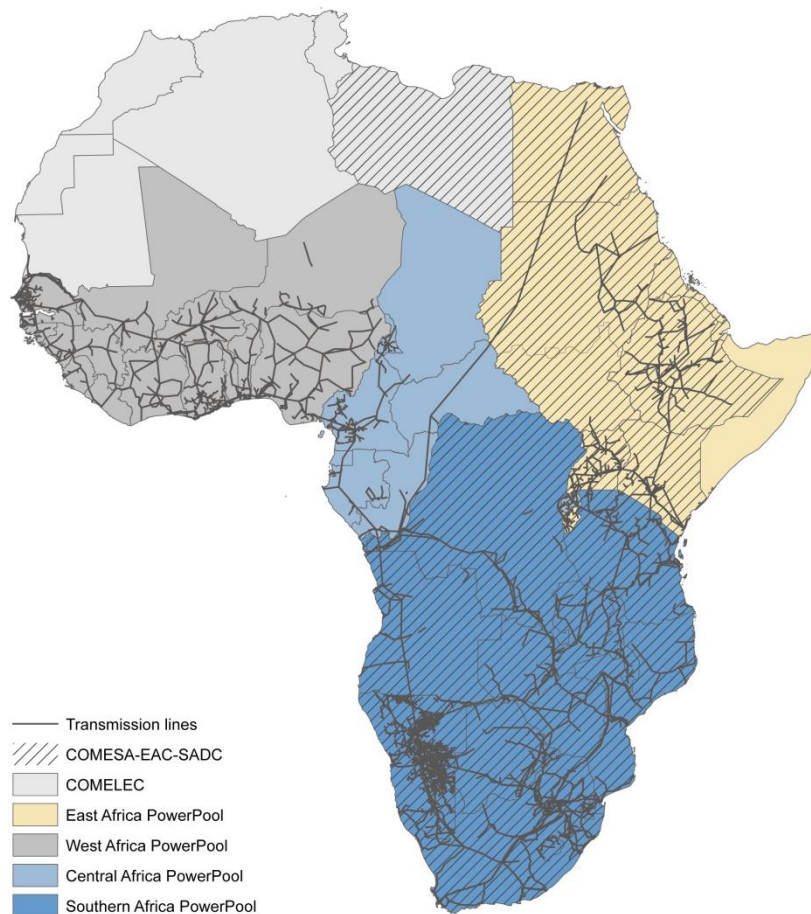


Figure 3 Regional Power pools in SSA¹⁰. It shows the transmission lines already installed and the planned transmission lines (2009). The nature of the existing power infrastructure in Africa is primarily due to the mining activities, which are expected to continue being the primary drivers of the economy in many countries. COMELEC (Comité Maghrébin de l'Electricité). Source: www.infrastructureafrica.org, www.trademarksa.org.

for Eastern and Southern Africa's (COMESA) Energy Programme (COMESA, 2009a); The East African Community (EAC) Regional Strategy on Scaling-up Access to Modern Energy Services (EAC 2009) and its Power Master Plan Study (BKS Acres, 2005); The Southern African Development Community's (SADC) Protocol on Energy (Kritzinger-van Niekerk, et al., 2002); and its Regional Indicative Strategic Development Plan (RISDP) (SADC, 2003).

3.2.2 Potential benefits and constraints of expanded regional power trade

The expected benefits from power pools include “reduction in capital and operating costs through improved coordination among power utilities, optimization of generation resources with large units, improved power system reliability with reserve sharing, enhanced security of supply through mutual assistance, improved investment climate through pooling risks, coordination of generation and transmission expansion increase in inter-country electricity exchanges, and development of regional markets for electricity” (Sebitosi, et al., 2009). However, most of the utilities that planned to participate in the regional power trading have a history of fundamental structural and administrative weaknesses, intensified by an ineffective management due to political patronage and continuous regional civil conflicts (Sebitosi, et al., 2009). Specifically, challenges identified in the SAPP include lack of trust and confidence among pool members (e.g., some SAPP utilities have imposed confidentiality requirements on their generator cost data despite the provisions about disclosing information agreed by the pool members), failure to deliver adequate generating capacity and failure to properly design reserve margins (Sebitosi, et al., 2009). Moreover, most of the power utilities of the power pool lack a legal framework for electricity trading, rules for access to the transmission grid, a regional regulation and correct mechanisms for dispute resolution (Sebitosi, et al., 2009). So far, despite numerous successes in promoting regional power trade, overall trading volume in the region is very limited (Eberhard, et al., 2011).

Regardless of the several constraints, if regional power trading was pursued to its full economic potential could save to SSA around USD 2.7 billion per year, roughly the 5 percent of the annual cost of meeting power needs (Rosnes , et al., 2009). However, current savings of regional power trading are not equitably shared between users and large power suppliers (Sebitosi, et al., 2009). This is exemplified by the deadlock of Cahora Bassa hydro plant and the company Eskom in South Africa, 2003. Regional power trading from the Cahora Bassa hydro project provided some savings for the energy users (particularly to Eskom); however, the benefits for the large power supplier (Hidroelectrica de Cahora Bassa, HCB) were not apparent. This led to disputes over tariffs and to overnight suspensions of the power supply (Africa, 2002).

4

Evolving electricity system

4.1 Electricity system needs

Power systems vary hugely in terms of scale, generation, interconnection, storage, transmission and distribution, and demand and market behavior (IEA, 2010). Therefore, there is no specific way to handle large-scale variable energy shares, but it depends on the characteristics of the system into which the variable energy is being supplied. In Africa, variable energy in the power mix is increasing (IRENA, 2011) and according to the BLUE Map scenario of the International Energy Agency (IEA, 2010); variable electricity is expected to continue increasing over the next decades (see Figure 4). A power system flexible enough to efficiently handle a mix of both centralized and distributed, non-variable and variable generation, coupled with new demand profiles due to changes in lifestyles (Foster, et al., 2010) will be required. However, as penetration rates of variable electricity increases, it becomes more difficult to ensure reliability of supply relying merely on conventional grid architectures and limited flexibility (IRENA, 2011).

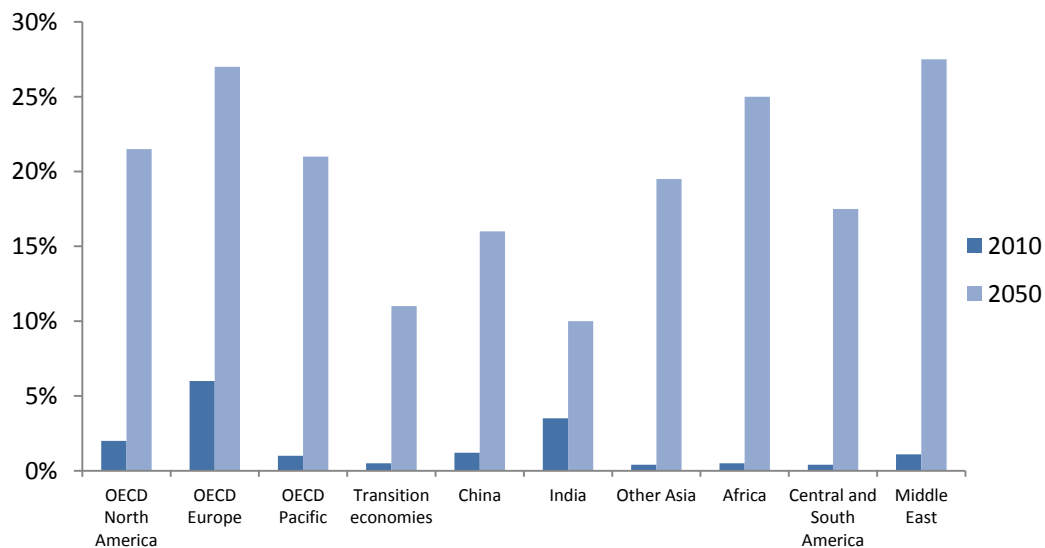


Figure 4 Portion of variable generation of electricity by region (BLUE Map Scenario, IEA 2010). In Africa, variable electricity is expected to account for 25 percent of total share in 2050. Variable generation technologies produce electricity dependent on climatic or other conditions; therefore there is no guarantee that electricity can be dispatched when required. This includes electricity generation from wind, photovoltaic, run-of-river hydro, combined heat and power, and tidal technologies. Adapted from (IEA, 2010) and (IEA, 2011).

The output from variable renewable energy such as wind, solar PV or run-of-river hydropower, requires greater power system flexibility than in the case of most conventional sources to supplement quickly periods of low output and to manage production peaks (IEA, 2010). Low shares of variable renewables have shown to have slight or no impact on power system operation (IEA, 2010). But in Africa, local stability issues can arise when renewables account for as little as 10 percent of total generation (IRENA, 2011). To increase flexibility of the electricity system and allow the integration of larger proportions of variable renewables (IEA, 2010), a number of options and technologies including smart grids and additional storage capacity will need to be considered into modern energy planning:

- i. Large scale interconnections - Super Grid
Increasing the size of balancing areas i) to supply energy from a geographically wider area to rely on a smaller proportion of reserve generation capacity ii) to maintain system reliability iii) to enable imbalances to be resolved where electricity cost less iv) and to take advantage of the smoother average generation that is likely to result from a large geographic spread of variable energy (IEA, 2010).
- ii. Demand side response - Smart Grid
Multiple scattered generation sources aggregated together and managed by “smart technologies” to improve energy services. With the adoption of smart meters, for instance, consumers can be aware when to shift electricity consumption from periods of high demand to periods of lower demand. Therefore, consumers will be able to monitor their electricity use and thus decrease the electricity bills (IEA, 2011).
- iii. Storage
The fluctuating nature of generation from renewable energy requires new methods of balancing load and generation. Cross-regional Super Grid infrastructure will help to level out differences in the local generation. This can be done through connections to large scale energy storages like CSP plants or pumped hydro power plants.

4.2 Rationale for Super Grid technology

As previously mentioned, power trade at full potential can save to SSA countries an estimated USD 2.7 billion in annual costs of power system operation and development (Rosnes , et al., 2009). However, regional planning, harmonization of standards and procedures, equitable commercial terms and coordination at power pools level are essential elements of successful regional integration (IRENA, 2013). To enhance regional trade, a structure capable to connect abundant and low-cost renewable energy resources to centers of demand is required. The concept of an electricity transmission system designed to facilitate large-scale sustainable power generation in remote areas to centers of consumption is known as Super Grid (Blarke, et al., 2013). A Super Grid approach relies on an electricity exchange mechanism (export and import) across systems with different intermittent resources, balancing technologies and demand patterns. This mechanism may allow to handle large-scale penetration of intermittent resources in a wide supply area without any short to medium-term need for storage, and to deliver high quality balancing services to the grid (Blarke, et al., 2013). An ideal candidate to deliver balancing services and to bridge electricity needs during nights and cloudy days is CSP with thermal storage. Selecting the best solar sites across Africa will help keeping the costs of applying this smoothing option within a reasonable margin. To deal with the remaining fluctuations, hydropower with storage is already a cheap technology with sufficient capacity installed mainly in Central Africa. Fast and flexible biomass power stations may do the rest to complement the balancing system. Therefore, the rationale behind the Super Grid is to create efficient energy transfer corridors across African states to accrue the following benefits:

- i. Improving security of supply through diversification of electricity generation

- ii. Reducing regional isolation
- iii. Increasing cross-border capacity for the import and export of electricity
- iv. Improving the reliability of the electricity network
- v. Promoting renewable energy sources

4.2.1 HVDC for long transmission distance

Electricity can be transmitted at high voltages either in alternating current (HVAC) or direct current (HVDC). Therefore, the selection of each technology must be assessed depending on the characteristics of a particular project. However, HVDC (High Voltage Direct Current) technology offers notorious benefits when it comes to import electricity across large distances¹¹. The HVDC technology is used to transmit electricity over long distances by overhead transmission lines or submarine cables. In a HVDC system, electric power flows from one point in a three-phase AC network, is converted to DC in a converter station and transmitted to the receiving point by an overhead line or cable (ABB, 2013). There, electric power is converted back to AC in another converter station and injected into the receiving AC network (see Figure 5).

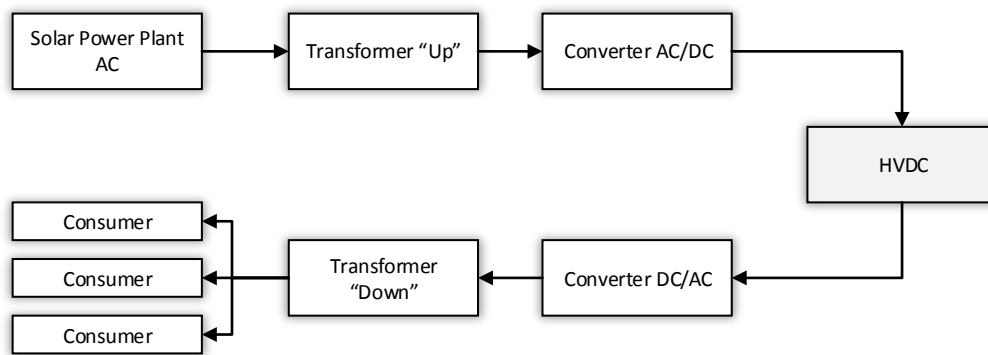


Figure 5 Schematic diagram of interconnecting HVDC and AC technologies.

To illustrate the benefits of HVDC lines in contrast with HVAC, a comparison of the system losses and investment costs of AC and DC transmission lines for comparable voltage levels is done in Table 5. First, the investment costs of HVDC line are considerably lower than for HVAC lines. However, because generation plants and consumer goods were developed using AC technology, a conversion (AC-DC rectifier / DC-AC inverter) is required at both extremes of the line. Moreover, the cost of the converter stations is far more expensive than transformer stations used for AC. Second, the cost of AC and DC overhead lines is similar for low voltage levels, but at high voltage levels i.e. 800 kV, DC lines are less expensive than comparable AC lines (see Table 5) (Siemens, 2012). Moreover, AC power links become uneconomic for long distance transmissions because of the need to supply reactive power,¹² especially for underground or undersea cables (see Figure 6). From an economic

¹¹ "High-voltage" is defined for transmission facilities operating at 230 kV or higher (Mason, et al., 2012).

¹² Reactive power flow in the network is an annoying side-effect of AC transmission because: i) Reactive power cannot be transformed into other forms of energy but causes loading of the transmission line and other components. ii) Causes extra power losses. iii) Contributes to most of the fluctuations of the AC voltage level. When reactive power is not well balanced, it can lead to voltage collapse or excessive over-voltages threatening system stability (Schettler, et al., 2012).

perspective, for shorter distances and lower voltages AC is often the preferred option, but for distances above 600-800 km¹³, HVDC lines become more economically attractive (Siemens, 2012).

In addition to the cost of components, the transmission losses of HVDC overhead lines are roughly a half than those of HVAC lines of similar scope. This difference in losses occurs over the increase of voltage which proved to have a stronger impact on DC technology (Trieb, et al., 2009). To illustrate, we compare a 500 kV HVDC with a 500 kV HVAC overhead line. The HVDC line has losses around 7 percent each 1000 km, while the HVAC line has losses around 10 percent each 1000 km. Also, the different losses in the converter and transformer station have to be considered. Converter stations have losses in the range of 0.8 percent in front 0.2 percent losses in the transformer stations (Greenwire, 2011) (Trieb, et al., 2009). However, this is not such a big difference and do not influence much in the selection.

Table 5 Cost comparison of transmission alternatives.

	Line voltage (kV)	Rated power (MW)	Transmission Line Cost (USD/Mile)	Line losses (%)	Lifetime (years)
±800 kV HVDC Bi-pole	800	3000	1,582,000	3.63	40
±600 kV HVDC Bi-pole	600	3000	1,456,000	5.29	40
±500 kV HVDC Bi-pole	500	3000	1,400,000	6.96	40
765 kV HVAC Single circuit	765	3000	3,600,000	4.56	40
500 kV HVAC Double circuit	525	3000	2,720,000	9.69	40
500 kV HVAC Two single circuits	525	3000	1,700,000	9.69	40

Source: (ABB, 2007)¹⁴. Line losses given by (ABB, 2007) may be subject to converter losses. Lines of 2500 MW and ±800 kV HVDC may achieve efficiencies up to 2.6 percent each 1,000 km (Siemens, 2012). 1 Mile = 1.609 km.

Moving from the technical to the environmental level, the extent of influence on the environment of the two technologies becomes crucial when assessing viability of transmission projects. The most important point to remark is that a transmission line with the same voltage level in HVAC and HVDC, HVDC lines are capable to transport considerably more capacity than HVAC lines. Therefore, the land coverage and the associated right-of-way cost for a HVDC overhead transmission line is not as high as for an AC line (WECC, 2013). Moreover, electricity is usually transmitted using three-phase AC systems and therefore needs to apply three lines. In DC systems, only two conductors are necessary to transmit electricity. Thereby, DC long distance transmissions require a narrower power corridor than AC lines, and consequently produce less impact on the environment (see Figure 7). However, there still some environmental issues that must be considered for the converter stations, such as: visual impact, audible noise, electromagnetic compatibility and use of ground or sea return lanes in monopolar operation (Larruskain, et al., 2005).

¹³ This number depends on several factors (both for submarine cables and overhead lines) and an analysis is needed for each individual case. 600 km to 800 km is an approximate threshold and the importance of the break-even-distance should not be over-stressed, since several other factors such as controllability are important in the selection between AC or HVDC technologies.

¹⁴ Black & Veatch together with the Western Electricity Coordinating Council (WECC) developed updated assumptions on transmission line and substation costs. The results were peer review by regional transmission experts and are within the range of the costs given by ABB: 500 kV HVDC Bi-pole = 1,484,000 \$/Mile, 500 kV Double Circuit = 2,967,000 \$/Mile, 500 kV Single Circuit = 1,854,000 \$/Mile (Mason, et al., 2012).

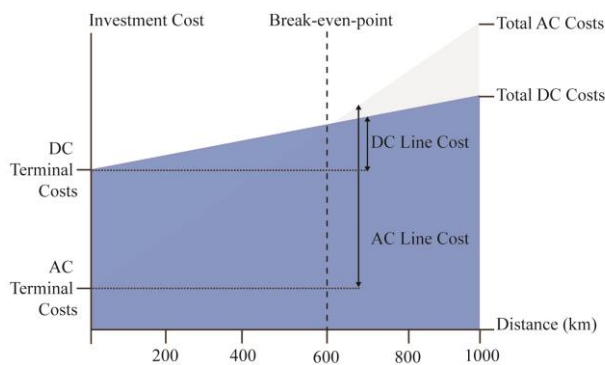


Figure 6 Comparison of AC and DC transmission costs. HVDC has higher initial costs –the converter stations- but because the overhead lines and the cables are less expensive per kilometer for DC, there is a break-even-point usually around 600 km–800 km. Source: Adapted from <http://www.abb.com>

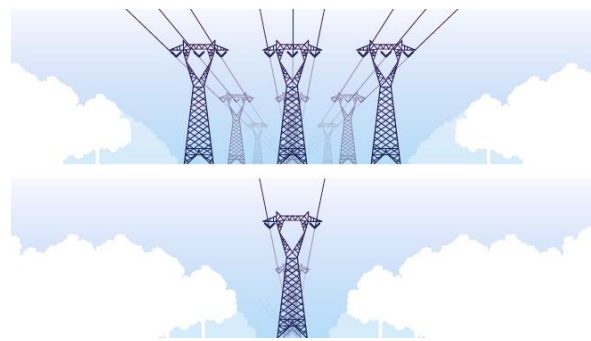


Figure 7 Comparison of the number of lines required and the difference of land required to transmit the same power amount. DC systems (down) have less land coverage requirements than AC systems (up). Source: Adapted from <http://www.abb.com>

Regarding technical security, sometimes it is difficult or impossible to connect two AC networks due to stability reasons and HVDC is the only way to make the exchange of power between the two networks (Larruskain, et al., 2005). With HVDC, it is also possible to connect grids that otherwise could not be connected, e.g., networks with different nominal frequencies -50 Hz and 60 Hz- (Larruskain, et al., 2005). Moreover, it results very easy to control the active power in the transmission link¹⁵. Recent projects to transmit renewable energy from remote hydro generation plants (e.g., Xiangjiaba-Shanghai transmission project in China), have led to the development of Ultra High Voltage Direct Current (UHVDC) at 800 kV DC (UHVDC refers to systems with a voltage level above ± 600 kV) that transmit capacities up to 5,000 MW–8,000 MW. Therefore, 800 kV UHVDC is an appropriate application for bulk power transmissions over long distances. This is because unlike AC transmission, HVDC is not constraint by stability limits and does not require intermediate substations for voltage and reactive power control. Nevertheless, reactive power is also required at an HVDC link. Specifically, it is required at both terminal stations to support converter requirements and maintain appropriate AC voltage level at the terminals (ABB, 2013).

4.2.2 Super Grid – Global approach

In China, India, Brazil and Africa, the demand for energy is growing dramatically. In China, the power is needed in the East and South; therefore large amounts of electricity need to be transmitted by exploiting the hydropower reserves in the West (ABB, 2013). To benefit from the large hydropower potential, lines over 2,000 km from one side of the country to the other are required (ABB, 2013). Commonly, 500 kV DC has been used for long-distance transmission for distances up to 1,000 km. However, new transmission lines of 800 kV HVDC allow to transmit power as far as 3,000 km with reasonable transmission losses (ABB, 2013). As a result of the high electricity demand, China is planning to build one 800 kV HVDC line per year over the next ten years, with line capacities between

¹⁵ An HVDC transmission link is composed by: a station to convert the AC to DC, the transmission equipment itself in the form of cables and overhead lines, and another station that converts DC back into AC so to be supplied to consumers.

5,000 MW and 6,400 MW (ABB, 2013). Additionally, India has plans to expand hydropower in the North-Eastern part of the country and expects to build one 800 kV line every two years over the next decade, with a capacity of 6,000 MW per line. In Africa, the Inga-Kolwezi interconnexion transmits power from the Inga falls in the Congo River to the Democratic Republic of the Congo. It consists of a ± 500 kV HVDC, 560 MW transmission line of 1,700 km length with two monopolar lines in case of a monopolar line outage. The Caprivi Link interconnector connects the electricity grids in Namibia and Zambia along a transmission line of ± 350 kV HVDC, 300 MW and 950 km length. To ensure reliable power transfer capability and help to stabilize the weak networks of East and West of the Southern African Power Pool, an HVDC Light system using an overhead line was selected for the first time. Lastly, the Cahora Bassa interconnexion provides an important power import facility to the South African grid by transmitting power from the Cahora Bassa generating station on the Zambezi River in Northern Mozambique. It consists of a ± 533 kV, 1,920 MW transmission line of 1,420 km length with two parallel monopolar lines (ABB, 2013).

4.3 Rationale for Smart Grid technology

Demand-side actions such as energy efficiency measures will be increasingly important in a gradual transition to system more dependent on renewable sources. Multiple scattered generation sources aggregated together and managed by intelligent technologies to efficiently match the volatile supply with the demand, are known as Smart Grid (IEA, 2011). In the Smart Grid approach, the actors involved vary widely regarding their use of the grid: some actors produce energy, others consume energy and some do both. The actors behave largely autonomously but have to communicate and work together to balance energy supply and demand (Sebitosi, et al., 2009).

Electricity is the component that is growing faster in the global energy demand (IEA, 2011). OECD countries expect to have modest demand growth rates, and the growth will be based on the high levels of current demand. In developing countries, however, with high energy demand growths, Smart Grid technologies can be incorporated in new built infrastructure, offering better market-function capabilities at the same time that more efficient operation (IEA, 2011). Independently of the country, Smart Grid technologies may enable to effectively integrate higher amounts of variable resources into the grid and help to reduce electricity demand by providing consumers with the information required (increase awareness) to reduce energy use or use electricity more efficiently (IEA, 2011). Moreover, as decentralized systems focus to locate power sources as close as possible to the end users, Smart Grid technologies can considerably reduce transmission and distribution losses at the same time than optimizing the use of existing infrastructure (IEA, 2011).

4.3.1 Smart Grid – South Africa approach

All countries across Africa strive to increase security of supply to their households and industry (IRENA, 2013). In some countries as South Africa, failures in the electricity supply are due to the strong growth of industrial demand and the lack of investment in new generating capacity (KPMG, 2011). Moreover, national utility's reserve margin on their generation capacity is steadily decreasing and may present a problem in the near future. Exemplary, in 2008, South Africa had such low reserves that, to satisfy the needs of industrial customers, could only intermittently supply electricity to residential customers, depriving them temporarily from energy needs (KPMG, 2011).

To overcome these problems, South Africa has become the only country in SSA with plans to implement Smart Grids in their near future as mentioned in the South Africa IRP 2010 (Africa, 2011). The South Africa IRP 2010 is a 20-year national electricity plan that set the basis of the generation

portfolio of the country. For next IRP revision settled by 2030, South Africa aims to implement Smart Grids and storage solutions to better integrate growing variable renewable sources (Africa, 2011).

Other countries are also engaged in programs and pilot projects of Smart Grids. This is the case of Yokohama City, Japan, where a large scale management project is assessing the effects of energy consumption and distribution infrastructure for old and new-built houses in a certain area (IEA, 2011). Other examples can be found in the US, Europe, South Korea, India, Australia, Brazil and China (IEA, 2011) (Giordano, et al., 2011).

5

Concentrating solar power technologies

5.1 Technology selection

After a gap in interest between 1990 and 2000, CSP deployment has been growing over the past decade (IRENA, 2012). CSP plants offer an integrated solution by concentrating the solar radiation and integrating low-cost thermal energy storage to provide power on demand (see Figure 8) (IRENA, 2012). CSP plants are classified into four main categories depending on the design, configuration of mirrors and receivers, heat transfer fluid and whether or not heat storage is involved. Table 6 compares the major features of the types of CSP technologies –parabolic trough, solar tower, linear fresnel and dish-stirling–. While the first three types are used mostly for centralized electricity generation, solar dishes are more suitable for distributed generation. Moreover, parabolic trough and solar tower are the technologies which present major grid stability and benefit from a wider deployment in the market. Therefore, as the use of proven technologies will ensure the technical feasibility of a sustainable power system, this study has selected the parabolic trough and solar tower technologies for being widely used today.

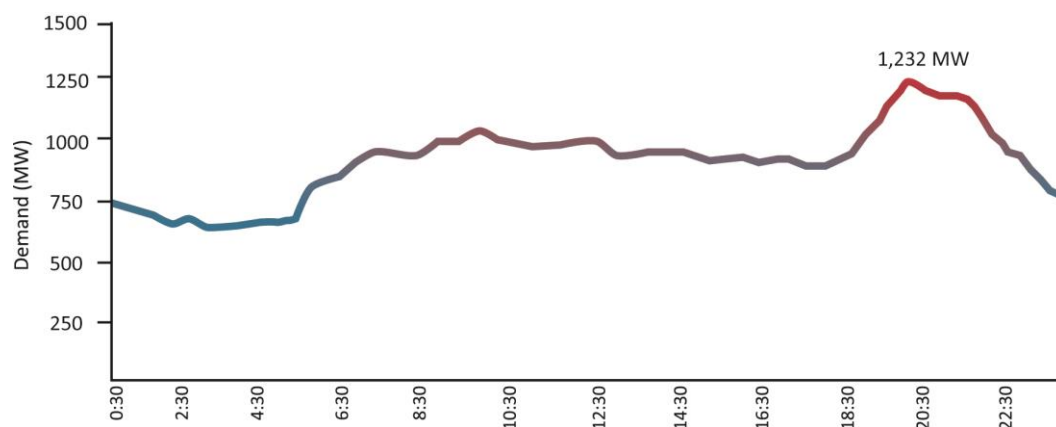


Figure 8 Daily demand curve for Kenya (February 22nd 2012) (Lighting, 2012). CSP plants can supply electricity to match the load during light time. To supply peak times when sun is not shining (e.g., peak demand at 20:30 pm), electricity is delivered from the thermal storage equipment.

When equipped with thermal storage¹⁶, the capacity factor of the CSP plant and the dispatchability of the generated electricity increases, thus providing grid integration and economic competitiveness of such power plants (IRENA, 2013). A recent study demonstrates that large-scale deployment of CSP plants has the ability to provide base load and dispatchable electricity with an appropriate dimensioning and management of the plants, providing very stable power output over long periods of time (Pfenninger, et al., 2012). The base-load generation mode can assume the critical role of balancing other intermittent sources, therefore reducing both the need of expensive storage and the reliance on hydropower to handle the system balancing (Battaglini, et al., 2010) (Trieb, et al., 2011). To increase the value of CSP technology, solar energy-based thermodynamic power cycles can also operate combined with fossil fuels such as oil, gas and coal and renewable fuels such as biomass¹⁷. Hence, by increasing power availability of CSP plants, it results in lower costs by making a more effective use of the power block (Viebahn, et al., 2011).

CSP plants require high direct irradiance to operate and are a very interesting option for installation in the Sun Belt region (between 40 degrees North and South of the equator), covering the North Africa and South Africa regions (IRENA, 2013). Additionally, the technical potential of concentrating solar generation in these regions is typically higher than their electricity demand, resulting in opportunities for electricity export to regions with higher demand shares of renewable electricity. However, these regions are partially affected by desert environments with water availability constraints. In contrast to other renewable technologies like PV or wind, CSP needs considerable amounts of water mainly for cooling purposes, a condition also applicable to other thermal power technologies. CSP plants present wet, dry or hybrid cooling systems. Depending on the annual mean temperature, the difference from wet to dry cooling amounts to approximately 3 percent annual output loss in Southern Spain (Szabó, undated), roughly 4.5 percent in the Mojave Desert (Szabó, undated), and 5–10 percent in North Africa with annual mean temperatures between 20 °C and 30 °C (Damerau, et al., 2011).

The US and Spain strongly dominate the CSP market, and regions such as North Africa and Middle East, Australia and Asia, are developing their first projects. If implementation is successful, further projects are expected in these countries. Currently, 2,672 GW are operating worldwide (see Table 26 and Table 25 in Appendix A) with installations that have been in continuous use for more than two decades. Furthermore, nearly 17 GW of capacity is planned to be installed by 2015. The US leads the ranking with about 8 GW, Spain stays second with 4.46 GW and China follows with 2.5 GW (World, 2013). The African layout remains prosperous; Egypt has plans for a 140 MW CSP plant in Kuraymat, Tunisia has plans to develop 40 solar power projects between 2010 and 2016, Morocco has allocated almost EUR 7 billion for the development of 2 GW of CSP by 2020 and Algeria is planning to supply 10 per cent of its electricity from solar power by 2050 (IRENA, 2011).

¹⁶ Solar heat collected during daytime can be stored in storage systems based on molten salt, concrete, ceramics or phase change materials. When sun is not available, the heat is extracted from the storage to run the power block on base load or on balancing mode.

¹⁷ CSP plants can be integrated in coal- or gas-fired power plants to produce fully dispatchable electricity. Projects based on this concept are in operation in Algeria, Australia, Egypt, Italy and the United States (IRENA, 2013).

Table 6 Comparison of CSP technologies. Technologies differ significantly from one another with regard to technical and economic aspects and in relation to their reliability, maturity and operational experience in utility scale conditions. Parabolic trough and solar tower are commercially proven technologies used worldwide.

CSP technology	Parabolic trough	Solar tower	Linear Fresnel	Dish-Stirling
Solar collector	Line focus	Point focus	Line focus	Point focus
Solar receiver	Mobile	Fixed	Fixed	Mobile
Concentration ratio	70-80	>1000	>60	>1300
Solar field slope (%)	<1-2	<2-4	<4	10 or more
Working temperature (°C)	Medium	Higher	Relatively lower	Highest
Operating temperature (°C)	350-550	250-565	390	550-750
Steam conditions (°C/bar)	380 to 540/100	540/100 to 160	260/50	n.a.
Current efficiency (%)	15-16	16-17	08-10	20-25
Plant peak efficiency (%)	14-20	23-35	18	30
Typical capacity (MW)	10-300	10-200	10-200	0.01-0.025
Annual capacity factor (%)	25-28 (no TES) 29-43 (7h TES)	55 (10h TES)	22-24 (no TES)	25-28 (no TES)
Grid stability	Medium to high (TES or hybridization)	High (large TES)	Medium (back-up firing possible)	Low
Development status	Commercially proven	Commercial	Pilot project	Demonstration projects
Technology development risk	Low	Medium	Medium	Medium
Outlook for improvements	Limited	Very significant	Significant	Via mass production
Efficiency with improvements	18	25-28	12	30
Relative rise of efficiency after improvements (%)	20	40-65	25	25
Water requirement (m³/MWh)	3 (wet cooling) 0.3 (dry cooling)	2-3 (wet cooling) 0.25 (dry cooling)	3 (wet cooling) 0.2 (dry cooling)	0.05-0.1 (mirror washing)
Application type	On-grid	On-grid	On-grid	On grid/Off-grid
Suitability for air cooling	Low to good	Good	Low	Best
Storage with molten salts	Commercially available	Commercially available	Possible, but not proven	Possible, but not proven

Sources: (IRENA, 2012) (Behar, et al., 2013). TES, Thermal Energy Storage.

5.2 Parabolic Trough technology

Parabolic trough collector is the most mature CSP technology, and further improvements in performance and cost reductions are expected. Parabolic trough technology accounts for more than 90 percent of the capacity installed worldwide (see Table 25 in Appendix A), and it shows the lowest development risk among the other CSP technologies. Parabolic trough CSP plant consists on parabolic-shaped mirrors that concentrate the sun's rays on heat receivers (e.g., stainless steel tubes) placed at the focal line of the collector. The absorber tube has a spectrally selective coating that maximizes energy absorption (solar short wave), while minimizing long wave radiation emitted by the absorber. The absorber tubes work in vacuum glass cover to avoid convection heat losses. Solar heat is

transmitted by a heat transfer fluid (HTF, e.g., synthetic oil, molten salt) through the receiver tube and transferred to a steam generator to produce super-heated steam required to run the turbines or be transferred to the heat storage system, if any. Mirrors and receivers are usually aligned North-South and track the sun's path with a single-axis tracking mechanism as the sun moves from East to West. An array of mirrors can be up to 100 meters long with a curved aperture of 5 meters to 6 meters (IRENA, 2012).

At the beginning of 2013, around 2,600 MW of installed CSP capacity used the parabolic trough technology accounting for virtually all CSP capacity installed nowadays. Among them, the three 50 MW Andasol units by ACS/Cobra Group and Marquesado Solar SL together with the two 50 MW (Valle I and II) plants by Torresol Energy in Spain are particularly interesting, as they use synthetic oil as the heat transfer fluid and molten salts as the thermal storage fluid (IRENA, 2013). Large parabolic trough plants constructed include the Shams 1 100 MW project in the United Arab Emirates (March, 2013), and the Godawari 50 MW project in India (June, 2013). Large parabolic trough plants under construction include the 250 MW Mojave project in California due to start operation in 2014, and the 280 MW Solana project in Arizona due in 2013 (IRENA, 2013).

5.3 Solar Tower technology

Solar tower systems use computer-controlled mirrors, so-called heliostats, to focus direct solar irradiation on a receiver located on a central tower where the light is captured and converted into heat (IRENA, 2012). The heat drives a thermo-dynamic cycle, usually a water-steam cycle, and moves a conventional turbine to generate electric power. Heliostats track the sun individually in two axes and reflect the sun onto the central receiver where a fluid is heated up. Normally current solar towers use water/steam, air or molten salt to transport the heat to the heat-exchange/steam-turbine system achieving high temperatures.

The typical size of a current solar tower plants ranges from 10 MW to 50 MW (see Table 26 in Appendix A). The required size of the solar field increases with the annual electricity generation desired, leading to greater distances between the receiver and the outer mirrors of the solar field. This results in an increase of optical losses due to atmospheric absorption, inevitable angular mirror deviation due to imperfections in the mirror and minor errors in mirror tracking (IRENA, 2012).

Solar tower has higher concentration factors than other CSP technologies. It means that more sunlight can be concentrated on a single receiver and thus the heat losses can be minimized (IRENA, 2013). This reduction in heat losses allows solar tower plants to achieve even higher temperatures than parabolic troughs plants (IRENA, 2013). By achieving higher temperatures, solar towers offer potential advantages over other technologies in terms of efficiency, performance, water consumption, heat storage, capacity factors, and costs. Higher temperatures make the use of thermal storage more attractive to achieve dispatchable power generation and allow greater temperature differentials in the storage system. By achieving greater temperature difference, it is possible either to reduce the costs or to allow greater storage for the same cost. In the long run, if costs are reduced and operating experience gained, solar tower could provide the cheapest CSP electricity among other technologies and achieve a significant market by displacing parabolic trough systems (IRENA, 2012).

Among the current solar tower projects, there are the PS10 and PS20 demonstration projects in Seville (Spain), with capacities of 11 MW and 20 MW respectively. Both plants are equipped with one hour of steam-based thermal storage to ensure power production despite varying solar radiation. The PS10 consists of 624 heliostats over 75,000 m². The receiver converts 92 percent of solar energy into saturated steam at 250°C and generates 24.3 GWh a year, with 17 per cent efficiency. In Spain, a 19 MW molten salt-based solar tower plant (Gemasolar) with a 15 hour molten salt storage system started

operation in April 2011. Gemasolar plant is expected to run for almost 6,500 operation hours per year, reaching a 74 percent capacity factor and produce fully dispatchable electricity. Larger solar tower plants are under construction, e.g., the Ivanpah project in Primm, California, with a capacity of 370 MW and 29 percent efficiency; and the Supcon project in Delingha, China, with a capacity of 50 MW. Additionally, several solar tower plants are under development, e.g., this is the case of eight units with a total capacity of 1.5 GW planned to be installed in California, United States (IRENA, 2013).

6

Spatial analysis of potential solar fields

6.1 Irradiation potential in Africa

Renewable energy technologies –except for biomass and biogas– have no fuel costs and relatively small O&M costs. As fuel costs are zero, the capital costs account for the greatest share of generation costs. However, for fossil fuel-based technologies with significant fuel cost, both fuel cost and capital cost affect the generation costs. Hence, for some renewable technologies, the windiness or the irradiation level become the main variables to determine the production costs, and thus the most important variables for siting decision. To illustrate the importance of the variable resources, a comparison is done between capital costs of renewable and non-renewable technologies: 84 percent of the LCOE of a parabolic trough and solar tower plants accounts for the initial capital investment of the plants (IRENA, 2012). While roughly the 60 percent of the LCOE of a conventional coal plant, and the 30 percent of the LCOE of a combined gas cycle plant accounts for the initial capital investment of the plants (EIA, 2013). Thus, the high capital cost of a solar plant makes the quality of the production site –the irradiance level– the main variable for determining the production costs.

In Africa, the size of the largely unused land and the favorable climatic conditions make the continent ideal for solar electricity production. The Northern and Southern regions (the Sahara and the Kalahari Deserts respectively), benefit from excellent solar resources up to 3,000 kWh/m²/year (see Figure 9). These vast solar and land resources allow for “cherry-picking” of the best sites, discarding the marginal sites. This results in an increase of the efficiency in the power system.

Analysts estimated that the LCOE of a CSP plant declines by 4.5 percent for every 100 kWh/m²/year that the DNI exceeds 2100 kWh/m²/year (IRENA, 2012). For a sense of comparison, the cost of electricity production by parabolic trough systems in Southern Europe is in the order of EUR 0.18 kWh to EUR 0.20 kWh for a DNI of 2,000 kWh/m²/year (CSP Today, 2008). Consequently, for DNI of 3,000 kWh/m²/year, the LCOE decreases to approximately EUR 0.11-0.13/kWh. Therefore, selecting good production sites is essential to achieve generation capacities ready to compete with traditional generation technologies. E.g., for high irradiance values (DNI > 2,800 kWh/m²/year), the generation potential of the concentrating solar plants is 100-300 GWh_e/Km²/year. This is roughly the same electricity generated annually by a 20 MW coal-fired power plant with 75 percent capacity factor (IRENA, 2013). Thereby, by adequately utilizing the excellent solar resources in Africa, solar power technology may play an important role in meeting renewable energy targets and balancing the energy system at a reasonable cost.

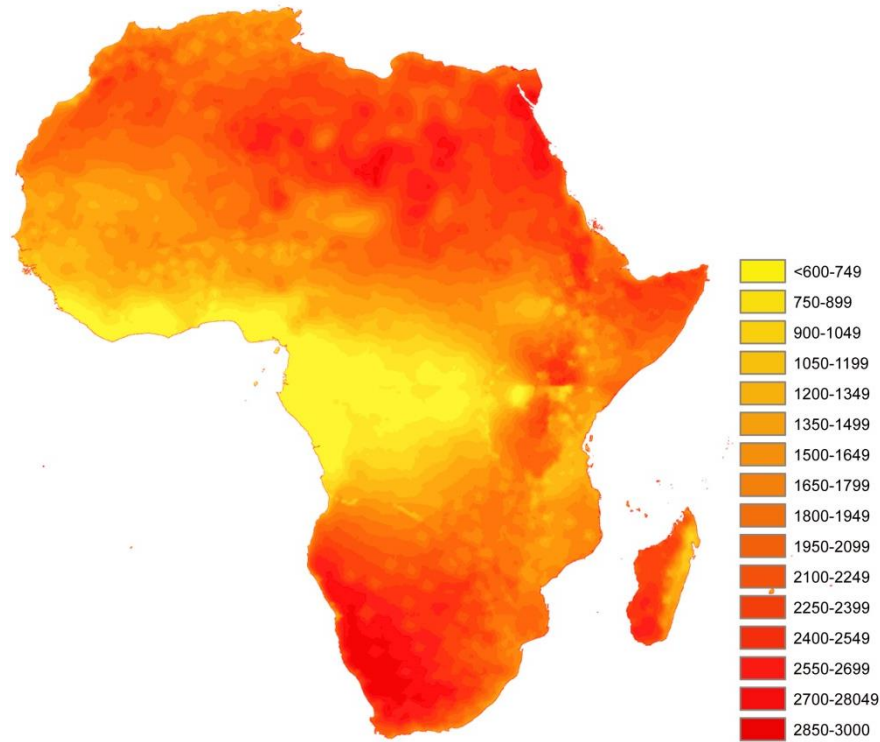


Figure 9 Direct Normal Irradiance of Africa in kWh/m²/year. Solar resource is evenly distributed in Northern and Southern regions where the main deserts are located and thus there is a high availability of land for sitting decision.

Africa has an enormous potential for generating renewable electricity by means of solar power. Nevertheless, only strong direct sunlight can be concentrated to reach required temperatures for electricity generation in concentrating solar plants. The generally accepted minimum technical operational limit of a CSP plant is 1,800 kWh/m²/year, and the generally accepted threshold for commercial CSP projects is 2,000 kWh/m²/year, which accounts for slightly more than 5 kWh/m²/day (Breyer, et al., 2009) (IRENA, 2011) (IRENA, 2012). Recent demonstration projects, however, reveal that a lower solar radiation threshold is technically viable for solar power operation. E.g., Beijing Badaling and Jülich solar tower plants operate at the lowest irradiation ranges (1,290 kWh/m²/year and 902 kWh/m²/year respectively) among the current CSP plants in operation (see Table 7).

Table 7 Central receiver power plants operating at the lowest solar radiation among other central receiver plants in operation.

Name/type	Country	Latitude, longitude	Solar radiation (kWh/m ² /year)	Land area (ha)	Field area (m ²)
Beijing Badaling solar tower/demonstration	China	40°40' North, 115°90' East	1,290	5	100,000
Jülich solar tower/demonstration	Germany	50°54' North, 6°23' East	902	17	17,650

Source: www.nrel.gov

Even more, the minimum daily DNI level for a SEGS parabolic trough plants is 2 kWh/m². This results in a minimum annual DNI level of 750 kWh/m²/year, below which a SEGS plant has no production (IEA, 2010) (Pharabod, et al., 1991).

6.2 Identification of potential solar fields

The next step of the analysis is the detection of the most appropriate land resources to install concentrating solar plants. This detection was performed by means of suitability mapping involving a variety of geographical and meteorological data according to specific siting criteria (Janke, 2010). The identification of potential solar fields in Africa was done with the Geographic Information System (GIS) ArcGIS 10.1® from ESRI, California (USA). Geographic information systems for CSP site targeting has become more advanced over the past decade as data quality has improved and the performance of modeling techniques has increased. All layers of the analysis were converted into the projected coordinate system African Equidistant Conic in meters with datum WGS84¹⁸ and resampled to a standard resolution of 1 km x 1 km at the Equator (0.008333333 degrees).

6.2.1 Evaluation of the suitability of land for solar fields

Current literature does not specify a specific method using GIS analysis to assess site suitability of a CSP plant and the necessary transmission corridors. The extent of the spatial analysis included in the vast majority of studies assessing CSP site suitability lies in an excluding criteria approach (Gastli, et al., 2010) (DLR, 2009) (Christian Breyer, et al., 2009) (Fluri, 2009) (Mehos, et al., 2007) (Broesamle, et al., 2001). This results in the development of an exclusion mask of non-suitable areas for CSP location. Other studies have employed a weighting criterion to the different variables that determines CSP location (Dawson, et al., 2012) (Clifton, et al., 2010) (Figueira, et al., 2002). This weighting criterion results in a ranking of the variables in terms of importance and therefore assesses suitability of land. These studies used diverse methods and assumptions depending on the scope of each particular investigation. However, sufficiently strong classification certainty to identify common criteria for suitability mapping was not found.

The ideal location for electricity plants is as close as possible to centers of demand. This is important to reduce the cost of transmission between supply and demand points, and to reduce losses experienced during transmission. Therefore, to decrease the uncertainty given by the variability of weighting criteria for CSP site location, this study relies on excluding and non-excluding criteria to identify potential solar sites. Moreover, builds in previous assumptions and considers closeness to centers of demand as the common criteria to connect the transmission corridors to the centers of demand. Suitability mapping, either relying in a weighting system or an exclusion mask of non-suitable areas, entails compulsive criteria and optional criteria. Compulsive criteria are e.g., direct normal irradiation, slope of the terrain and shifting sands; while optional criteria for site exclusion are croplands, irrigated croplands or tree crops which would be competing with the solar fields on land use. This analysis has applied both compulsive criteria and optional criteria for the detection of suitable land resources for the installation of solar plants, which results in a conservative estimate of the available land resources.

¹⁸ WGS84: World Geodetic System 1984 when the Earth is considered an ellipsoid. The origin of the WGS84 Cartesian system is the Earth's centre of mass.

6.2.2 Suitability parameters

Direct Normal Irradiance

The primary resource for CSP technologies is the DNI. It refers to the “amount of electromagnetic energy at the Earth’s surface on a flat surface perpendicular to the sun’s beam with surrounding diffuse sky radiation blocked, and it is equivalent to the solar constant¹⁹ minus the atmospheric losses due to absorption and scattering” (NASA, 2013).

GIS data of annual direct normal solar irradiance for solar concentrating systems was obtained from NREL. The isolation values were derived from a Climatological Solar Radiation (CSR) model²⁰, which considers information on cloud cover, atmospheric water vapor and trace gases, and the amount of aerosols in the atmosphere in order to calculate the monthly average daily total insolation falling on a horizontal surface. Grid cells have an output resolution of 40 km x 40 km (~22 arc-minute). The values of each cell are accurate to approximately 10% of the true measured value due to uncertainties related with meteorological parameters considered in the model. Overall uncertainty of the model may increase with distance from reliable measurement sources and with the complexity of the terrain.

Ground slope

Parabolic trough and solar towers are particularly limited by ground slope to the design of the solar field. Areas with ground slope > 2 percent are deemed to be unsustainable for parabolic trough technologies. Solar towers are less sensitive to slope and can be built on slopes up to 4 percent (IRENA, 2012). Building on land with slopes > 4 percent may result in increased design requirements and thus higher construction costs in order to decrease the slope of the terrain.

The 30 arc-sec (1 km x 1 km) digital elevation model (DEM) obtained from the NASA Shuttle Radar Topography Mission (SRTM) was used to calculate slope values in terms of percentage. For the determination of the land slope model, slopes higher than 4 percent from the DEM were excluded for the construction of CSP plants.

Land cover

The Land Cover map of Africa is one regional component of the Global Land Cover 2000 (European Commission, 2003) dataset of 1 km x 1 km resolution, conceived and coordinated by the European Joint Research Centre. It includes information regarding forest coverage, woodlands, shrub lands and grasslands, agriculture, bare soil, and water bodies. Suitable land covers for CSP location are shown in Figure 23 Appendix A. Special focus receives the analysis of dune mobility (excluded parameter in Figure 23), as long as areas of high irradiation resources are usually in arid or semi-arid areas.

Shifting sands

Large areas particularly of the Sahara Desert were identified as potentially covered by shifting dunes. Shifting dunes may incur in high costs for earth removal and in the creation of a suitably stable foundation both for solar plant construction and erection of transmission pylons (Dawson, et al., 2012)

¹⁹ Is the amount of solar power flux that passes through the mean Earth orbit. The currently accepted value is 1,366 W/m². Source: www.nrel.com

²⁰ This CSR model covers data of 6 years measurements. A one year basis is not sufficient to compute an analysis for the development of large CSP projects, as the annual climatic fluctuations may be in the range of $\pm 15\%$. For project development purposes, at least 5 to 15 years of data should be processed (DLR, 2005).

(Trieb, et al., 2011). Furthermore, solar plants are often associated with high risks of sandstorms, which may be prone to abrasion phenomena in the mirror field²¹.

This study considers that neither the geographic information layer *sandy desert and dunes* of the Global Land Cover 2000 dataset nor specified literature such as (Sharaky, et al., 2002) (Ashkenazy, et al., 2011)²², are considered to be of sufficiently high spatial resolution or sufficiently strong classification certainty to positively identify shifting sands in the Sahara Desert. This study recognizes that it is highly probable that areas within regions identified for being affected by shifting sands, are in fact free of shifting sands; thus, areas not identified as shifting sands may contain some areas with shifting dunes. Given the lack of reliable data, suitability results for areas potentially containing shifting sands should be treated with caution. Nevertheless, other parameters in addition to DNI potential were taken into consideration for the evaluation of CSP plants. Therefore, sites not located in the deserts but in the periphery with lower irradiation levels are also potentially suitable for solar plant location regarding proximity to loads.

The worst case scenario regarding dune mobility in the Sahara Desert is given by (Embabi, 1981). It states that dune mobility in some particular areas of the desert achieves 100 m/yr and is mainly directed to the South. In the Namib Desert, dune mobility achieves 0.1 m/yr (Lancaster, 2008), and in the Kalahari Desert dunes achieve a rate of 0.5 m/yr²³ (Lancaster, 2008). Considering an average CSP life plant of 30 years (Turchi, 2010), we have created a protecting buffer of 3 km around Sahara moving dunes to eliminate the endangering of the facility during the operation lifetime. This study neglects mobility dune in the Namib and Kalahari deserts due to its low moving rates²⁴. Furthermore, considering current moving rates, this is equivalent to create a buffer of 20 meters around, and it cannot be shown in the map as the size of one pixel in the *sandy desert and dunes* Global Land Cover 2000 dataset is of 1 km.

Salt hardpans

Salts pans soils may be an indicator of low drainage conditions and thus, be highly corrosive for solar plant installations (Trieb, et al., 2009). Sites exhibiting high salinity are expected to have higher corrosion risks resulting in an accelerated replacement schedule for components subject to degradation and thus resulting in higher operating and material costs. The main hardpans are Etosha and Magadikgadi Pans in Southern Africa, the Natron Lake in East Africa, and the urge Chotts in Northern Africa

Water bodies

All water bodies, regardless of their ephemerality, were deemed unsuitable for CSP plants and were identified for exclusion. However, narrow water bodies (i.e. rivers), were classified as highly unsuitable (but possible) for the installation of transmission infrastructure. This criterion allows transmission infrastructure to cross rivers.

²¹ In the case of parabolic trough, mirrors can be protected from the sand simply by turning them upside down. In the case of the solar tower, the heliostats turn out of the wind.

²² Analyst (Ashkenazy, et al., 2011) studies the mobility of sand dunes related with the average minimum precipitation and identifies areas where precipitation is below the 50 mm/year and considers this rate as a threshold for dune reactivation.

²³ Dunes in the Kalahari Desert are considered stable dunes fixed by vegetation (Sharaky, et al., 2002) (Ashkenazy, et al., 2011).

²⁴ When a feasibility study is needed to construct a CSP plant in a particular location, areas identified as potentially containing shifting sands may benefit from a downscale study of the site.

Protected areas

The World Database on Protected Areas (WDPA) is the most extensive dataset on protected areas worldwide. The WDPA is a joint project between the United Nations Environment Programme-World Conservation Monitoring Centre (UNEP-WCMC) and the International Union for Nature Conservation (IUCN) World Commission on Protected Areas (WCPA).

According to IUCN, a protected area is defined as “a clearly defined geographical space, recognized, dedicated and managed through legal or other effective means, to achieve the long term conservation of nature with associated ecosystem services and cultural values” (Dudley, 2008). The classes of the protected areas considered in the analysis are shown in Figure 24 Appendix A according to the classification of the IUCN. All protected areas have a buffer of 2 km in order to provide a safety region for conservation. The areas in the table below were deemed unsuitable for CSP plants. For more information of each specific area, see Table 27 Appendix A.

Table 8 Protected areas unsuitable for CSP plant location.

I	Strict protection [a) Strict nature reserve and b) Wilderness area]
II	Ecosystem conservation and protection
III	Conservation of natural features
IV	Conservation through active management
V	Landscape/seascape conservation and recreation
VI	Sustainable use of natural resources

Industrial locations and population

Information about population settlements was taken from the dataset the Land Cover map of Africa, a regional component of the Global Land Cover 2000 dataset of 1 km x 1 km resolution. The Land Cover map of Africa provides information regarding the location and size of the settlements including commercial and industrial areas. Location of airports is provided with a safety zone of 3 km to avoid the collisions of airplanes with conductor wires.

Proximity to load

The location of loads is of considerable importance in determining suitable locations for all centralized generation plants. Ideal location of electrical plants is as close to loads as possible to reduce the cost of transmission between supply and demand points, at the same time than reducing losses experienced during transmission. As one of the objectives of this study is to analyze land adequacy to supply CSP electricity to the main populated urban centers of SSA, these urban centers were considered as the loads in the next steps of this study.

Access to network infrastructure

The presence of infrastructure does not necessarily mean the availability of that infrastructure. However, this study assumes that where infrastructure exists there is also the possibility to use it. Moreover, capacity restriction in transmission lines may also involve significant obstacles for the establishment of CSP plants as auxiliary capacity may not always be available for all lines. Information regarding transmission infrastructure was obtained from the African Development Bank Group database. This database includes all power lines in SSA implemented and under construction up to the year 2009.

Availability of land

Land requirements for a CSP plant may attain for 7,500 m²/MW_{cap} (NREL, 2012), 10,000 m²/MW_{cap} (NREL, 2013) or even 15,000 m²/MW_{cap} (NREL, 2011). It is broadly known that CSP plant at utility scale requires substantial amounts of land for electricity production, and land is often abundant and available at relatively low cost in the areas where CSP is highly suitable (such as arid or semi-arid areas). Although land requirements per MW_{cap} for CSP production are high, they are often lower than most of the other renewable generation technologies or fossil fuel technologies when considering the requirements for drilling, mining, refinement and transportation. Furthermore, interviews with stakeholders of the solar industry conducted by (Dawson, et al., 2012), revealed that some developers may purchase larger areas of land than are required for given plant capacities. This is due to the relatively low cost of land and the possibility of expansion of plant capacity in the future.

After applying the exclusion criteria, land classified as suitable for CSP plants whose area was less than 1 km², was also excluded for CSP plant location.

Availability of auxiliary fuel supply

This study considers that CSP plants incorporate sufficient thermal storage capacities and thus eliminates dependence on the availability of auxiliary fuel sources such as natural gas. Current storage units show efficiencies greater than 90 percent, and increasing storage capacity, at least up to 15–16 hours capacity, does not influence negatively the levelized cost of the electricity (LCOE) (Trieb, et al., 2009).

The Sahara Desert experiences several sand storms every year. In the course of these storms, visibility drops close to zero and CSP plants have to rely on their thermal storage to keep producing during the storm (Patt, et al., 2010). When power plant is equipped with 16 hours of thermal storage, this results in continuous power production; while longer periods (>24h) would allow to smooth out the effects of large sand storms (Trieb, et al., 2009). Only in the case that the storm lasts significantly or if the power plant is not equipped with enough thermal storage, power generation will cease.

Natural Hazards

Natural hazards such as earthquakes, tsunamis, volcanoes and high wind loads affect both coastal and inland areas. Long-term data from these events was used to establish the past record of natural hazard event occurrences in order to diminish the impact during the lifetime of the plants and overhead lines.

Earthquake activity

The United Nations Environmental Programme/Global Resource Information Database (UNEP/GRID) provides information regarding earthquake intensity zones by specifying intensity and locations. The dataset was obtained according to the probability of occurrence of earthquakes of 5 general intensity levels. These intensity levels are associated with the 1956 version of the Modified Mercalli Intensity Scale (MMI). The MMI scale consists of 12 intensity levels describing potential damage to the Earth's surface and built environment. Each zone illustrates an area where the associated MMI level is the maximum likely intensity of an event with an estimated return period of 50 years.

Volcano activity

Global Volcano Hazard Frequency and Distribution is a 2.5 by 2.5 minute gridded dataset based upon the National Geophysical Data Center (NGDC) Volcano Database spanning the period 79 AC through 2000. This database includes nearly 4,000 volcanic events categorized from 2 to 8 according to the Volcano Explosivity Index (VEI). To get the final output, the frequency of a volcanic hazard is computed for each grid cell, with the dataset consequently being classified into deciles (10 classes of

approximately equal number of grid cells). The higher the grid cell value in the final output, the higher the relative frequency of hazard posed by volcanoes.

Cyclone activity

Global Cyclone Hazard Frequency and Distribution is a 2.5 by 2.5 minute grid based on more than 1,600 storm tracks for the period January, 1 1980 through December, 31 2000 for the Atlantic, Pacific and Indian Oceans that were assembled and modeled at UNEP/GRID-Geneva PreView. Wind speeds around storm tracks were modeled to assess the grid cells likely to have been exposed to high wind levels. The higher the value of the grid cell, the higher the decile ranking and the greater the frequency of the hazard relative to other cells. The final map classifies the range of frequency into deciles, 10 classes of approximately an equal number of grid cells. 10 represent cells that are the most hazardous, while 1 denotes the least hazardous.

Floods

Global Flood Hazard Frequency and Distribution is a 2.5 by 2.5 minute grid derived from a global listing of extreme flood events between 1985 and 2003 compiled by Dartmouth Flood Observatory and georeferenced to the nearest degree. The resulting flood frequency grid was then classified into 10 classes of approximately equal number of grid cells. The greater the grid cell value in the final dataset, the higher the relative frequency of flood occurrence.

Landslides

Global Landslide Hazard Distribution is a 2.5 by 2.5 minute grid of global landslide and snow avalanche hazards based upon work of the Norwegian Geotechnical Institute (NGI). The hazards mapping of NGI integrates a range of data including slope, soil, soil moisture conditions, precipitation, seismicity, and temperature. Hazards values 4 and below were considered negligible and only values 5 to 9 were utilized in the analyses. To ensure compatibility with other datasets, 1 is added to each of the values to provide a hazard ranking ranging 6 to 10 in increasing hazard.

Visual impact

Visual impact was ranked as the least important parameter in a series of interviews with stakeholders of the solar industry conducted by (Dawson, et al., 2012) when considering CSP implementation. One of the stakeholders suggested that the visual impact of solar plants could be even considered to be a positive value in the area. Considering the low and relative importance, this study excludes visual impact from CSP sites and transmission corridor's assessment.

Table 9 Suitability parameters considered for the assessment of CSP sites and HVDC transmission corridors. A final resolution of 1km x 1 km was defined to facilitate the implementation of algebraic calculations.

Variable	Ideal Conditions	Type	Original Resolution	Final Resolution
Solar potential	Maximize kWh/m ² /year	Point	40km x 40km	1km x 1km
Ground slope	Slope < 4% to reduce construction costs	Grid	1km x 1km	1km x 1km
Land cover	Not classified as land use with high environmental cost	Grid	1km x 1km	1km x 1km
Protected areas	Not classified as protected area	Line	-	1km x 1km
Settlements	Not located in build-up residential/commercial/industrial areas	Grid	1km x 1km	1km x 1km
Distance to transmission lines	Closer to transmission lines	Line	-	1km x 1km

Water bodies	Not located in water bodies or wetlands regardless of ephemerality	Grid	1km x 1km	1km x 1km
Earthquake	Minimum hazard frequency	Line	-	1km x 1km
Volcano	Minimum hazard frequency	Grid	2.5 x 2.5 min	1km x 1km
Cyclone	Minimum hazard frequency	Grid	2.5 x 2.5 min	1km x 1km
Floods	Minimum hazard frequency	Grid	2.5 x 2.5 min	1km x 1km
Landslides	Minimum hazard frequency	Grid	2.5 x 2.5 min	1km x 1km

6.3 Solar power production sites

The selected sites for the installation of concentrating solar plants are a compromise between solar energy intensity, land resource availability, inclination of the terrain and closeness to the main SSA centers of demand²⁵. These sites do not necessarily represent optimal sites –as the closeness to mine sites was not under the scope of this study²⁶–, but potential sites for the production of solar electricity from the technical and economic point of view.

The combination of the suitability parameters shown in Table 10 results in an exclusion mask (Figure 10) that yields the potential fields for the installation of CSP plants. Figure 11 shows the potential sites that have been identified in Africa for the location of CSP plants using the geographic and weather information available. The final decision of plant installation, however, may lie primarily in the host regions which will decide on the location of sites and construction time to export electricity to the demand centers.

Table 10 Selected and excluded criteria for selection of CSP potential fields.

	Selected	Excluded	Considerations
Direct Normal Irradiation			
DNI < 2,000 kWh/m ² /year		x	
DNI > 2,000 kWh/m ² /year	x		
For the identification of potential transmission corridors in following section, the DNI ranges from 1,000 to 2,800 kWh/m ² /year			
Slope			
Slope < 4%	x		
Slope > 4%		x	
Land cover			

²⁵ This study focuses on strategic assessment of site suitability at a continental/sub-continental scale. However when assessing site suitability at a narrow geographical level, other criteria for site selection such as land properties and national policies may play an important role.

²⁶ CSP site selection can be associated with proximity to mine sites. It reflects the development possibility of a CSP plant to supply electric power to the mine and heat for industrial processes. Regarding the relatively high capital cost of CSP plants, it is important to assess its suitability considering mine lifetime. However, as mine lifetime is often difficult to predict due to inadequate exploration of the underlying resources and corporate confidentiality restrictions (Dawson, et al., 2012), this study considers that there is not enough mine site information available for the whole Africa to consider mine location into discussion.

Forests			
Closed evergreen lowland forest		x	
Degraded evergreen lowland forest		x	
Montane evergreen forest (> 1,500 m)		x	
Sub-montane forest (> 900 m)		x	
Swamp forest		x	
Mangrove		x	
Mosaic Forest / Croplands		x	
Mosaic Forest / Savanna		x	
Closed deciduous forest		x	
Woodlands, shrub lands and grasslands			
Deciduous woodland	x		
Deciduous shrub land with sparse trees	x		
Open deciduous shrub land	x		
Closed grassland	x		
Open grassland with sparse shrubs	x		
Open grassland	x		
Sparse grassland	x		
Swamp bushland and grassland	x		
Agriculture			
Croplands (> 50%)	x		
Croplands with open woody vegetation	x		
Irrigated croplands	x		
Tree crops	x		
Bare soil			
Bare rock	x		
Stony desert	x		
Sandy desert and dunes		x	Buffer 3 Km around shifting sands
Salt hardpans		x	
Other land-cover classes			
Water bodies		x	
Protected areas		x	
Natural Hazards			
Earthquake	x		Excluded MMS ≥ IX
Volcano	x		Excluded class 10
Cyclone	x		Excluded class 9 to 10
Floods	x		Excluded class 10
Landslides		x	
Industrial locations and population			
Airports		x	Buffer 3 Km around airports
Population settlements		x	

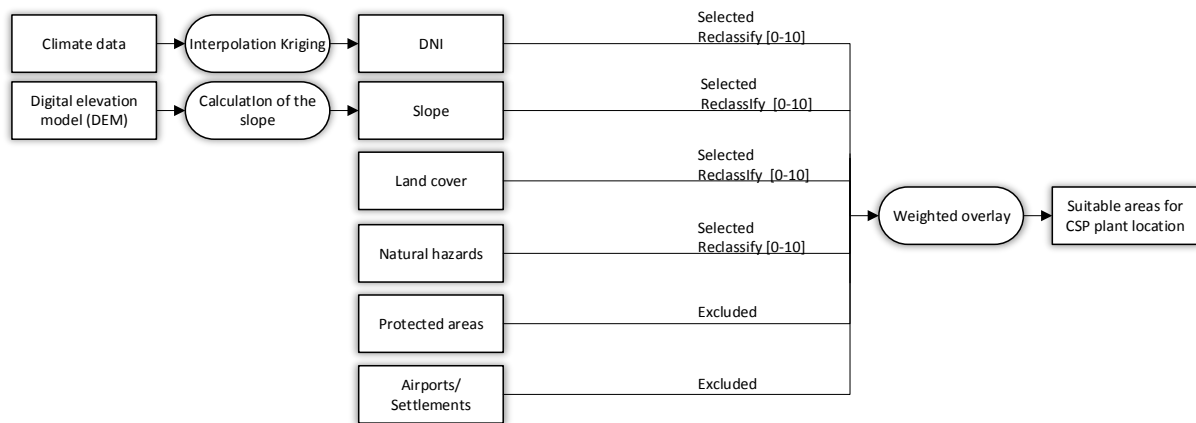


Figure 10 Methodological flowchart for the GIS-based site suitability identification.

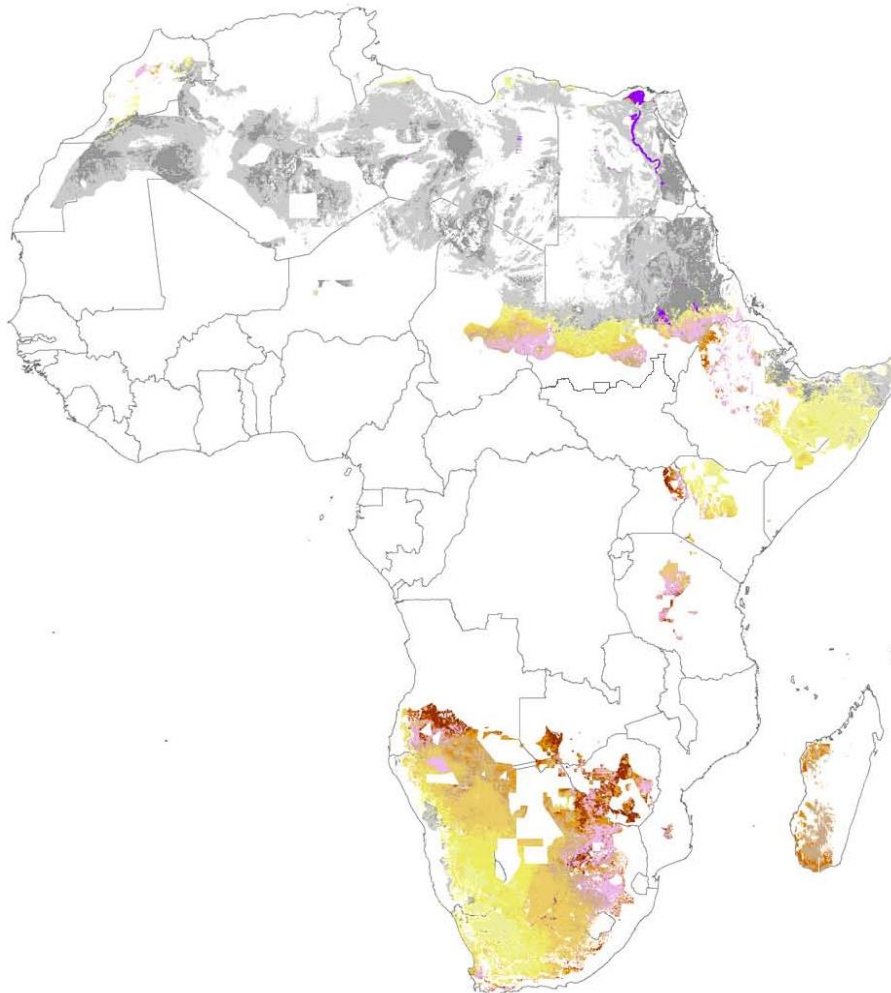


Figure 11 Potential fields for CSP plants in Africa for $\text{DNI} \geq 2,000 \text{ KWh/m}^2/\text{year}$. This figure shows that most suitable land for CSP plant installation is located in the North, North-East, and South regions. While in the North land is mainly bare rock and stony desert, in the South is sparse grassland, open grassland and open grassland with sparse shrubs.

7

Spatial analysis of import corridors

7.1 Sub-Saharan Africa energy demand nodes

World population is expected to be 67 percent urban in 2050 (United Nations, 2012), and urban agglomerations across Africa will play an important role in that share. Among them, SSA million-plus cities are expected to expand by an average 32 percent between 2010 and 2020 (UN-HABITAT, 2010), being an important focus of population growth and energy demand, especially for commercial and industrial sectors. This exponential growth will increase the demand for shelter, infrastructure and services. Although urban demographic growth is generally considered a positive force for economic development, very rapid urbanization can pose great challenges for urban economies (e.g., massive urban slum proliferation may enhance inequality and human misery). When demographic expansion is coupled with economic progress, then it turns to opportunities for the dwellers.

Table 11 lists major sub-Saharan million-plus centers of demand selected to analyze the role of introducing CSP in SSA energy system and the subsequent subset of envisioned grid developments. The selection is done such as although one single country may have several million-plus cities, only one million-plus city per country is selected. When mapping selected centers of demand, we obtain an uneven distribution of growing nodes across SSA; with the major concentration located in the Gulf of Guinea (see Figure 12).

Table 11 Selected SSA cities with total population over one million.

Demand center	Country	Total population
Lagos	Nigeria	10,578,000
Kinshasa - capital	Democratic Republic of the Congo	8,754,000
Khartoum - capital	Sudan	5,172,000
Luanda - capital	Angola	4,772,000
Abidjan - seat of government	Côte d'Ivoire	4,125,000
Johannesburg	South Africa	3,670,000
Nairobi - capital	Kenya	3,523,000
Dar es Salaam - capital	United Republic of Tanzania	3,349,000
Dakar - capital	Senegal	2,863,000

Accra - capital	Ghana	2,342,000
Douala	Cameroon	2,125,000
Ouagadougou - capital	Burkina Faso	1,908,000
Bamako - capital	Mali	1,699,000
Harare - capital	Zimbabwe	1,632,000
Maputo - capital	Mozambique	1,655,000
Lusaka - capital	Zambia	1,451,000
Mogadishu - capital	Somalia	1,500,000
Brazzaville - capital	Congo	1,323,000
Niamey - capital	Niger	1,048,000
Monrovia - capital	Liberia	827,000*

Source: (UN-HABITAT, 2010). Reference year 2010, United Nations own projections. * Current estimations (2013) account for more than 1.2 million inhabitants (Kaiser, 2013).



Figure 12 Selected high populated SSA demand centers. From July 9 2011, South Sudan is officially a new country. Final boundary between the Republic of Sudan and the Republic of South Sudan has not yet been determined.

An important feature to determine future electricity needs of selected urban nodes is the urbanization trend. Figure 13 shows the population projections of selected SSA million-plus cities from 1990 to 2025. Selected Africa's urban agglomerations are expected to increment 29 percent from 1990 to 2025 projections. This will lead to higher energy demand in a country currently facing insufficient generation capacity.

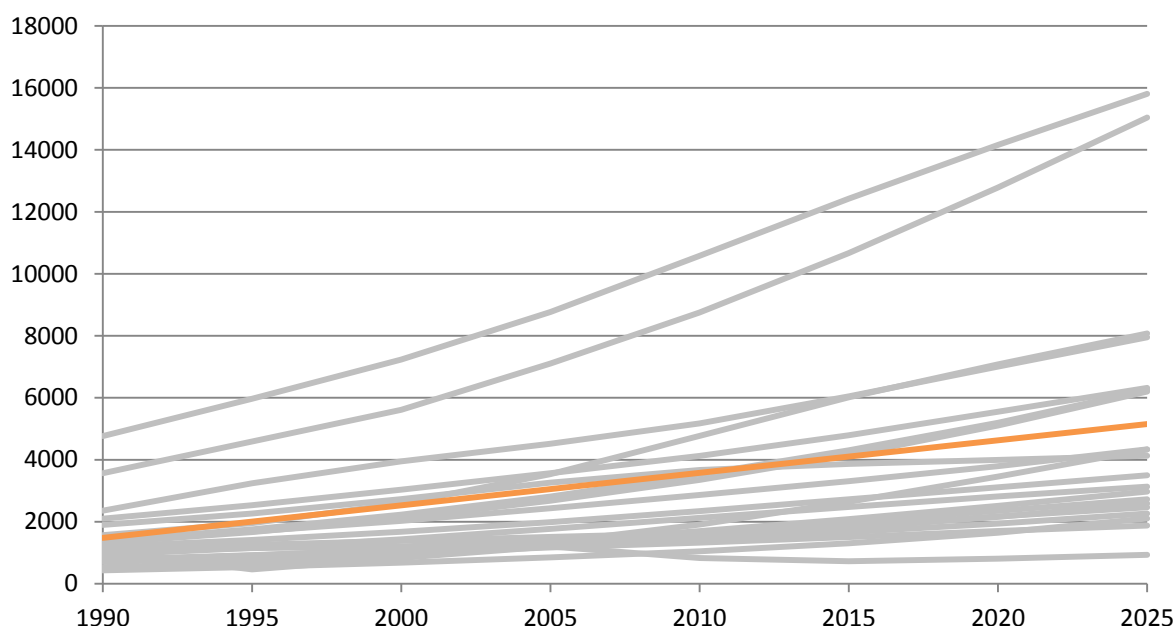


Figure 13 Population projections (in thousands) of selected SSA million-plus cities. An increment of 29 percent in the population is expected from 1990 to 2025 estimations. Source: (UN-HABITAT, 2010).

Another important feature that needs to be considered in the electricity demand analysis is the GDP growth rate. In developing countries, electricity demand tends to grow at least in the same range as the GDP (IRENA, 2011). Energy demand modeling performed by (IRENA, 2011) considers that given the low starting point of electricity consumption per capita in SSA, electricity demand in Central and Eastern regions is expected to grow at a rate of GDP growth plus 2 percent (so more than 7 percent electricity demand growth per year between now and 2030). In the Western region, electricity demand is expected to grow at 1 percent above GDP. While in Southern Africa, it is expected that electricity demand will match the rate of GDP growth (it means not more than 3-4 percent electricity demand growth per year) (see Table 12). In a country level, we observe that almost all selected countries also face an increment of their GDP rates, thus a general increment in the consumption of electricity is expected (see Table 13).

Table 12 GDP growth projections.

Real GDP Growth rate	2008-2015 (%/year)	2015-2030 (%/year)	2030-2050 (%/year)
Central	8.0	7.0	6.0
East	6.0	5.0	4.0
North	6.0	5.0	4.0
South	4.0	3.5	3.0
West	8.0	6.0	5.0
Total Africa	5.8	5.0	4.3

Source: (IRENA, 2011)

Table 13 GDP growth rates of the countries of selected centers of demand (Group, 2013). (e) estimations, (p) projections.

Real GDP Growth rate	2011	2012(e)	2013(p)	2014(p)	Increment (%)
Nigeria	7.4	6.6	6.7	7.3	-1%
D. Republic of the Congo	6.9	7.2	7.4	11.5	66%
Sudan	2.7	-0.6	2.2	3.4	25%
Angola	3.9	7.9	8.2	7.8	100%
Côte d'Ivoire	-4.7	8.6	8.9	9.8	308%
South Africa	3.5	2.5	2.8	3.5	0%
Kenya	4.4	4.2	4.5	5.2	18%
U. Republic of Tanzania	6.4	6.4	6.9	7	9%
Senegal	2.1	3.7	4.3	5.1	142%
Ghana	14.4	7.1	8	8.7	-39%
Cameroon	4.1	4.9	5	5.2	26%
Burkina Faso	4.4	8	6.7	6.8	54%
Mali	2.7	-1.5	5.4	5.1	88%
Zimbabwe	10.6	4.4	5	5.7	-46%
Mozambique	7.3	7.4	8.5	8	9%
Zambia	6.8	7.3	7.5	7.8	14%
Somalia	-	-	-	-	-
Congo	3.4	4.9	5.1	5.3	55%
Niger	2.1	13.1	5.5	6.5	209%
Liberia	8.2	8.9	7.7	5.4	-34%

Source: (Group, 2013)

Finally, consumer's ability and willingness to pay for electricity is also an important factor that determines electricity demand growth. In countries like Guinea, rural surveys found that the willingness to pay for basic electricity services was just USD 1.6/month, which would just cover the cost of 12 kWh/month if we consider that the average tariff in SSA region is USD 0.14/kWh (Ncube, 2013). Here we illustrate the low electricity consumption that can be afforded with USD 1.6/month: a 100 Watt light bulb burning for one hour uses 0.1 kWh, then when the bulb burns for 100 hours uses 10 kWh. Therefore, 12 kWh/month results in approximately 4 hours of burning a single bulb per day. Under such conditions, any form of electricity is a major expense.

7.2 Identification of import corridors

The next step of the analysis is the detection of land resources which allow the deployment of the transmission corridors. This detection was performed by means of suitability mapping involving a variety of geographical data at which was assigned different weights according to specific siting criteria. The identification of the transmission corridors was done with the Geographic Information System ArcGIS 10.1® from ESRI, California (USA). All layers of the analysis were converted into the projected coordinate system African Equidistant Conic in meters with datum WGS84 and resampled to a standard resolution of 1 km x 1 km at the Equator (0.008333333 degrees).

7.2.1 Evaluation of the suitability of land for import corridors

For identification of transmission corridors purposes, (May, 2005) and (Trieb, et al., 2009) use a sectorized method to identify the suitable land for transmission line location by evaluating separately isotropic and anisotropic map images. The evaluation of land for import corridors is built on previous analysis and distinction between isotropic and anisotropic features during the weighting method is taken into account. Moreover, a weighted system for slope variability, land cover typology and proximity to current transmission lines as described in (May, 2005) and (Trieb, et al., 2009) is considered. The result is a multi-criteria analysis in a raster data model with continuous grid-based representation. This multi-criteria analysis is built on the combination of numerical weight values that logically influence the modeling results so that a final weight-based image results.

In the analysis, non-excluded areas for implementation of transmission lines are assigned different weights regarding its suitability for power line location in order to identify the least-cost interconnection path between supply and demand sites. By summing the different weights, the model calculates the friction costs (relative costs) for each pixel of the image that must be raised to get from one pixel to the adjacent one. Then, the model generates a cost-distance image that identifies the least-cost HVDC transmission corridors to the specified centers of demand.

7.2.2 Suitability parameters

To determine the exact pathways of the HVDC electricity lines to interconnect the points of solar power generation with the points of demand in SSA, the following technical criteria was considered: need to minimize line length and angle points, avoid steep terrain and avoid to the maximum extent water bodies²⁷. An exclusion mask (Figure 14) similar to the CSP plant suitability map was created excluding all sites not suitable for the construction of HVDC lines to minimize adverse economic and environmental side effects.

Isotropic features

Transmission network

Considering the interest to install CSP plants as close as possible to existing power lines in order to reduce further construction costs of line extension, this study considers proximity to network line as a

²⁷ For a further downscaling, more criteria for site selection will play an important role. Land properties, regional and national policies, environmental, resettlement and land compensation aspects have to be considered in an effort to minimize negative impact from transmission corridors.

weighting factor. This is selected in the way that it increases linearly with increasing distance to the power line up to a maximum value of 10.0.

Land cover

Friction costs are also assigned to land cover considering its suitability for line location. When transmission lines cross urban terrain and thus an increment of 6 meters of tower height is required, costs increase around 11 percent (Parsons Brinckerhoff, 2012). When transmission lines cross wet peaty ground, costs increase around 16 percent to 24 percent over the base case of drained arable ground. Unstable/mined/quarried ground requires larger and deeper tower foundations to avoid subsidence during operation foundation. In this case, costs may increase in 24 percent to 48 percent. For large river crossing, associated structures are needed, and the increment in costs rises up to 60 percent to 100 percent (Parsons Brinckerhoff, 2012).

A value of 1.0 is specified for the base cost value of the friction image up to a value of 100.0 depending on the typology of land. Those factors correspond to the cost that has to be raised to pass from one pixel to the adjacent one. Thus, areas with a friction factor of 1.0 have high priority for the location of HVDC power lines.

Anisotropic features

On the contrary than the isotropic features, the anisotropic features receive weighting values in a specific direction, as applied to the slope of the terrain (z axis). Generally, it is assumed that the line costs rise linearly when increasing the slope. Thus, an increment of 5 percent in costs is normally given over the base case of normal ground conditions when line may cross rolling hills where typically 3meters extra of tower height is required (Parsons Brinckerhoff, 2012).

ArcGIS® considers slopes of 100 percent for terrains with 45° of inclination and of 200 percent for terrains with 90° of inclination. As additional expenditures are expected with the increment of slope, this study considers a base factor equal to 1.0 for slopes up to 20 percent and ranges the slope in steps of 45 percent up to a friction factor of 10.0.

7.3 Transmission corridors for selected energy demand nodes

The combination of the suitability parameters shown in Table 14 results in an exclusion mask (Figure 14) that yields the potential transmission corridors to connect the solar fields with the SSA major centers of demand (Figure 15). The analysis was done such as a potential transmission corridor was mapped from the center of demand to the nearest solar field for each DNI level, ranging from 1,000 kWh/m² per year to 2,800 kWh/m² per year. This results in ten possible corridors for each center of demand in dependence of the DNI level.

Table 14 Weighting criteria for the evaluation of land for HVDC import corridors. Weighting criteria was based on (DLR, 2009).

Feature	
Land cover	Weight
Forest	
Closed evergreen lowland forest	10.0
Degraded evergreen lowland forest	10.0
Montane evergreen forest (> 1500 m)	10.0
Sub-montane forest (> 900 m)	10.0
Swamp forest	10.0

Mangrove	10.0
Mosaic Forest / Croplands	1.0
Mosaic Forest / Savanna	1.0
Closed deciduous forest (Miombo)	1.0
Woodlands, shrub lands and grasslands	
Deciduous woodland	1.0
Deciduous shrub land with sparse trees	1.0
Open deciduous shrub land	1.0
Closed grassland	1.0
Open grassland with sparse shrubs	1.0
Open grassland	1.0
Sparse grassland	1.0
Swamp bushland and grassland	1.0
Agriculture	
Croplands (> 50%)	1.0
Croplands with open woody vegetation	1.0
Irrigated croplands	5.0
Tree crops	1.0
Bare soil	
Bare rock	5.0
Stony desert	5.0
Sandy desert and dunes	5.0
Salt hardpans	Excluded
Water bodies	100
Protected areas	10
Transmission network (km)	Weight
< 10	1.0
10-20	2.0
20-30	3.0
30-40	4.0
40-50	5.0
50-60	6.0
60-70	7.0
70-80	8.0
80-90	9.0
> 90	10.0
Slope (%)	Weight
0-20	1.0
20-65	2.0
65-110	6.0
110-155	10.0
155-200 (90°)	10.0

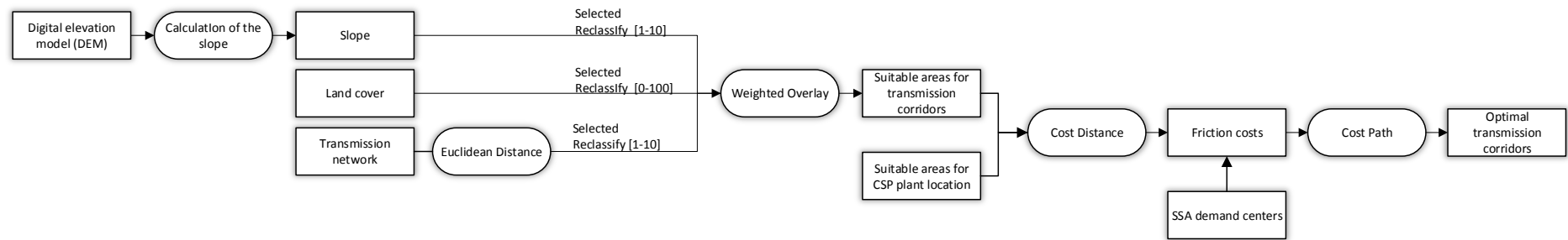


Figure 14 Methodological flowchart for the GIS-based HVDC identification of import corridors.

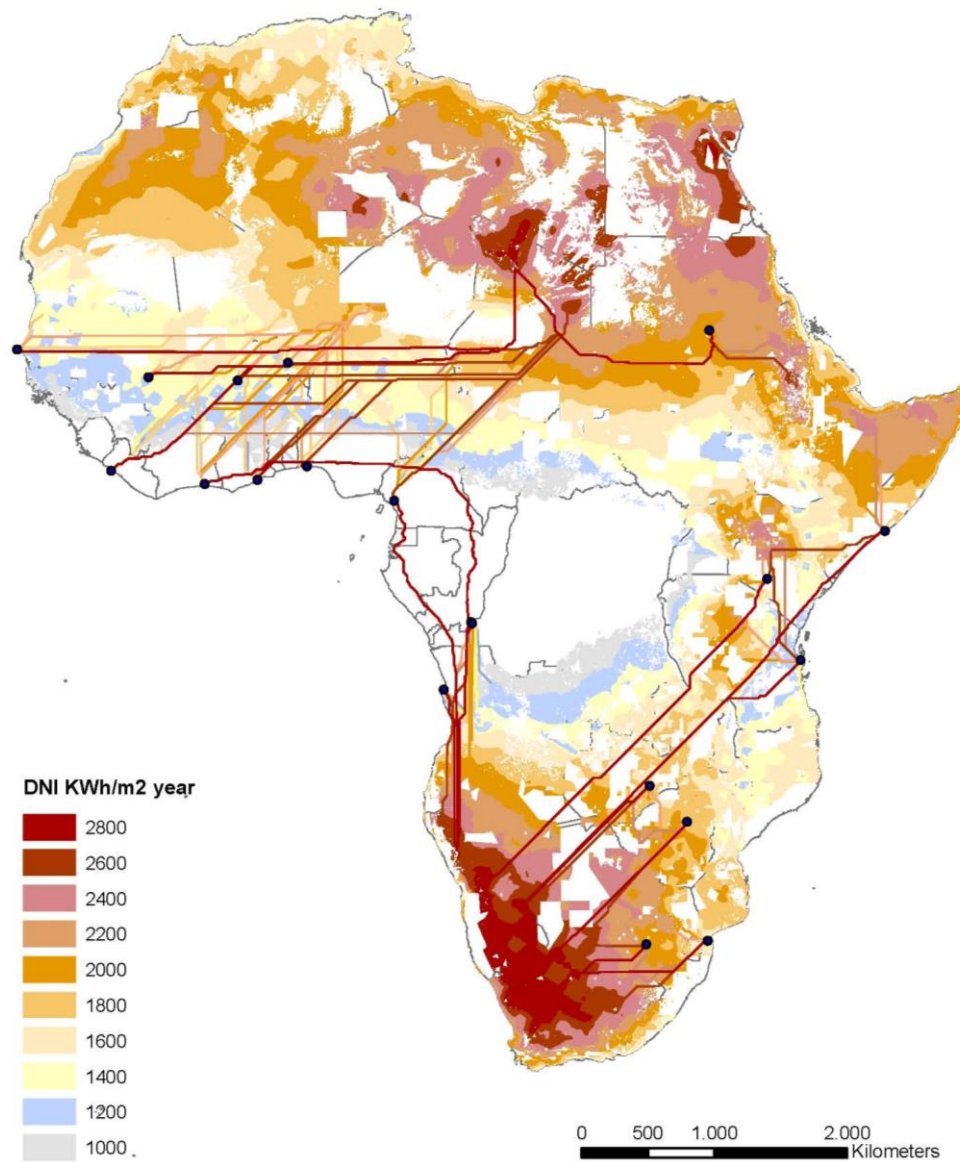


Figure 15 Transmission corridors to the nearest potential solar fields for each DNI level to the main SSA centers of demand. For individual maps for each DNI level, see Appendix B.

8

CSP Site Suitability Model

8.1 CSP Site Suitability Model

Total technical potential for CSP plants present in the different African regions described in section 3.1.1, shows that concentrating solar power technologies may have a very important role in the supply of renewable electricity to growing African centers of demand. Nevertheless, affordability of CSP technology at different irradiation levels for the whole Africa in relation with selected million-plus cities is still uncertain. Thus, this study analyses the cost evolution of generating electricity by means of CSP both in geographic and temporal terms in relation with major SSA urban centers.

For this purpose, we have developed the **CSP site suitability model** that simulates the generation and transmission costs of providing electricity from a CSP plant at any point in Africa to the main SSA centers of demand. The model has been developed using Geographic Information System ArcGIS 10.1® and implements the algorithm developed in Equation 6 Appendix C. All layers of the analysis were converted into the projected coordinate system African Equidistant Conic in meters with datum WGS84 and resampled to a standard resolution of 1 km x 1 km.

The model calculates the electricity production and total costs of parabolic trough and solar tower CSP plants and HVDC transmission installations on a yearly basis. It considers the value of the DNI for each pixel of the GIS map and the difference in distance from each pixel of the map to the pixel corresponding to the nearest center of demand. Among the top twenty SSA centers described in section 7.1, we have targeted four urban centers regarding their importance for each geographical region and their differences in the energy production portfolio: Lagos (Nigeria, Western African Power Pool), Kinshasa (DR of the Congo, Southern African Power Pool), Nairobi (Kenya, Eastern African Power Pool) and Johannesburg (South Africa, Southern African Power Pool).

The main value of the model is that it can be run under a number of different scenarios and reflect the reduction costs of different CSP generation technologies and transmission systems under different growth deployments. For example, by comparing a scenario under which reduced transmission capacity is built worldwide with reduced learning rates²⁸ with an expansion scenario under which cross-border transmission capacity is widely deployed. The model can also be used to evaluate the feasibility (costs and site implementation) of a CSP plant when exclusion masks to identify the suitable land to install CSP plants are also included into the model.

²⁸ Due to laws of mass production, technology development and experience learning, future costs of electricity produced by means of CSP are expected to decrease substantially in next decades (Hernandez-Moro, et al., 2013).

Final outputs are total investment costs, total electricity delivered to the grid and the average annual levelized electricity costs. It considers different learning rates for CSP and HVDC technologies, different interest rates, insurance rates, and efficiency improvements for solar tower and parabolic trough plant along the period under study.

Figure 16 represents the methodological flowchart for the **CSP site suitability model**. The subject of modeling is a 100 MW solar power tower plant with dry water cooling and 12 hours of thermal storage. Dry cooling is assumed as it can be applied everywhere without constraints of water availability. However, dry cooling will reduce plant efficiency as specified in section 5.1. 100 MW of capacity was selected in accordance with the largest solar thermal plant, Shams 1, installed in Abu Dhabi in March 2013. In 2016, South Africa plans to install another solar plant of 100 MW under the portfolio of the South Africa's integrated resource plan (IRP). In accordance with the breakthrough HVDC lines being built in China (ABB, 2013), a transmission system with a line capacity of 5 GW and ± 800 KV HVDC lines with double dipole were used as reference to simulate the future transmission system. Bipolar lines were assumed in order to increase security of supply, as in case of failure half of the capacity will remain available. The year 2020 was defined as temporal reference for the construction of the solar plant and the transmission system.

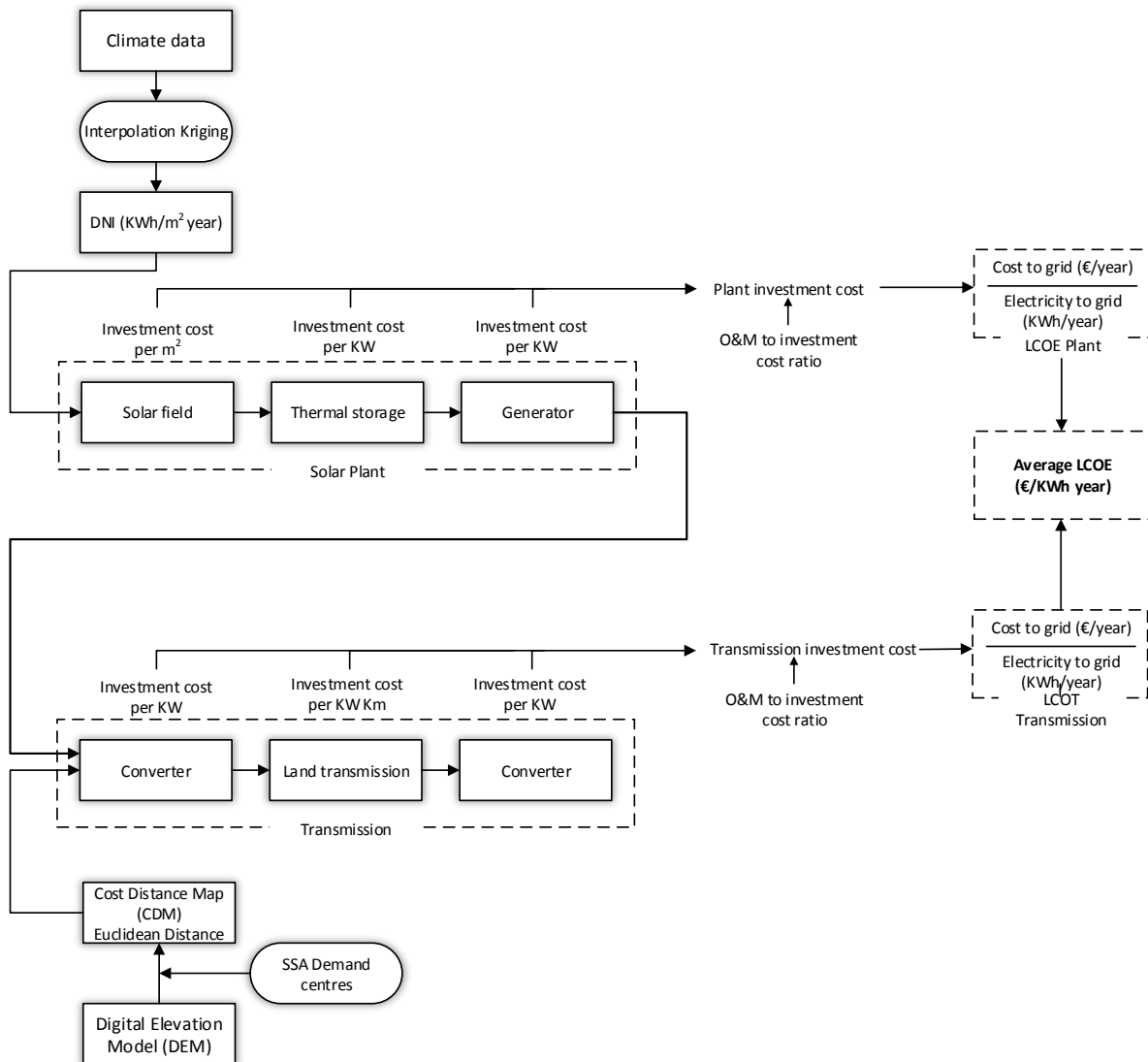


Figure 16 Methodological flowchart for the CSP site suitability model.

8.2 Methodology

The levelized cost of the electricity (LCOE) is the price of the electricity required for a project when revenues would equal to cost, including making a return on the capital invested equal to the discount rate. For an electricity price above this, it would result in a greater return on capital. While for an electricity price below, it would result in a lower return on capital, or even a loss. The LCOE of renewable energy technologies differs with technology, country and project, based on the renewable energy resource, capital and operating cost and the efficiency/performance of the technology (IRENA, 2012). Therefore, many potential trade-offs have to be considered when developing a LCOE modeling approach in a cross-country scale. Nevertheless, a more detailed LCOE analysis results in a significantly higher overhead cost in terms of the granularity of assumptions required. As this analysis is based in a continent scale, the differences among the countries do not allow to robustly populating the model with different assumptions because the accuracy of the approach may be misleading. Exemplary, among these differences arise country- and location-specific costs of doing business because of the uncertainty of present and future business practices in each specific country.

The most important components to calculate the LCOE are capital expenditures, i.e. investments for power plants and transmission lines. A 10 percent discount rate²⁹ is used to calculate the annuity at which capital expenditures are included in the system cost. Furthermore, fixed costs of operation and maintenance (O&M) for power plants and transmission lines are also considered. This approach, however, does not take into account dispatching aspects which would change specifics. Eq. (2) and Eq. (3) summarize the formula used in the model to calculate the LCOE of a CSP plant and the LCOT of the HVDC transmission lines:

$$\text{LCOE} = \text{LCOE}_{\text{plant}} + \text{LCOT}_{\text{transmission}} ; \text{Equation 1}$$

$$\text{LCOE}_{\text{plant}} = \frac{I_{\text{plant}} * (\text{FCR} + j) + O\&M}{E_t} ; \text{Equation 2}$$

$$\text{LCOT}_{\text{transmission}} = \frac{I_{\text{trans}} * (\text{FCR} + j) + O\&M_{\text{trans}}}{C_{\text{trans}}} ; \text{Equation 3}$$

$$\text{FCR} = \frac{i(1+i)^n}{((1+i)^n) - 1} ; \text{Equation 4}$$

²⁹ Discount rate in this context is the interest rate central bank charges when borrowing from the country's national reserves. Projects perceived to have relatively higher risks, are charged with higher discount rates. Exemplary, the lower the discount rate (e.g., 5%), shows that the project could be feasible immediately with government loan guarantees; while the highest the discount rate (e.g., 15%), shows that CSP is still perceived as a risk for the investors.

Specific formulas for the LCOE of the plant are listed below:

$$I_{plant} = I_{power\ block} + I_{power\ tower} + I_{storage} + I_{mirror\ field};$$

$$I_{power\ block} = C_{plant} * power\ block;$$

$$I_{power\ tower} = C_{plant} * power\ tower;$$

$$I_{storage} = C_{plant} * storage\ time * power\ tower;$$

$$I_{mirror\ field} = ((C_{plant} * 8760\ h * load\ factor) / (irradiance * efficiency)) * mirror\ field;$$

Specific formulas for the LCOT of the transmission system are listed below:

$$I_{trans} = \left(\left(distance\ km * HVDC_{line} \frac{\epsilon}{kWkm} * C_t kW \right) + \left(converter \frac{\epsilon}{kW} * C_t kW * 2 \right) \right) * CRF;$$

$$O\&M_{trans} = \left(converter \frac{\epsilon}{kW} * C_t * 2 * O\&M_{con} \right) + \left(distance\ km * HVDC_{line} \frac{\epsilon}{kWkm} * C_t kW * O\&M_{line} \right);$$

$$C_{trans} = (C_t\ kW * 8760\ h) - Losses;$$

$$Losses = (Converter_{losses} * 2 * C_t kW * 8760\ h) + (distance\ km * HVDC_{losses} * C_t kW * 8760\ h);$$

Where:

LCOE _{plant}	the average lifetime levelised cost of electricity generation
LCOT _{transmission}	the average lifetime levelised cost of transmission
FCR, annuity factor	the fix change rate or annuity (%/year)
i	discount rate 10%
j	insurance rate 1%
n	expected economic life of the system, 30 years for the CSP plant and 40 years for the HVDC transmission lines
E _t	electricity production (kWh/year)
C _{plant}	Plant capacity (kW)
storage time	Hours of thermal storage (h)
load factor	Percentage of time that the plant is operating at design capacity

C_t	transmission capacity (kWh/year)
C_{trans}	final transmission capacity kW
I_{plant}	investment expenditures for the CSP plant (€)
$I_{power\ block}$	investment expenditures for the power block (€)
$I_{power\ tower}$	investment expenditures for the power tower (€)
$I_{storage}$	investment expenditures for the thermal storage (€)
$I_{mirror\ field}$	investment expenditures for the mirror field (€)
I_{trans}	investment expenditures for the HVDC lines (€)
power block	investment for the power block (€/kWh _e)
power tower	investment for the tower and receiver (€/kWh _{th})
storage	investment for the thermal storage (€/kWh _{th})
mirror field	investment for the mirror field (€/m ²)
efficiency	Solar to electric efficiency of the solar plant (%)
$O\&M_{plant}$	operations and maintenance expenditures for the CSP plant assumed to be 4.8% ³⁰ of the total investment per year
$O\&M_{trans}$	operations and maintenance expenditures for the HVDC lines
$O\&M_{line}$	cost of operations and maintenance HVDC lines (% investment/year)
$O\&M_{con}$	operations and maintenance of the converter station (% investment/year)
$HVDC_{line}$	cost of the HVDC lines (€/kWkm)
Converter	cost of the converter station (€/kW)
$HVDC_{losses}$	losses produced in HVDC transmission lines % /1000 km
$Converter_{losses}$	losses in the converter station (%)
CRF	cost reduction factor from reference year to projected construction year
distance	distance from each pixel of the Euclidean distance GIS map to the nearer center of demand (km)
irradiance	irradiance level for each pixel of the DNI GIS map (kWh/m ² /year)

8.2.1 Economic and technical parameters

One of the main reasons that have delayed a larger deployment of solar electricity by utilities so far is that, even for sites with high solar resource, electricity generation costs by mean of CSP are substantially higher than the cost of the electricity provided by the current grid. Indeed, CSP cost ranges between USD 0.2-0.33 kWh for parabolic trough and USD 0.16-0.27 kWh for solar tower

³⁰ O&M cost for a solar tower power plant of 100 MW rise up to USD 0.05 W according to (Turchi, et al., 2010). If the lifetime of a solar tower plant is 30 years, total O&M costs rise up to USD 1.5 W. If installation costs are meant to be USD 7.3 W according to (Turchi, et al., 2010). The ratio of total investment cost by total O&M cost is 4.8 percent.

plants depending on location (DNI), energy storage, interest rate and other assumptions (IEA, 2010) (Fichtner, 2010) (ESTELA, 2010) (Hinkley, et al., 2011); whilst the average tariff in sub-Saharan region is USD 0.14 kWh (Ncube, 2013).

Reality shows that is very difficult for developing countries to invest in renewable technologies because of the high upfront investment, and many countries still relying on traditional fossil fuel technologies for the production of electricity. This is the major reason why large amounts of traditional fossil fuel technologies continue to be invested in developing countries, despite the threats posed by climate change, the risks of increasing fuel costs and the risk of an increasing carbon price (Battaglini, et al., 2010). Thereby, achieving cost reductions in solar power generation is not only important for the cost-efficient implementation of renewable electricity worldwide, but specifically for the cost-efficient implementation of renewable electricity in Africa where investment competition among different sectors is high. Consequently, keeping a sustained level of solar power expansion worldwide is crucial to make renewable technologies competitive with conventional sources, to ultimately become the cheapest and least risky solution to satisfy the rapid growing electricity demand in developing countries. This may be achieved by several ways such as technology development, economies of scale (technology becomes cheaper as total production increases) and learning effects (costs are reduced through “learning by doing”).

To calculate future LCOE of CSP and HVDC transmission system as part of the **CSP site suitability model**, the evolution of the cost of the systems and thus the concept of “learning by doing” is described by a log-linear relation between the accumulated experience and the technical and economic performance. This log-linear relation is known as the learning curve and is described in Eq. (5).

$$C_t = C_0 * \left(\frac{P_t}{P_0}\right)^{\frac{\log PR}{\log 2}} ; \text{Equation 5}$$

According to Eq. (5), the cost of the system C_t at a future t year can be calculated from the initial cost C_0 of the CSP system ($C_{0\ 2012} = \text{USD } 4.60_{\text{REAL}} \text{ W}$ (Hernandez-Moro, et al., 2013)), and the cumulative installed capacities P_0 at the starting year 2012, and at the year t , P_t (Table 15). The exponent PR , the so-called progress ratio, is used to express the progress of cost reductions for different technologies. E.g., a progress ratio of 80 percent means that costs are reduced by 20 percent each time the cumulative production is doubled. Thus, considering the expression ($LR = 1 - PR$), and given a learning rate (LR) of 10 percent suggested by (Agency, 2010), we calculated C_t for different P_t forecasts. The ratio of C_t and C_0 along the years provides the cost reduction factor to be applied to the present cost of the different CSP plant components to calculate the relative costs of such components in the future.

Table 15 CSP cumulative installed capacity at the beginning of 2012 and objectives for decades 2020-2050 according to the Roadmap Scenarios published by the IEA in 2008 (IEA, 2008).

	2012, P_0	2020, P_{2020}	2030, P_{2030}	2040, P_{2040}	2050, P_{2050}
Cumulative installed capacity (GW)	1.76	148	337	715	1089

Economic and technical parameters used to calculate the LCOE of the power plant and the LCOT of the transmission system are listed in Table 16.

Table 17 lists the data sources where the economic and technical parameters were extracted.

Table 16 Description of the economic assumptions.

Factor	Description
Annual irradiation data	Correspond to the DNI value of each of the pixels of 1km ² of the DNI GIS map
Starting year and worldwide capacity	2012 with a worldwide estimated capacity (for learning rate purposes) of 1.76 GW
Year of construction of the facilities	2020
Plant capacity:	100 MW
Storage capacity:	12 h of thermal storage
Heat transfer fluid:	Molten salts
Lifetime CSP plant:	30 years
Efficiency parabolic through plant 2020/2050:	0.16/0.17
Efficiency power tower plant 2020/2050:	0.18/0.19
Cost of parabolic trough plants:	Power block: USD 1007 _{REAL} kW _e Thermal storage: USD 25 _{REAL} kWh _{th} Mirror field: USD 190 _{REAL} m ²
Cost of power tower plants:	Power block: USD 1140 _{REAL} kW _e Tower and receiver: USD 200 _{REAL} kWh _{th} Thermal storage: USD 20 _{REAL} kWh _{th} Mirror field: USD 143 _{REAL} m ²
HVDC connection length:	Correspond to the distance from each pixel of the Euclidean-Distance GIS map to the nearest center of demand
HVDC connection type:	± 800kV HVDC double-dipole
Transmission system capacity:	5 GW
Lifetime transmission infrastructure:	40 years
HVDC connection cost:	EUR 0.06 _{REAL} kWkm (cables installed same year as CSP)
HVDC transmission losses:	2.6%/1000 km
Cost converter station (x2):	EUR 67.90 _{REAL} kW
Converter station losses:	0.8%
Learning rate:	10% for CSP plant and HVDC transmission. Indicates the cost reduction percentage per cumulative doubling of future installed capacities (Agency, 2010).
Financing:	Interest rate 10%; insurance rate 1%
Exchange rate used USD/EUR:	0.763

Table 17 Data types and sources used in CSP site suitability model.

Type	Source(s)	Comments
Costs parabolic trough plant	(Turchi, et al., 2010)	Power block, mirror field, storage and O&M costs
Costs solar tower plant	(Turchi, et al., 2010)	Power block, mirror field, storage and O&M costs
HVDC transmission costs	(Institute, 2010) (Wind, 2013)	HVDC line, converter station costs
HVDC transmission costs	(Siemens, 2012) (Greenwire, 2011)	Line losses, converter losses

9

Economic screening assessment

9.1 Key benefits of CSP energy in sub-Saharan African context

Recent external factors have exacerbated the already precarious power situation in SSA. Sectorial assessment of the risks of climate change in Africa reveals a rainfall variability and increase incidence of drought (UNEP FI, 1202). Drying of lakes and rivers, coupled with persistent droughts, will decrease water flows to hydropower plants. So far, drought has seriously reduced the power available to hydro-dependent countries in Western and Eastern Africa (see Table 29 in Appendix C). Therefore, countries as Kenya, Ghana, Uganda, Tanzania and Zimbabwe, must rely on expensive emergency diesel generators to meet the demand. Moreover, high and variable international oil prices have put a huge burden on oil-importing countries, especially those dependent on diesel and heavy fuel oil for power generation (Eberhard, et al., 2011). Excluding the 6 largest oil-exporting countries in SSA (Nigeria, Angola, Sudan, Equatorial Guinea, Republic of the Congo and Gabon) and analyzing the situation of the remaining countries, total oil imports are significantly larger than oil exports. Furthermore, imports are growing faster than exports (UNEP FI, 1202). The soaring cost of importing refined oil constitutes a significant burden for SSA countries and can severely hinder their economic growth. Exemplary, in 2010, African countries imported USD 18 billion worth of oil; this value is higher than the amount they received in foreign aid (IRENA, 2013). Further, SSA countries are, on aggregate terms, net importers of coal. In 2010, the total imports of coal exceeded 1.4 million short tons, while total exports were 325,000 short tons (UNEP FI, 1202). Renewable energy generation can increase security of supply by diversifying the existing energy portfolio and reducing economy vulnerability to fossil fuels' price volatility. For oil-importing countries, acceptance of renewable technologies may be an opportunity to redirect foreign exchange flows by promoting imports of goods that cannot be produced locally, instead of depending on conventional energy imports (Edenhofer, et al., 2011). Consequently, oil-importing countries will need to decide whether they continue on the traditional path or strive for energy sufficiency through renewables.

Increasing awareness of some SSA countries in renewable energy by including renewable energy policies and renewable energy targets in their portfolios (Edenhofer, et al., 2011), and the increasing incidence of droughts, makes it essential to evaluate the importance of renewable energy sources other than hydropower. Among wind, biomass, photovoltaics or other exotic technologies as waves and tidal, CSP has a huge potential due to the excellent solar resources and its intrinsic characteristics in terms of supplying power on demand. In next section, we examine the possibility of SSA urban centers to benefit from CSP in economic and geographical terms. Additionally, we examine into which extent imports of CSP for the different SSA regions are an affordable replacement for the electricity produced from traditional fossil fuel sources; and whether or not imports of CSP are an interesting option in those countries already reliant on other renewable sources.

9.2 Case studies

With the objective to assess the role of CSP plants for major SSA centers of demand, the **CSP site suitability model** results in a GIS map that provides the total LCOE (generation + transmission) of a parabolic trough and solar tower plant for the years 2020 and 2050 (see Figure 40 to Figure 43 in Appendix C for examples of the model results of a solar tower plant). Total LCOE is calculated by adding the cost of the electricity generation by a CSP plant for each pixel of the map in relation with the DNI level to the cost of deploying an HVDC transmission line from each pixel of the map to the selected center of demand. We have selected four SSA centers of demand likely to be representative regarding their geographical location and variability of the energy consumption and production sources.

Financial viability of projects will depend upon the resources, technology and project costs, and the extent of government-driven financial support. Current technology costs and constraints on financial support indicate that only projects that are located in the areas with the highest irradiance level (with annual average DNI values greater than 2,000 kWh/m²/year) are expected to be viable in the near future (IRENA, 2012). When applied this constraint, the model results in a spatial distribution of the LCOE up to a maximum acceptable cost level (see Figure 17 to Figure 20).

In this section we focus on the four selected centers (Lagos, Kinshasa, Nairobi and Johannesburg) to allow for a more in-depth discussion at a narrow geographical level. From the model results, we can observe that the least-cost electricity by means of CSP is achieved both in domestic and trans-boundary level. For a $\text{DNI} \geq 2,000 \text{ kWh/m}^2/\text{year}$, Lagos achieves CSP electricity at a cost of > cEUR 9.0/kWh within the borders of Niger and Chad (see Figure 17). However, for a $\text{DNI} \geq 2,800 \text{ kWh/m}^2/\text{year}$, Lagos achieves least-cost electricity < cEUR 8.0/kWh within the Sahara Desert at a cross-border level between Chad and Libya. When a transmission corridor across different countries is needed, power stations and transmission lines may be an intrusion into different jurisdictions, tribal lands, lands in conflict (e.g., religious conflicts) or sensitive military zones, among others. Therefore, a number of specific issues other than solar resources must be addressed in a regional level. From a foreign perspective, Sahara Desert may seem arid and worthless, but is inhabited by several Bedouin tribes and therefore the people's right must be included in the planning framework. And if the right of the local people is not recognized, security of supply can turn into a greater risk and the threat of attacks against power plants and transmission lines may increase (Battaglini, et al., 2010).

Regions with low irradiance levels as Central Africa will have to rely on transmission lines either towards the Sahara Desert or the Kalahari Desert to achieve least-cost CSP electricity < cEUR 8.0/kWh at $\text{DNI} \geq 2,800 \text{ kWh/m}^2/\text{year}$. (e.g., from a least-cost distance perspective, Kinshasa will have to rely on imports from the Northern part of the Kalahari Desert) (see Figure 18).

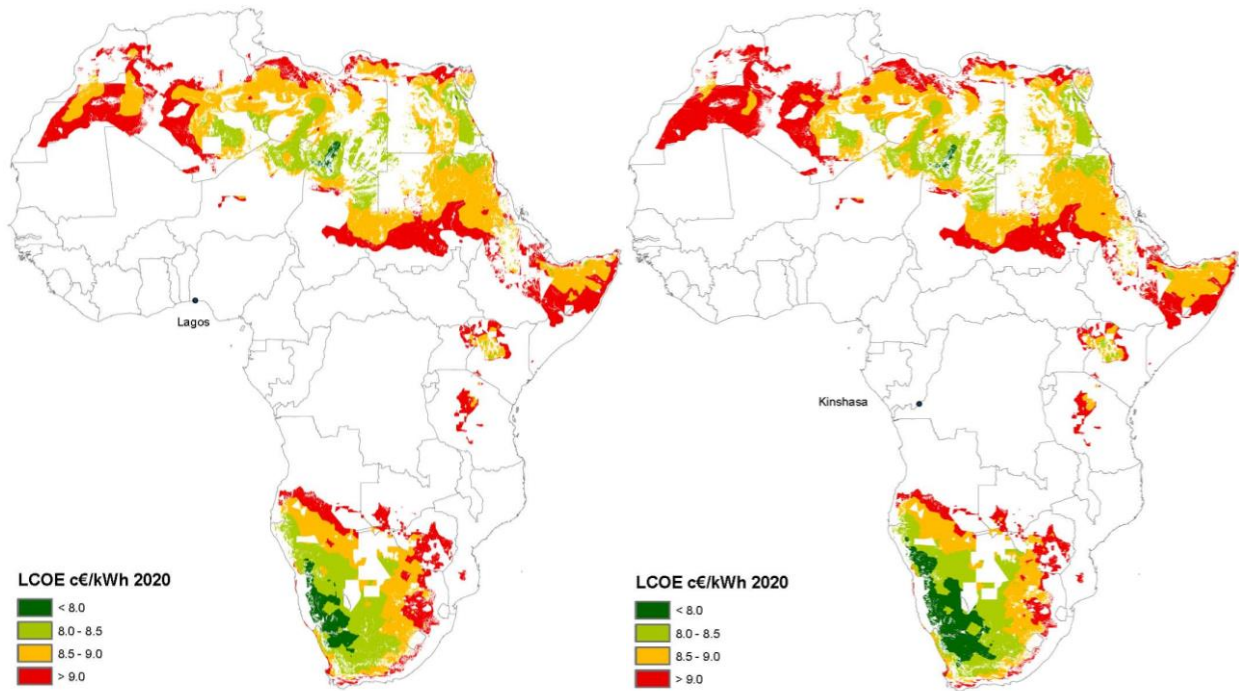


Figure 17 (Left) Spatial distribution of the levelized cost of the electricity (cEUR/kWh year) in Africa by 2020 of a solar tower plant for $\text{DNI} \geq 2,000 \text{ kWh/m}^2/\text{year}$ in relation with Lagos as center of demand (Western Africa Power Pool).

Figure 18 (Right) Spatial distribution of the levelized cost of the electricity (cEUR/kWh year) in Africa by 2020 of a solar tower plant for $\text{DNI} \geq 2,000 \text{ kWh/m}^2/\text{year}$ in relation with Kinshasa as center of demand (Southern Africa Power Pool).

Eastern African is well endowed with irradiation resources and thus has more probabilities to produce electricity from CSP domestically. A solar electricity supply within the region will create independency from trans-boundary electricity exchange and therefore the political risk of countries in conflict is minimized. E.g., Nairobi achieves domestic electricity between cEUR 8.0 kWh and cEUR 8.5 kWh, for a $\text{DNI} \geq 2,000 \text{ kWh/m}^2/\text{year}$. However, to achieve lower prices and ensure a better implementation of CSP in the energy sector, the region will have to rely on imports from distant countries (see Figure 19).

South Africa is one of the major CO_2 emitters worldwide (Boden, et al., 2010). Moreover, the country plans to commission two of the largest coal-fired plants in the world. However, is the only country among the twenty million-plus cities identified in this study that can achieve least-cost CSP electricity domestically at < cEUR 8.0 kWh without depending on third countries (see Figure 20). This is also the case of other South African countries that can achieve least-cost electricity domestically, i.e. Namibia and Botswana.

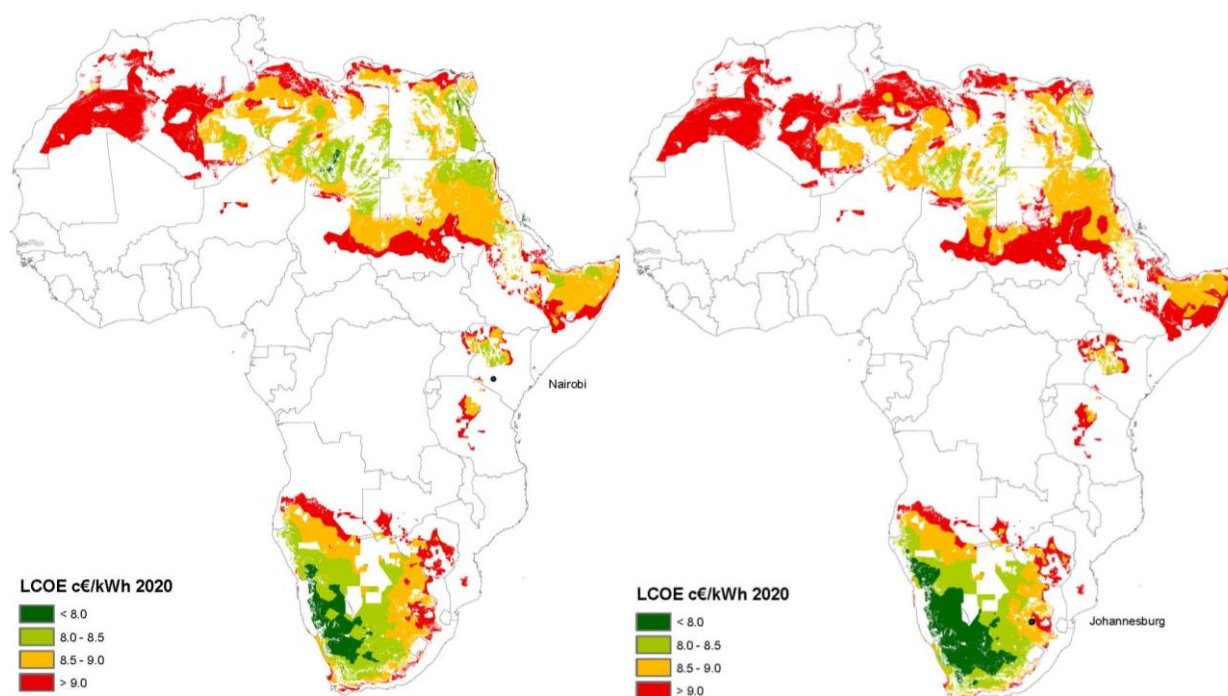


Figure 19 (Left) Spatial distribution of the levelized cost of the electricity (cEUR/kWh year) in Africa by 2020 of a solar tower plant for $\text{DNI} \geq 2,000 \text{ kWh/m}^2/\text{year}$ in relation with Nairobi as center of demand (Eastern Africa Power Pool).

Figure 20 (Right) Spatial distribution of the levelized cost of the electricity (cEUR/kWh year) in Africa by 2020 of a solar tower plant for $\text{DNI} \geq 2,000 \text{ kWh/m}^2/\text{year}$ in relation with Johannesburg as center of demand (Southern Africa Power Pool).

The decision whether to benefit from best solar resources or from siting sites with lower solar resources will depend mainly on the country and the political risk. This not only includes the transition factors mentioned before but also the associated risk of expropriation or breach of contract. These factors often act as an initial selection filter in many financial decision-making processes (UNEP FI, 1202). Table 18 represents an overview of the potential cost-opportunities and transition countries dependent on importing CSP to selected regions. In addition to country and political risk, currency risk (enhanced by weak and volatile currencies and financial markets), and commercial risk (induced by the poor creditworthiness of state owned utilities), are also factors that worsens the risk profile and will influence the final decision on CSP investment (UNEP FI, 1202).

Table 18 Overview of the different costs and line transition factors to import CSP to the selected SSA regions. Johannesburg and some countries in Southern Africa are the ones that can benefit from least-cost CSP within their boundaries. Hence, most of the SSA countries will have to depend on geopolitical situations in transition countries to achieve least-cost CSP.

	Spatial location	Domestic production	Cross-border production
		Cost (cEUR/kWh)	Transition countries to achieve nearest least-cost CSP (<cEUR 8.0 kWh)
Lagos, Nigeria	Western Africa	-	Chad
Kinshasa, DRC	Central Africa	-	Angola, Namibia
Nairobi, Kenya	Eastern Africa	8.0 – 8.5	Sudan, Chad
Johannesburg, South Africa	Southern Africa	< 8.0	-

9.2.1 Cross-cutting economic assessment

Coal is currently the predominant fuel for electricity generation worldwide (Epstein, et al., 2011). In 2005, coal generated 7,334 TWh of electricity, which accounts for the 40 percent of total electricity generated. In 2005, coal-derived electricity was responsible for 7,856 Gt of carbon dioxide emissions, the 30 percent of CO₂ emissions worldwide (Epstein, et al., 2011). By 2030, global CO₂ emissions are projected to grow 1.8 percent per year up to 41,905 Gt, with emissions from the coal-power electricity sector projected to grow 2.3 percent per year up to 13,884 Gt (Epstein, et al., 2011). The scenario in Africa is not expected to be much better as coal and coal products account for 40 percent of total power generation (IRENA, 2011).

South Africa recently decided to build two coal-fired (Medupi and Kusile) power plants with a total capacity of 96,000 MW. Eskom has already announced that financing the Kusile plant may lead to an increase in prices by 40 percent over the period 2013/2018 (Eskom, 2013), which it will directly affect the economies of low-income classes. According to the South African organization GroundWork, “poor people will have to carry the financial burden that is connected to the construction of the plant. These plants will aggravate energy inequality: in order to cope with it, these households will have to restrict their energy consumption or stop using electricity altogether” (Petz, 2013).

In previous step, this study has examined the possibilities of selected urban demand nodes to benefit from least-cost electricity from CSP. In this section, we examine into which extent importing electricity from CSP to selected urban nodes is an affordable replacement to the electricity produced from coal. By downscaling the CSP costs given in previous section, we provide accurate ranges of the LCOE from CSP achieved by selected centers of demand along different years and for different technologies (see Table 19). Moreover, we provide total average LCOE of generating electricity by means of coal and natural gas-based plants for a wide range of technologies (see Table 20). The economic decision regarding capacity additions, however, would need more complex tools besides the leveled costs to simulate the operation of the power system.

Table 19 Minimum and maximum LCOE (cEUR/kWh) (generation + transmission) of a parabolic trough and solar tower plant for 2020 and 2050 in relation with Lagos, Kinshasa, Nairobi and South Africa as centers of demand. Prices are achieved in areas which land characteristics fulfill the requirements of CSP plant installation. Direct transmission was considered between the solar field and the center of demand. An irradiance threshold of $\geq 2,000$ kWh/m²/year was considered to be the minimum acceptable limit for a CSP plant.

Technology / Centre of demand	Levelized electricity costs (cEUR/kWh)			
	2020		2050	
	Minimum	Maximum	Minimum	Maximum
Parabolic Trough				
Lagos, Nigeria	8.26	11.09	6.12	8.22
Kinshasa, DRC	8.14	10.98	6.03	8.14
Nairobi, Kenya	8.22	11.10	6.09	8.23
Johannesburg, South Africa	8.03	11.26	5.94	8.35
Solar Tower				
Lagos, Nigeria	7.65	9.71	5.54	7.00
Kinshasa, DRC	7.53	9.60	5.45	6.92
Nairobi, Kenya	7.61	9.71	5.51	7.01
Johannesburg, South Africa	7.42	9.87	5.36	7.13

Table 20 Total generation costs of fossil fuel technologies in (cEUR 2012 kWh) (EIA, 2013). Exchange rate for (EIA, 2013) (REAL cUSD 2012 kWh): 1 USD = 0,763 EUR.

	Levelized capital cost (cUSD 2012 kWh)	Fixed O&M (cUSD 2012 kWh)	Variable O&M (including fuel) (cUSD 2012 kWh)	Total system levelized cost (cUSD 2012 kWh)	Total system levelized cost (cEUR 2012 kWh)
Coal-fired					
Conventional coal	6.66	0.41	2.97	10.1	7.71
Advanced coal	8.59	0.69	3.13	12.4	9.46
Advanced coal with CCS	9.00	0.89	3.79	13.7	10.45
Natural gas-fired					
Conventional combined cycle	1.61	0.17	4.93	6.71	5.12
Advanced combined cycle	1.77	0.20	4.58	6.56	5.01
Advanced CC with CCS	3.46	0.41	5.51	9.39	7.16
Conventional Combustion Turbine	4.50	0.27	8.14	12.9	9.84
Advanced Combustion Turbine	3.09	0.26	6.94	10.3	7.86

Nigeria

In Nigeria, the majority of electricity generation derives from thermal power plants (77 percent), with about two-thirds of thermal power derived from natural gas (64 percent) and the rest from oil (13 percent) (EIA, 2012). However, hydroelectricity (23 percent), the only other source of power generation, has decreased regularly from its peak of 8.2 billion KWh in 2002 to 4.5 billion KWh in 2009 (EIA, 2012).

Nigeria is the largest oil producer in Africa and the tenth largest worldwide (Outlook, 2013); however, the contribution of Nigeria's oil and gas sector to the national GDP in 2012 was roughly 14 percent. This is the result of importing 80 percent of the goods and services needed for projects in the sector (Outlook, 2013). Moreover, issues like the low capacity utilization and the common breakdowns of the refineries obliged the country to import about 85 percent of the refined petroleum products (Outlook, 2013). The economic growth associated to the oil sector has not translated into job creation or poverty alleviation; intensifying the need to diversify the economy into non-oil alternatives. Moreover, pollution from oil exploration activities and gas flaring in the Niger Delta (where Lagos is located), stressed by the heavy concentration of industries in locations highly vulnerable to climate change are an increasing concern (Outlook, 2013). Other factors that severely affect the power sector are the impacts of periods of drought. To illustrate the severity of these impacts, in 2007 in Ghana, the water level at the Akosombo dam fell below the minimum level of 73 meters. The results were a reduction of the hydroelectricity generation and therefore load shedding of electricity in the country (UN, 2007). Table 21 shows that for 2020, electricity from conventional gas combined cycle (CGCC) plants is still more economical than from concentrated solar tower plants. However, in 2050, concentrated solar tower plants are expected to produce electricity roughly at the same price than CGCC plants; thereby, becoming a potential candidate if Nigeria wants to diversify the electricity supply.

Democratic Republic of the Congo

Although the Democratic Republic of the Congo has substantial oil potential, the exploration and output still minor because of low investment and outdated equipment (Outlook, 2013). However,

electricity production is based on its huge hydropower potential as 60 million of its 100 million hectares of dense tropical forest can be sustainably used for electricity production (Outlook, 2013). Nevertheless, problems with outdated transmission network and equipment of the national water firm Ragideso and electricity company SNEL severely aggravated the performance of the energy sector since 1960 (Outlook, 2013). Currently, SNEL only serves to 10 percent of the population, with a poor electricity supply that cannot meet the demand; with lines in Kinshasa and from Inga to Katanga overloaded (Outlook, 2013). However, hydropower potential in Central Africa is huge. The Grand Inga project in the lower Congo river has a potential of around 44 GW, with just 1,775 MW currently installed (IRENA, 2011). The electricity from hydropower is affordable when compared to renewable sources or even fossil fuel sources and is very flexible operating both in peak and base load. However, recent droughts suggest that an alternative source of electricity can be required. Instead, Central Africa may decide to increase its hydropower capacity.

Kenya

Kenya mainly relies on import of crude oil that refines for both domestic and export use. Moreover, these imports are increasing over time. From 2007 to 2011, the quantity of imported petroleum products grew almost 20 percent; however, during the same period, the quantity of petroleum exports only grew by 5 percent (Outlook, 2013). In 2011, total electricity generation grew 8.4 percent; however, due to low rainfalls, hydropower generation declined by 0.21 percent compared to 2010. Kenya heavily relies on hydro and thermal power for electricity production; nevertheless, hydropower potential is continuously affected by the impacts of drought (see Table 29 in Appendix C). E.g., from 1992 to 2001, Kenya experienced periods of power rationing as a consequence of drought, and had to rely on more expensive fuel-based generation. Therefore, with the introduction of renewable electricity from CSP, Kenya may decrease the dependency on hydro potential and on the production of electricity from imports of oil.

South Africa

South Africa's total electricity consumption increased by 20 percent from 2000 to 2010, while installed capacity increased at a slower rate of 7 percent during the same period (EIA, 2013). In late 2007 and early 2008, the country experienced a power crisis that resulted in several blackouts and endangered the power supply to many businesses. This was the result of high economic growth and increasing electricity demand in the absence of new power plants (EIA, 2013). Due to the shortage in power capacity, South Africa is seeking for alternatives to supply the electricity gap. However, the main focus of the country has not been in renewable energy but in fossil fuel sources. CSP may lead the shift of this situation. In 2020, costs of importing CSP electricity with parabolic trough plant, considering production and transmission costs, will be just around 4 percent higher than conventional coal plants (CC) (see Table 21). Moreover, these costs will be even roughly lower than CC plants when considering production with solar tower plants. Additionally, in 2050, importing electricity from CSP could be roughly 20-30 percent more economical than producing it with CC if no fuel price variability is considered (see Table 22).

Variable capacity in South Africa impacts on system security and stability (Africa, 2011). The Policy-adjusted IRP proposes 10 percent penetration for wind and PV capacity as a share of total installed capacity in 2020 and 20 percent in 2030 (Africa, 2011). This results in impacts on the system from the fluctuating renewable capacity. Hence, the IRP 2010 considers the benefits of flexible dispatchable generation as back-up for this capacity to smooth the impact on the system (Africa, 2011). For this purpose, simple peak gas turbines are commonly used. However, other technologies as CSP with storage can be a complement to the back-up system. In 2020, importing CSP electricity will be far more expensive than using CGCC plants and thus is expected that gas turbines will have a leading role

in balancing system fluctuation. However, in 2050, imports of CSP electricity especially from solar towers can be a close competitor to gas turbines as back-up technology (see Table 21 and Table 22).

In a near future, new technologies are expected to enter in South Africa energy portfolio. Carbon capture and storage technology (CCS) capture CO₂ emissions at the point of release, and stores the particles emitted deep underground, helping to decrease the amount of CO₂ emissions. In South Africa, first demonstration projects with CCS are expected to be on the ground in 2020, and its commercialization stage is expected in 2025 (SACCCS, 2013). As this analysis takes year 2020 as temporal reference for parabolic trough and solar tower power technologies' construction, costs of demonstration plants with CCS were also considered. When comparing CSP import with advanced coal with carbon capture and storage (A with CCS), in 2020-2050 imports from CSP are 30-40 percent lower than producing electricity from coal and may be a substitute of coal plants for electricity generation.

Lastly, South Africa also heavily relies on hydro imports. The main risks associated with these imports are a delay in the construction of both the necessary grid extensions and the power plants, and severe, long-lasting droughts. In both cases, other dispatchable sources of generation as CSP can overcome the missing hydro capacity (Africa, 2011). Additionally, national policies as the National Climate Change Response Policy adopted in 2011 are crucial in adapting a higher generation share of renewable sources. As the main objective of the policy is to cut CO₂ emissions by 34 percent over the next decade (Outlook, 2013), alternative carbon free sources will be needed. Nonetheless, existing fossil-fuel plants have a life expectancy of about fifty years, and the impact of replacing this infrastructure prematurely may be harmful to African economy. A gradual implementation of CSP technology would be therefore a sustainable way to overcome the increasing electricity demand.

Table 21 Cost difference when compared the total system levelized cost from fossil fuel technologies without internalizing external costs with the levelized electricity cost in 2020 by means of CSP for selected centers of demand.

	Min. CSP cost 2020	CC	AC	AC with CCS	CGCC	Max. CSP cost 2020	CC	AC	AC with CCS	CGCC
Parabolic Trough										
Lagos, Nigeria	8.26	7%	-13%	-21%	61%	11.09	44%	17%	6%	117%
Kinshasa, DRC	8.14	6%	-14%	-22%	59%	10.98	42%	16%	5%	114%
Nairobi, Kenya	8.22	7%	-13%	-21%	61%	11.10	44%	17%	6%	117%
Johannesburg, SA	8.03	4%	-15%	-23%	57%	11.26	46%	19%	8%	120%
Solar Tower										
Lagos, Nigeria	7.65	-1%	-19%	-27%	49%	9.71	26%	3%	-7%	90%
Kinshasa, DRC	7.53	-2%	-20%	-28%	47%	9.60	25%	1%	-8%	88%
Nairobi, Kenya	7.61	-1%	-20%	-27%	49%	9.71	26%	3%	-7%	90%
Johannesburg, SA	7.42	-4%	-22%	-29%	45%	9.87	28%	4%	-6%	93%

CC: Conventional coal, AC: Advanced coal, AC with CCS, Advanced coal with Carbon Capture and Storage, CGCC: Conventional gas combined cycle.

Table 22 Cost difference when compared the total system levelized cost from fossil fuel technologies without internalizing external costs with the levelized electricity cost in 2050 by means of CSP for selected centers of demand.

	Min. CSP cost 2050	CC	AC	AC with CCS	CGCC	Max. CSP cost 2050	CC	AC	AC with CCS	CGCC
Parabolic Trough										
Lagos, Nigeria	6.12	-21%	-35%	-41%	20%	8.22	7%	-13%	-21%	61%
Kinshasa, DRC	6.03	-22%	-36%	-42%	18%	8.14	6%	-14%	-22%	59%
Nairobi, Kenya	6.09	-21%	-36%	-42%	19%	8.23	7%	-13%	-21%	61%
Johannesburg, SA	5.94	-23%	-37%	-43%	16%	8.35	8%	-12%	-20%	63%
Solar Tower										
Lagos, Nigeria	5.54	-28%	-41%	-47%	8%	7.00	-9%	-26%	-33%	37%
Kinshasa, DRC	5.45	-29%	-42%	-48%	6%	6.92	-10%	-27%	-34%	35%
Nairobi, Kenya	5.51	-29%	-42%	-47%	8%	7.01	-9%	-26%	-33%	37%
Johannesburg, SA	5.36	-30%	-43%	-49%	5%	7.13	-7%	-25%	-32%	39%

CC: Conventional coal, AC: Advanced coal, AC with CCS, Advanced coal with Carbon Capture and Storage, CGCC: Conventional gas combined cycle.

Prices from conventional fossil fuel sources, however, do not merely stand for the costs of power generation and distribution. Conventional energy has an impact on health, the environment and the climate that leads to several external effects, and the impact of GHG and the health damage caused by air pollutants are especially relevant for quantifying these effects. By including the cost of the external effects in the bills, it may imply that clean and advanced technologies increase their share in global electricity production. Mainly, natural gas combined cycles, natural gas fuel cells, wind, solar and biomass technologies may gain significant market shares at the expense of traditional coal-fired and gas-fired plants.

For governments pursuing for sustainable growth and investment, a secure and competitively priced supply of electricity is a pre-requisite. Due to numerous hidden subsidies and non-internalized environmental costs, there is a problem of how to sell renewable power if power from coal and gas remains less expensive. A joint study of the German Aerospace Centre (DLR) and Fraunhofer Institute for System and Innovation Research (ISI), estimates that the external cost for hard coal and natural gas are cEUR₂₀₁₂ 7.17 kWh and cEUR₂₀₁₂ 4.40 kWh respectively. Estimations on climate change damage costs are based on CO₂ emissions of EUR 70 tCO₂ (Krewitt, et al., 2006). The external costs of renewables (with the exception of solar photovoltaic and biomass) are significantly below cEUR 1 kWh (Krewitt, et al., 2006), thus the external costs of CSP generation are neglected in this analysis. If external costs per unit of energy produced are charged to consumers, generation of electricity by means of CC could be roughly 40 percent more expensive in 2020 than generation of electricity by means of CSP. For CGCC, prices from CSP are slightly less than generation by gas, which may suggest that even if external costs were accounted, natural gas would remain as a long-term back-up technology for the generation of electricity (see Table 30 and Table 31 in Appendix C).

It can be thought that internalizing or partially internalizing the external cost of fossil fuels in our bills is the most reasonable way towards a decarbonization of the energy sector, but including external costs in prices or through permits is not the only policy tool that exists to incorporate environmental impacts into decision making (Spalding-Fecher, et al., 2002). In industrialized countries, particularly in the US, the use of IRP has been an important instrument in environmental regulation of public utilities. External costs can be added to financial costs when making a decision about resource selection in IRP, although finally not adding the external costs to the price of the electricity (Bakken, et al., 1996). This can encourage socially optimal resource selection without major price increases (Bakken, et al., 1996).

10

Technical assessment

10.1 Super Grid to benefit from CSP potential

As a result of mapping the interconnections between the CSP suitable sites according to different DNI levels in relation with the SSA centers of demand (see Chapter 7), we found the specific distances (geographical distances, considering z axis) for each transmission corridor (see Table 28 in Appendix C). With the distances of the transmission corridors (km) and the irradiance value ($\text{kWh}/\text{m}^2/\text{year}$) for each CSP site, we have calculated the LCOE (cEUR/kWh) for a solar tower plant of 100 MW with 12 hours of storage with 2020 as the year of construction of the plant and the transmission facilities. Table 23 shows the minimum LCOE considering the cost of electricity production and transmission, and the distances at which the least cost option of electricity generation by means of CSP for each center of demand is achieved.

For distances longer than 600 - 800 km, overhead lines using HVDC transmission technology are more cost-effective than AC technology (Siemens, 2013). This figure depends on several factors and an analysis is required for each particular case. However, analysts from the DLR found that the break-even-point for a ± 800 kV 5 GW overhead line (same technology, voltage level and capacity than used in this study), in comparison with a 1150 kV HVAC line, was 830 km (DLR, 2006). Therefore, this figure was considered to be the technical limit in this analysis over HVDC reaches the break-even-point. Results show that except for the case of Johannesburg, all the other SSA centers of demand will have to rely on transmission lines longer than 800 km to benefit from the least-cost electricity from CSP (see Table 23).

Table 23 Minimum LCOE (cEUR/kWh) of a solar tower plant for 2020 in relation with SSA centers of demand and the transmission distance at which it is achieved. The table shows that most of the highest populated SSA centers of demand achieve the CSP least cost option at distances higher than 800 km. Johannesburg may benefit from least-cost CSP with AC lines.

Centers of demand/Distance (km)	< 800	800 – 2,000	2,000 – 4,000	> 4,000
Lagos, Nigeria				8.44
Kinshasa, Democratic Republic of the Congo		8.22		
Khartoum, Sudan			8.25	
Luanda, Angola		8.17		
Abidjan, Côte d'Ivoire				8.52
Johannesburg, South Africa	8.12			
Nairobi, Kenya			8.34	
Dar es Salaam, Tanzania			8.31	

Dakar, Senegal		8.45
Accra, Ghana		8.47
Douala, Cameroon	8.32	
Ouagadougou, Burkina Faso		8.48
Bamako, Mali	8.36	
Harare, Zimbabwe	8.18	
Maputo, Mozambique	8.16	
Lusaka, Zambia	8.18	
Mogadishu, Somalia		8.41
Brazzaville, Republic of the Congo	8.22	
Niamey, Niger		8.27
Monrovia, Liberia		8.41

Least-cost option for CSP production in SSA for the major populated centers of demand is achieved at high distances (> 800km) at which conventional grids have to deal with high inefficiencies and thus high electricity losses. If electricity generators, transmission system operators and planning agencies, operating in a cross-regional trade, want to increase the share of renewable electricity by operating with the least-cost CSP, transmission lines capable to transmit high shares of electricity over long distances with minimum electricity losses such as HVDC lines are required.

10.2 Whether or not Africa needs a Super Grid

Africa's vast renewable energy resources (i.e. hydro, geothermal, wind, solar and bioenergy), can cover most of the continent's demand by 2050 (IRENA, 2013). Despite a comparably better economic case, banks are either more disposed to finance fossil fuel plants instead of renewable plants, or banks agree to finance renewable plants but at premium rates (IRENA, 2013). Moreover, while oil subsidies in Africa cost approximately USD 50 billion per year, renewable projects can make good business sense even without subsidies (IRENA, 2013). Several regional and national initiatives have proven that the political will towards renewable projects exists; and not only exists but the will is rising. This is the moment for scaling-up Africa's renewable projects.

In previous section we have identified the need of constructing large scale corridors to benefit from economically attractive CSP technologies to compete with inlaid traditional sources of electricity production. However, the analysis is optimized for one single resource while vast renewable potential still on the ground. Tremendous renewable energy potential can be harnessed and be integrated into current power systems. This integration accrue significant benefits for participatory pools by sharing reserve capacity, by diversifying markets and generation mix; while providing access to lower cost generation facilities and potentially decrease power tariffs (IRENA, 2013).

A preliminary assessment for harnessing renewable potential is developed in this section. This preliminary assessment is not intended to answer specific questions but to raise the awareness about the vast renewable sources and its relation with centers of unceasingly increasing energy demand. Figure 21 shows the renewable energy potential in SSA in relation with current transmission grid and potential CSP corridors. The renewable energy potential is the result of overlapping CSP and wind potential and current and projected hydro plants. Information regarding wind potential (MWh/year) was provided by (Mentis, 2013). The wind potential was performed with a Vestas V90 (swept area: 6,362 m², rotor diameter: 90 m, power: 2 MW) wind turbine. The selection of this machine was due to its good performance at low and middle wind speeds. Moreover, wind turbines larger than 2 MW would face barriers in their transportation through the geographical variability of the continent.

Additionally, larger wind turbines would cause major visual disturbances and its installation may face an opposition from local inhabitants (Mentis, 2013).

Member States from the EAPP and SAPP are revising their transmission plans to accommodate the expected growing additions of renewable generation capacity (IRENA, 2013). This renewable generation capacity however, is based on geothermal, hydropower and wind. E.g., Ethiopia, Kenya, and Tanzania have identified jointly around 15 GW of cost-effective geothermal potential, 40 GW of cost-effective hydro potential and 8 GW of economically competitive wind potential (IRENA, 2013). As a result of previous analysis, we have identified that South African and North-Eastern African regions are well endowed with economically attractive CSP potential that can be competitive with traditional sources of power generation as coal. This potential can be exploited together with other renewable energy sources through energy corridors to places where high amounts of energy are needed. Moreover, current paths of transmission grids can serve as ideal location for the deployment of the potential energy corridors as interconnections with lower voltage level transmission grids are already on the ground.

In countries in the Eastern region as Sudan, Ethiopia and Kenya; there is a strong correlation between wind potential and current transmission grid that connects the hydropower plants. This correlation is complemented with the high solar resources in Southern Egypt and North Sudan that can transmit high quality solar resources directed to the South. In countries in the Southern region as South Africa, Botswana and Namibia; there are notorious wind and solar resources ready to deliver high quality energy to the grid. Additionally, there are a number of potential areas where current transmission grid is sensible for CSP energy transport either in the form of connecting directly to the grid or building high capacity transmission corridors connecting with other renewable sources.

Countries in the Western region as Nigeria, Benin, Togo, Ghana, Côte d'Ivoire, Cameroon or Senegal; have high wind resources in the Northern regions that together with the solar resources in areas as Chad, make it possible to transmit high amounts of renewable energy to the coastal regions where the most significant urban nodes in terms of population and energy needs are located. Additionally, when wind and solar potential is transmitted to the South, can be coupled with current hydropower potential. Countries in the Central region, with immense hydropower potential waiting to be exploited and acceptable levels of wind potential in the coastal areas, can benefit from the solar resources in the Northern part of the Kalahari Desert and directly transmit to central areas where less wind and solar resources are found.

Integration of renewable energy in energy planning is already in a developing stage. This is exemplified by the Ethiopia-Kenya power line project and other regional interconnectors (IRENA, 2013). The Ethiopia-Kenya project will deploy a high-voltage transmission line of 2 GW of capacity along 1,068 km with a total cost of USD 1.26 billion. The project is already partially financed by the African Development Bank and when completed, is expected to contribute to the regional integration and increase commerce (IRENA, 2013). Clean energy corridors can supply higher shares of renewable energy into Africa's rapidly expanding power grids. Therefore, strong transmission grids as HVDC lines to develop renewable potential are required. The stronger the grid, the greater is the reliability of the transmission grid and lowers the power losses when moving clean, low-cost renewable electricity to areas dominated by high-cost fossil fuels.

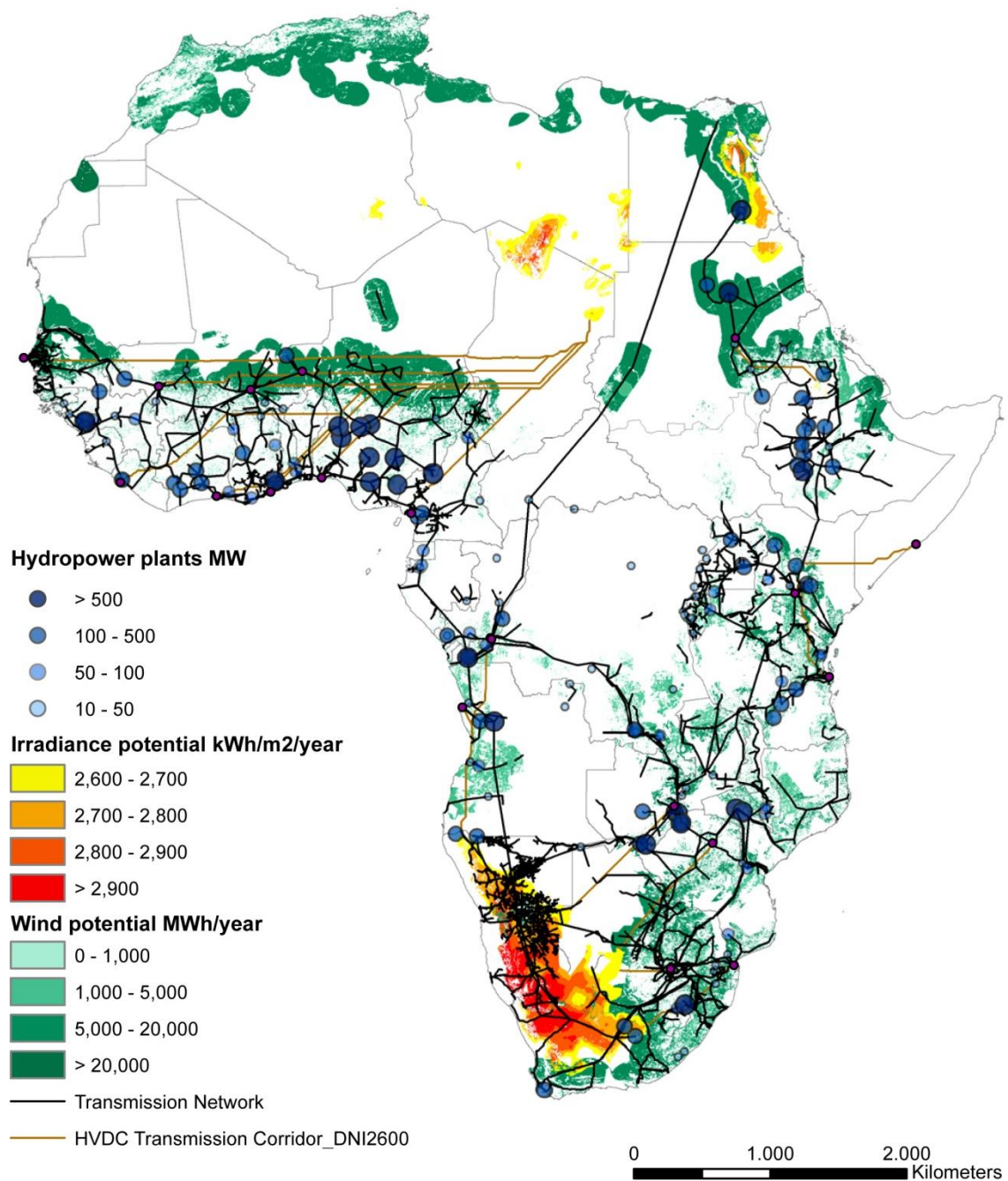


Figure 21 Overlapping map of the CSP and wind potential and installed and projected hydropower plants. Renewable energy potential is overlapped with the potential transmission corridors for $\text{DNI} \geq 2,600 \text{ kWh/m}^2/\text{year}$ and the existing transmission grid. This results in sensible areas for the deployment of transmission corridors to harness the renewable energy potential. Transmission corridors for $\text{DNI} \geq 2,600 \text{ kWh/m}^2/\text{year}$ were selected in preference of corridors associated with $\text{DNI} \geq 2,800 \text{ kWh/m}^2/\text{year}$ as they follow similar patterns with transmission corridors at lower DNI levels.

11

Conclusion

Africa is experiencing GDP increases higher than other continents at present. Investments within the region are rising sharply; however, unless consideration is paid to the potential provided by new renewable and transmission technologies, Africa might miss an opportunity to leapfrog the West to large scale low carbon energy systems. With the exception of hydro, the share of renewable energy for electricity generation in SSA is very low. Therefore, if cost effective, an important challenge in the region's energy sector is to enlarge the share of renewable energy capacity and to shift towards an environmentally sustainable nature of electricity generation.

This research suggests important implications when designing future African energy systems. The results show that SSA urban centers of demand can benefit from distant CSP that is economically attractive enough to compete with traditional sources of electricity generation as coal-based plants. At a discount rate of 10 percent, solar tower plants can achieve cost parity with conventional coal plants in 2020 when considering stable carbon prices. Further, as fuel prices for fossil plants can be volatile, and nuclear holds various social acceptance concerns, large-scale CSP generation may be a useful transition to more sustainable energy networks despite the complexities involved. It should be mentioned that the analysis was based on CSP plants endowed with dry cooling systems. However, when the selected location has no water availability constraints, CSP can rely on wet or hybrid cooling systems. These cooling systems are even less expensive than dry systems, resulting in lower LCOE and thus more attractive for integration in the power system.

Over the last decade, most of the growth in renewable energy has basically meant an increase in hydro-based electricity generation. However, under certain circumstances, hydroelectric development can result in environmental damages and social conflicts, principally in the case of large-scale dam-based generation. Additionally, it is immediately exposed to the effects of drought. For countries not benefiting from hydropower sources, in the short term, coal, gas and oil plants may continue to be attractive investments due to their low risk; but with that comes potential lock in to fuel price volatility as well as relatively intense GHG emissions. Moreover, they can also be easily configured to follow the demand for electricity (ramping production up and down when needed). Yet, if technology costs continue to decline as current rates, in the long term, CSP plants endowed with thermal storage may become close competitors for smoothing purposes in an energy system with increasing share of variable resources.

Governments, power pools and utilities seek the best approaches to enhance future power systems. Our results equally highlight the potential of SSA to move from elements of traditional power systems. First, Super Grids may provide an efficient mechanism to address the massive electricity infrastructure requirements, and benefit from the uneven distribution of renewable resources. Except for Johannesburg, major SSA centers of demand -in our analysis- might rely on a Super Grid approach to benefit from least-cost CSP potential (Johannesburg is the only demand node under study where least-cost CSP can be achieved nationally). Further, Africa is well endowed with solar and wind potentials,

and hydroelectricity is already being exploited (though limited by current grid infrastructure). This analysis shows the close correlation between the identified solar corridors and the places where other renewable resources and demand are located. For this reason, our results highlight the importance of integrating geographically-wide renewable resources into clean energy corridors supplying high quality energy as a way to enhance regional trade.

Currently, Africa is slowly facing a transition to a more integrated and interconnected energy system; however, future network regulations and design need to be carefully assessed for a proper integration into national and regional energy planning, to promote the implementation of the best solutions. The combination of measures that will lead to the most effective and cost-efficient results, is subject to i) the local socio-economic circumstances, ii) the endowment of renewable sources, iii) the best suited technologies, iv) and the national targets for renewable energy expansion. One particular obstacle for energy corridors is the high investment cost of transitional transmission assets, and how these costs must be allocated among the concerned parties. Moreover, critical opinions arise with huge infrastructure plans. Although new technology transmission expansion can be associated with a symbol of modernity and development, some analysts argue that this may not necessarily be translated into improved standards of living on the ground. Moreover, in the era of digital microelectronics, this concept may be even obsolete. Therefore, other ways of energy planning such as the Smart Grid approach need to be carefully assessed from a regional and local perspective. In this situation, other concentrating solar technologies such as dish-stirling, suitable for off-grid applications and for slopes higher than for parabolic trough and solar tower technologies, need to be considered and properly assessed with different solar power on-grid applications.

Finally, this study has identified significant gaps in research that need to be filled if large-scale renewable electricity projects are to succeed. Amongst others, the following areas of research should be developed further in future in-depth assessments: complementing previous research may entail detailed resource maps for different renewables generation technologies, both in terms of technical and economic potential. This may lead to a revision of the power pools' master plans to finally account for significant shares of renewable energy. Moreover, there is an immediate need for a detailed assessment of the capacity requirements of African countries and the cost of renewable generation deployment; and therefore, to design a planning system according to specific regional needs. Additionally, energy planning could start by answering questions such as: "How renewable energy integration should be developed taking advantage of advanced grid technologies?" "Does a Super Grid approach (and therefore massive expansion of power lines) make sense in the era of digital microelectronics?" "What political risks associated with a Super Grid approach are perceived as the most important barriers for infrastructure deployment among transition countries?" Making progress on these issues could bring benefits not only to countries seeking to increase the level of renewable sources, but also be an important step to achieve national and regional climate protection goals.

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A

Appendix

Table 24 Population of urban agglomerations in 1990-2025.

Demand center	Country	1990	1995	2000	2005	2010*	2015*	2020*	2025*
Lagos	Nigeria	4.764	5.966	7.233	8.767	10.578	12.427	14.162	15.810
Kinshasa - capital	Democratic Republic of the Congo	3.564	4.590	5.611	7.106	8.754	10.668	12.788	15.041
Al-Khartum (Khartoum) - capital	Sudan	2.360	3.242	3.949	4.518	5.172	6.046	7.005	7.953
Luanda	Angola	1.568	1.953	2.591	3.533	4.772	6.013	7.080	8.077
Abidjan - seat of government	Côte d'Ivoire	2.102	2.535	3.032	3.564	4.125	4.788	5.550	6.321
Johannesburg	South Africa	1.898	2.265	2.732	3.263	3.670	3.867	3.996	4.127
Nairobi - capital	Kenya	1.380	1.755	2.230	2.814	3.523	4.303	5.192	6.246
Dar es Salaam - capital	United Republic of Tanzania	1.316	1.668	2.116	2.680	3.349	4.153	5.103	6.202
Dakar - capital	Senegal	1.405	1.688	2.029	2.434	2.863	3.308	3.796	4.338
Accra - capital	Ghana	1.197	1.415	1.674	1.985	2.342	2.722	3.110	3.497
Douala	Cameroon	931	1.155	1.432	1.767	2.125	2.478	2.815	3.131
Ouagadougou - capital	Burkina Faso	537	667	921	1.328	1.908	2.643	3.457	4.332
Bamako - capital	Mali	746	910	1.110	1.368	1.699	2.086	2.514	2.971
Harare - capital	Zimbabwe	1.047	1.255	1.379	1.513	1.632	1.856	2.170	2.467
Maputo - capital	Mozambique	776	921	1.096	1.341	1.655	1.994	2.350	2.722
Lusaka - capital	Zambia	757	902	1.073	1.265	1.451	1.666	1.941	2.267
Muqdisho (Mogadishu)	Somalia	1.035	1.147	1.201	1.415	1.500	1.795	2.156	2.588
Brazzaville - capital	Congo	704	830	986	1.172	1.323	1.504	1.703	1.878
Niamey - capital	Niger	432	542	680	848	1.048	1.302	1.643	2.105
Monrovia - capital	Liberia	1.042	464	836	1.202	827	728	807	932
Average		1.478	1.794	2.196	2.694	3.216	3.817	4.467	5.150

Source: (UN-HABITAT, 2010). Reference data 2009. * Projections

Table 25 Parabolic trough power plants in operation.

Name	Country, location	Owners	Capacity (MW)	Starting year	Heliostat field area (m2)	Power cycle	Storage (h)
Aïn Beni Mathar ISCC	Morocco, Ain Beni Mathar	Office Nationale de l'Electricite Acciona	20	May 2010	183,120	Rankine	none
Alvarado 1 (La Risca)	Spain	Energia, Mitsubishi Corp	50	June 2009	352,854	Rankine	none
Andasol 1	Spain, Aldeire	Antin, Cobra, RREEF	50	November 2008	510,120	Rankine	7.50
Andasol 2	Spain, Aldeire	Antin, Cobra, RREEF	50	June 2009	510,120	Rankine	7.50
Andasol 3	Spain, Aldeire	Ferrostaal, RWE, Rhein E, SWM, Solar Millenium	50	August 2011	510,120	Rankine	7.50
Archimede	Italy, Priolo Gargallo	ENEL	5	July 2010	31,860	CC Rankine	8
Arcosol 50 (Valle 1)	Spain	Torresol Energy	50	December 2011	510,120	Rankine	7.50
Aste 1A	Spain, Alcázar de San Juan	Aries, Eiser, Elecnor	50	January 2012	510,120	Rankine	8
Aste 1B	Spain, Alcázar de San Juan	Aries, Eiser, Elecnor	50	January 2012	510,120	Rankine	8
Astexol II	Spain, Olivenza	Aries, Eiser, Elecnor	50	2012	510,120	Rankine	8
Colorado Integrated Solar Project	US, Palisade	Xcel Energy	2	2010	6,664	-	-
Extresol 1	Spain, Torre de Miguel Sesmero	Cobra	50	January 2010	510,120	Rankine	7.50
Extresol 2	Spain, Torre de Miguel Sesmero	Cobra, GE, KGAL	50	2010	510,120	Rankine	7.50
Extresol 3	Spain, Torre de Miguel Sesmero	Cobra	50	August 2012	510,120	Rankine	7.50
Godawari Solar Project	Naukh, India	Godawari Green Energy Limited	50	June 2013	392,400	-	none
Guzman (Termosolar Soluz Guzman)	Spain, Palma del Río	FCC, Mitsui	50	July 2012	310,406	Rankine	none
Hassi R'mel ISCC	Algeria, Hassi R'mel	Sonatrach	25	July 2011	183,860	Rankine	none
Helioenergy 1	Spain, Écija	Abengoa Solar, EON	50	September 2011	300,000	Rankine	none
Helioenergy 2	Spain, Écija	Abengoa Solar, EON	50	January 2012	300,000	Rankine	none

Helios 1	Spain, Puerto Lápice	Abengoa Solar	50	June 2012	300,000	Rankine	none
Helios 2	Spain, Puerto Lápice	Abengoa Solar	50	August 2012	300,000	Rankine	none
Holaniku	US	Sopogy	2	December 2009	-	Rankine	2
Ibersol Puertollano	Spain, Puertollano	Iberdrola	50	January 2009	287,760	Rankine	none
Kuraymat ISCC	Egypt, Kuraymat	NREA	20	June 2011	130,800	Rankine	none
La Africana	Spain, Posadas	Grupo Ortiz, Magtel, TSK	50	November 2012	550,000	Rankine	7.50
La Dehesa	Spain, La Garrovilla	Renovables SAMCA	50	February 2011	552,750	Rankine	7.50
La Florida	Spain, Badajoz	Renovables SAMCA	50	June 2010	552,750	Rankine	7.50
Lebrija 1	Spain, Lebrija	Siemens, Valoriza	50	December 2011	412,020	Rankine	none
Majadas	Spain, Majadas de Tiétar	Acciona Energia, Mitsubih Corp	50	October 2010	372,240	Rankine	none
Manchasol 1	Spain, Alcazar de San Juan	Cobra	50	January 2011	510,120	Rankine	7.50
Manchasol 2	Spain, Alcazar de San Juan	Cobra	50	April 2011	510,120	Rankine	7.50
Martin Next Generation Solar Energy Center	US, Indiantown	Florida Power & Light	75	December 2010	464,908	Rankine	none
Moron	Spain, Morón de la Frontera	Ibereolica	50	May 2012	380,000	Rankine	none
Nevada Solar One	US, Boulder City	Acciona Energia	75	June 2007	357,200	Rankine	0,5
Olivenza 1	Spain, Olivenza	Ibereolica	50	September 2012	402,210	Rankine	none
Orellana	Spain, Orellana	Acciona Energia	50	August 2012	405,500	Rankine	none
Palma del Rio I	Spain, Palma del Río	Acciona Energia, Mitsubih Corp	50	July 2011	372,240	Rankine	none
Palma del Rio II	Spain, Palma del Río	Acciona Energia	50	December 2010	372,240	Rankine	none
Saguaro	US, Red Rock	Arizona Public Service	1	January 2006	100,000	Rankine	none
SEGS I	US, Daggett	Cogentrix	14	December 1984	82,960	Rankine	-
SEGS II	US, Daggett	Cogentrix	30	1985	190,338	Rankine	-
SEGS III	US, Kramer Junction	NextEra	30	December 1985	230,300	Rankine	-

SEGS IV	US, Kramer Junction	NextEra	30	February 1989	230,300	Rankine	-
SEGS V	US, Kramer Junction	NextEra	30	February 1989	250,500	Rankine	-
SEGS VI	US, Kramer Junction	NextEra	30	February 1989	188,000	Rankine	-
SEGS VII	US, Kramer Junction	NextEra	30	February 1989	194,280	Rankine	-
SEGS VIII	US, Harper Dry Lake	NextEra	80	December 1989	464,340	Rankine	-
SEGS IX	US, Harper Dry Lake	NextEra	80	October 1990	483,960	Rankine	-
Shams 1	UAE, Madinat Zayed	Abengoa Solar, Masdar, Total	100	March 2013	627,840	Rankine	none
Solaben 3	Spain, Logrosán	Abengoa Solar, Itochu	50	June 2012	300,000	Rankine	none
Solacor 1	Spain, El Carpio	Abengoa Solar, JGC	50	February 2012	300,000	Rankine	none
Solacor 2	Spain, El Carpio	Abengoa Solar, JGC	50	March 2012	300,000	Rankine	none
Solnova 1	Spain, Sevilla	Abengoa Solar	50	January 2009	300,000	Rankine	none
Solnova 3	Spain, Sevilla	Abengoa Solar	50	January 2009	300,000	Rankine	none
Solnova 4	Spain, Sevilla	Abengoa Solar	50	January 2009	300,000	Rankine	none
Termesol 50 (Valle 2)	Spain, San José del Valle	Torresol Energy	50	December 2011	510,120	Rankine	7.50
Termosol 1	Spain, Navalvillar de Pela	NextEra Energy	50	March 2013	523,200	Rankine	9
Termosol 2	Spain, Navalvillar de Pela	NextEra Energy	50	March 2013	523,200	Rankine	9
Termosolar Borges	Spain	Abantia, Comsa Emte	22,5	December 2012	183,120	Rankine	none
TSE1 - PT	Thailand	Thai Solar Energy	5	January 2012	45,000	Rankine	none

Source: www.nrel.gov

Table 26 Solar tower power plants in operation.

Name	Country, location	Owners	Capacity (MW)	Starting year	Heliostat field area (m ²)	Power cycle	Storage
Acme solar thermal tower	India, Bikaner	ACME	2.5	April 2011	14,280	Rankine	none
Beijing Badaling	China Beijing	Academy of sciences	1.5	August 2012	10,000	Rankine	1 h
Gemasolar	Spain, Andalucía	Torresol energy	19.9	April	304,750	Rankine	15 h

Jülich	Germany, Jülich	DLR	1.5	2011 December 2008	17,650	Rankine	1.5 h
Lake Cargelligo	Australia, Lake Cargelligo	Graphite Energy	3	May 2011	6,080	Rankine	-
Planta solar 10	Spain, Sanlucar la mayor	Abengoa solar	11	June 25, 2007	75,000	Rankine	1 h
Planta solar 20	Spain, Sanlucar la mayor	Abengoa solar	20	April 22, 2009	150,000	Rankine	1 h
Sierra	United States, California	eSolar	5	July 2009	27,670	Rankine	no data

Source: (Behar, et al., 2013), www.nrel.gov

Table 27 Definition of protected areas according to the International Union for Nature Conservation (IUCN), (Dudley, 2008).

Category Ia:
Strict Nature Reserve: are strictly protected areas set aside to protect biodiversity and also possibly geological/geomorphological features, where human visitation, use and impacts are strictly controlled and limited to ensure protection of the conservation values. Such protected areas can serve as indispensable reference areas for scientific research and monitoring.
Category Ib:
Wilderness Area: protected areas are usually large unmodified or slightly modified areas, retaining their natural character and influence, without permanent or significant human habitation, which are protected and managed so as to preserve their natural condition.
Category II:
National Park: protected areas are large natural or near natural areas set aside to protect large-scale ecological processes, along with the complement of species and ecosystems characteristic of the area, which also provide a foundation for environmentally and culturally compatible spiritual, scientific, educational, recreational and visitor opportunities.
Category III:
Natural Monument: protected areas are set aside to protect a specific natural monument, which can be a landform, sea mount and submarine cavern, geological feature such as a cave or even a living feature such as an ancient grove. They are generally quite small protected areas and often have high visitor value.
Category IV:
Habitat/Species Management Area: protected areas aim to protect particular species or habitats and management reflects this priority. Many category IV protected areas will need regular, active interventions to address the requirements of particular species or to maintain habitats, but this is not a requirement of the category.
Category V:
Protected Landscape/Seascape: A protected area where the interaction of people and nature over time has produced an area of distinct character with significant ecological, biological, cultural and scenic value; and where safeguarding the integrity of this interaction is vital to protecting and sustaining the area and its associated nature conservation and other values.
Category VI:
Managed Resource Protected Area: protected areas conserve ecosystems and habitats, together with associated cultural values and traditional natural resource management systems. They are generally large, with most of the area in a natural condition, where a proportion is under sustainable natural resource management and where low-level non-industrial use of natural resources compatible with nature conservation is seen as one of the main aims of the area.

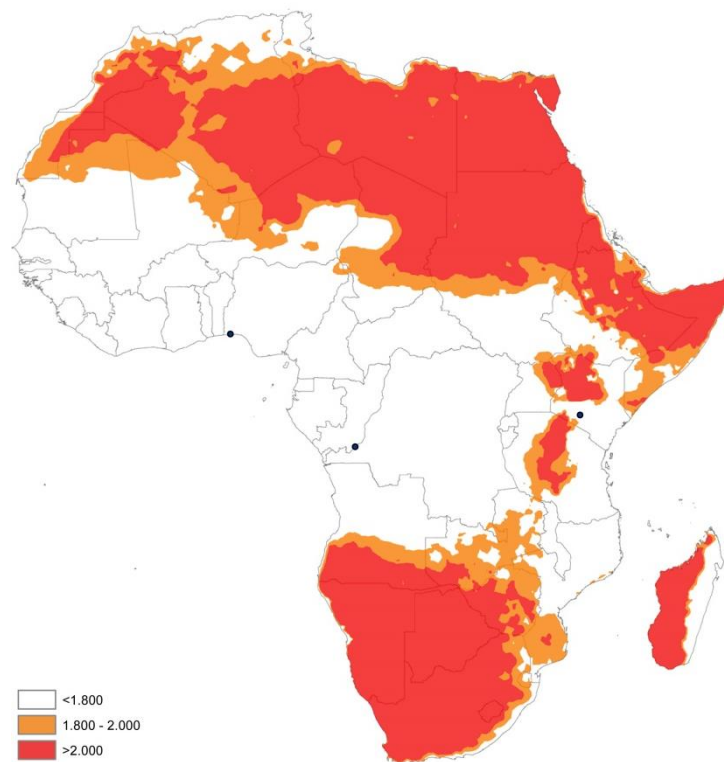


Figure 22 Areas in orange have an irradiance level $\geq 1,800$ KWh/m²/year defined as the minimum technical operational limit of a CSP plant. Areas in red have an irradiance level $\geq 2,000$ KWh/m²/year, defined as the accepted threshold for commercial CSP plants.

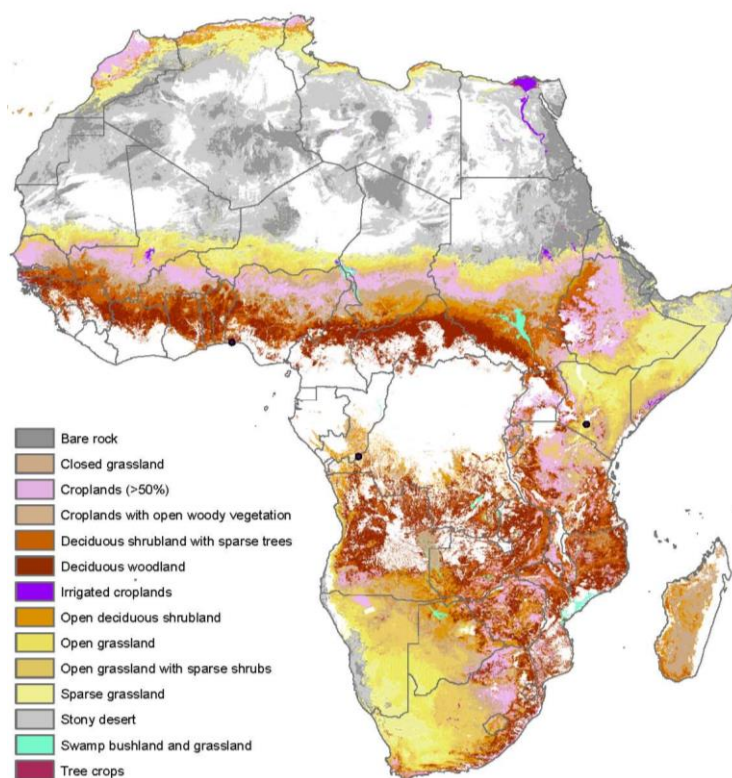


Figure 23 Suitable land cover for solar field location in Africa. Source: Global Land Cover 2000 (European Commission, 2003).

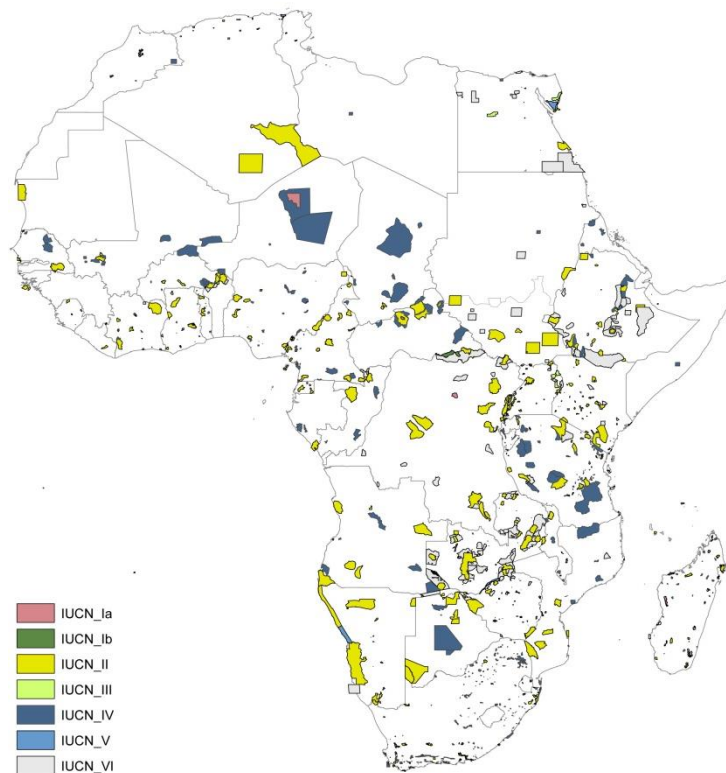


Figure 24 Protected areas of Africa. Source: World Database on Protected Areas (WDPA).

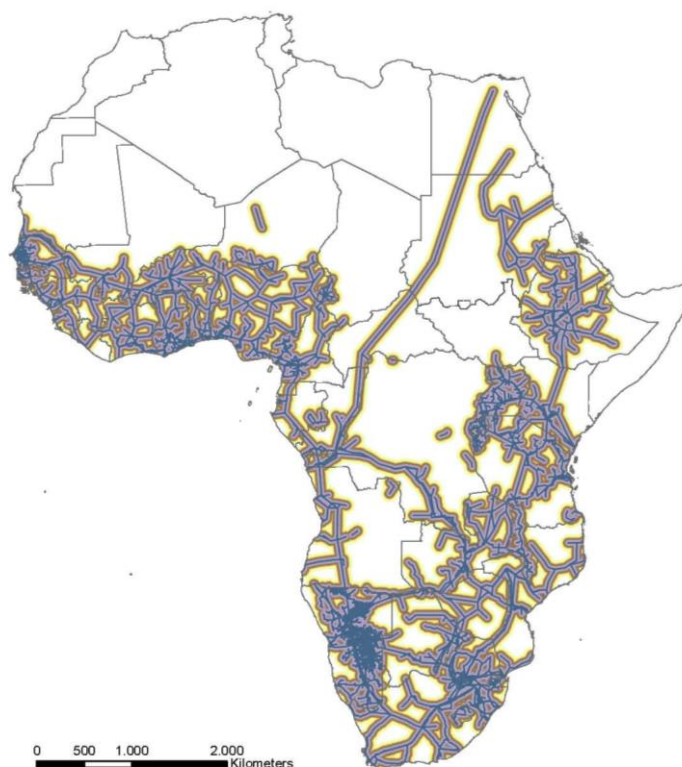


Figure 25 Transmission Network. Transmission lines already installed and under construction (2009). Source: African Development Bank Group.

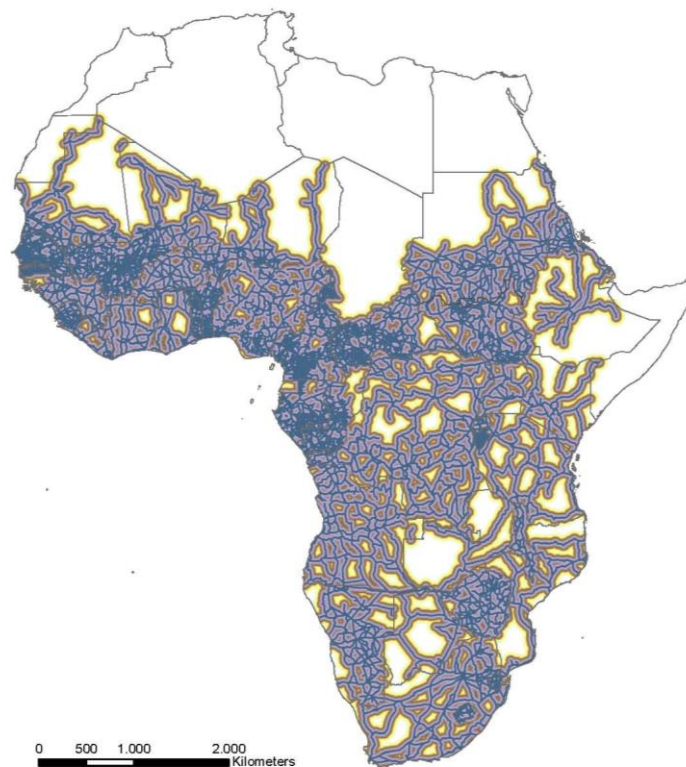


Figure 26 Road network. Source: African Development Bank Group.

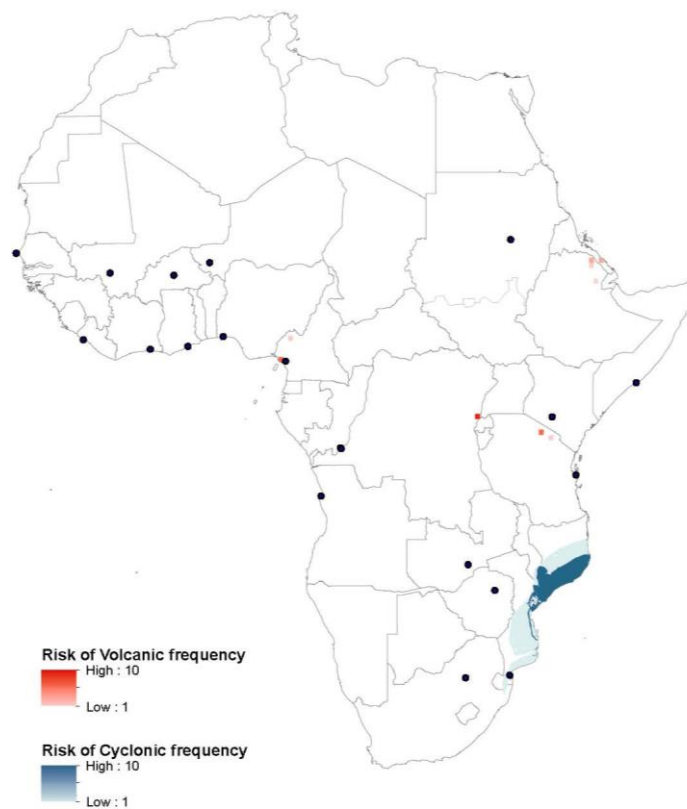


Figure 27 Cyclones and Volcanoes. Source: UNEP/GRID-Geneva PreView and National Geophysical Data Center (NGDC) Volcano Database.

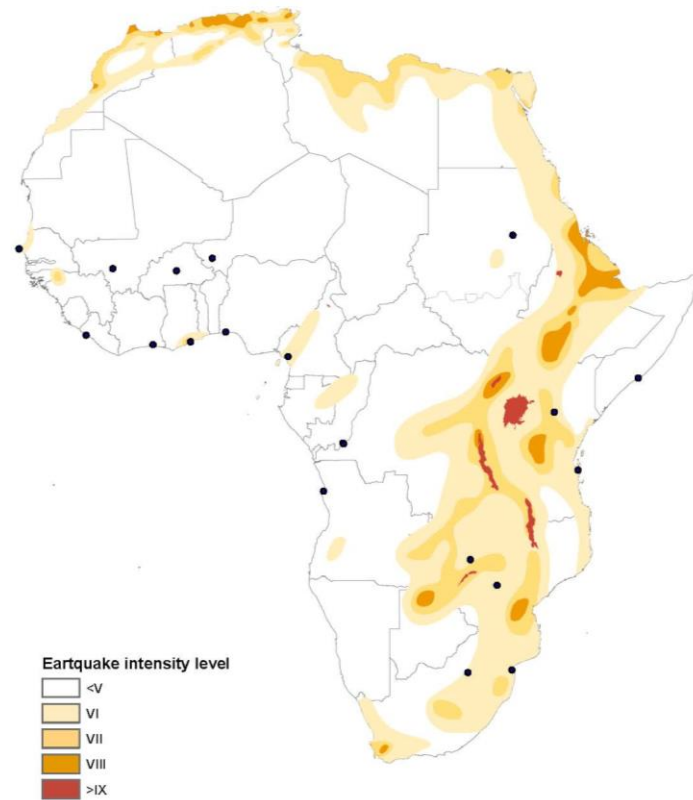


Figure 28 Earthquake. Source: United Nations Environmental Programme/Global Resource Information Database (UNEP/GRID).

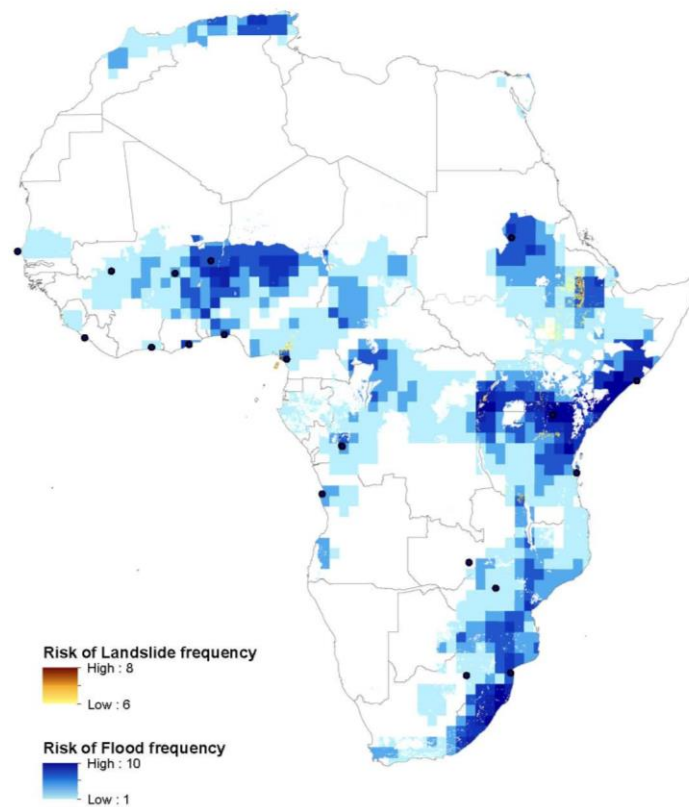


Figure 29 Flood and Landslide. Source: Dartmouth Flood Observatory and Norwegian Geotechnical Institute (NGI).

B

Appendix

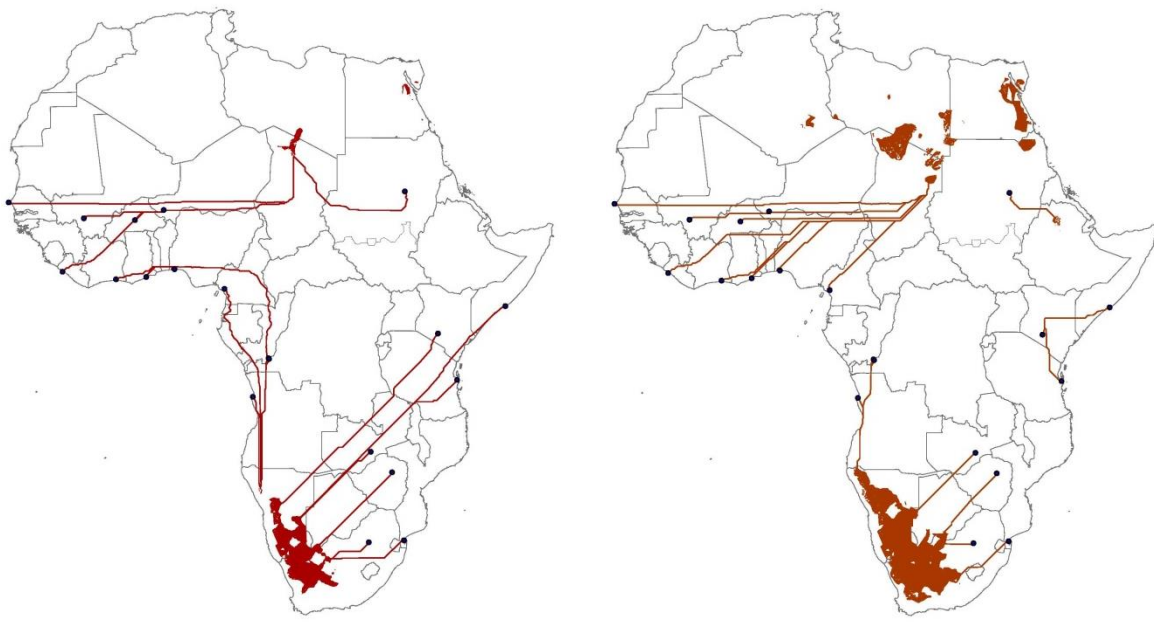


Figure 30 Suitable land for CSP plant location for $\text{DNI} \geq 2,800 \text{ KWh/m}^2/\text{year}$ and transmission corridors to demand centers

Figure 31 Suitable land for CSP plant location for $\text{DNI} \geq 2,600 \text{ KWh/m}^2/\text{year}$ and transmission corridors to demand centers

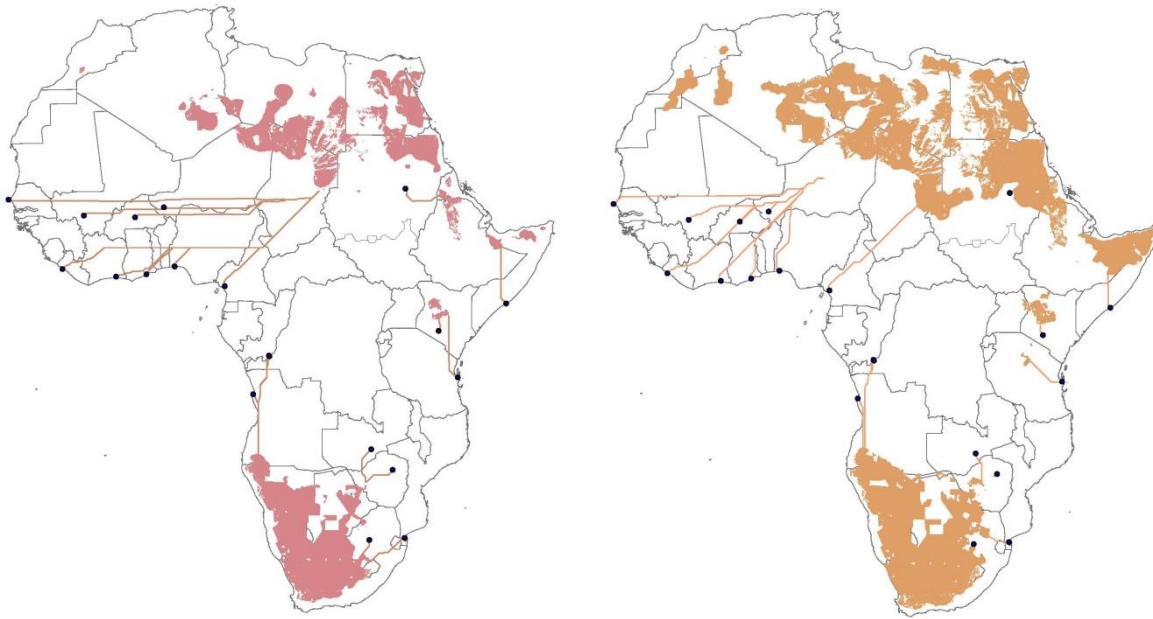


Figure 32 Suitable land for CSP plant location for $\text{DNI} \geq 2,400 \text{ KWh/m}^2/\text{year}$ and transmission corridors to demand centers.

Figure 33 Suitable land for CSP plant location for $\text{DNI} \geq 2,200 \text{ KWh/m}^2/\text{year}$ and transmission corridors to demand centers.

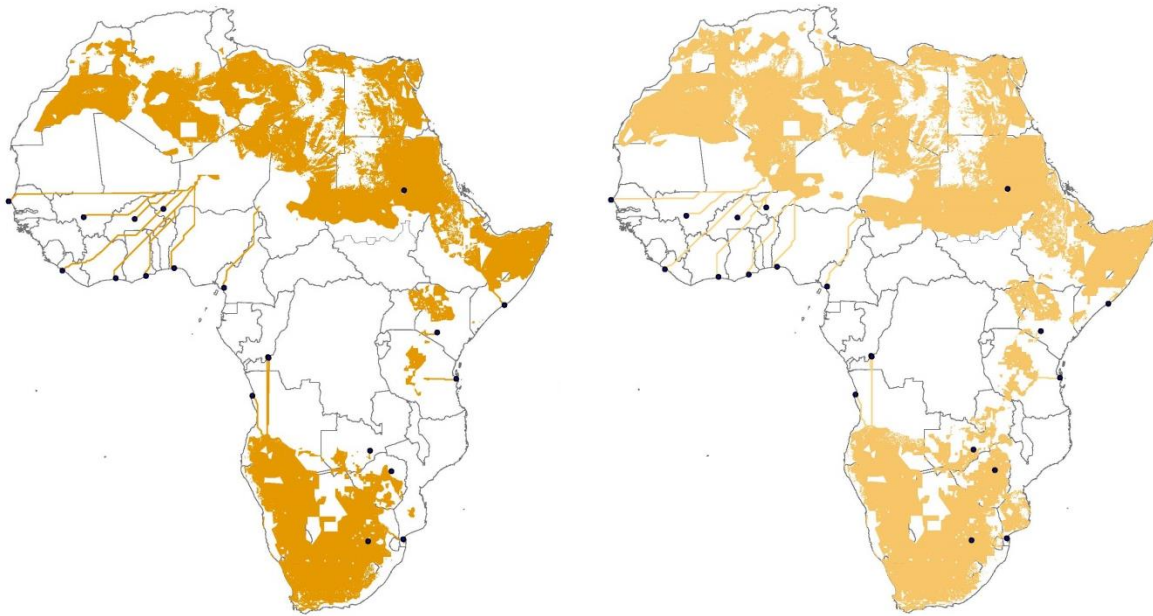


Figure 34 Suitable land for CSP plant location for $\text{DNI} \geq 2,000 \text{ KWh/m}^2/\text{year}$ and transmission corridors to demand centers.

Figure 35 Suitable land for CSP plant location for $\text{DNI} \geq 1,800 \text{ KWh/m}^2/\text{year}$ and transmission corridors to demand centers.

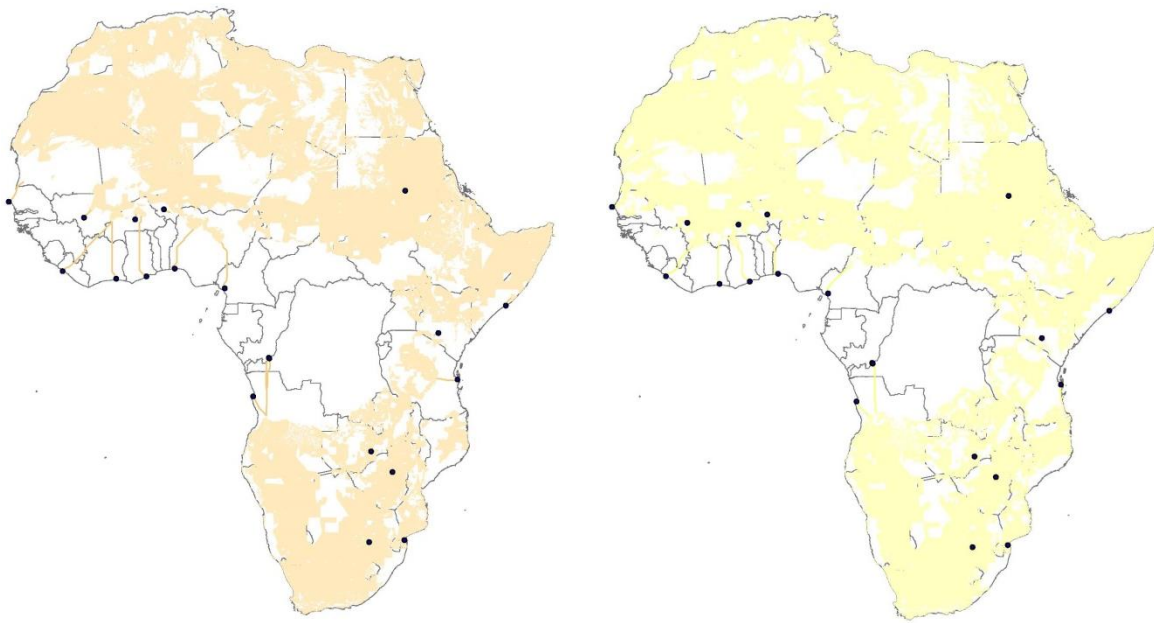


Figure 36 Suitable land for CSP plant location for $\text{DNI} \geq 1,600 \text{ KWh/m}^2/\text{year}$ and transmission corridors to demand centers.

Figure 37 Suitable land for CSP plant location for $\text{DNI} \geq 1,400 \text{ KWh/m}^2/\text{year}$ and transmission corridors to demand centers.

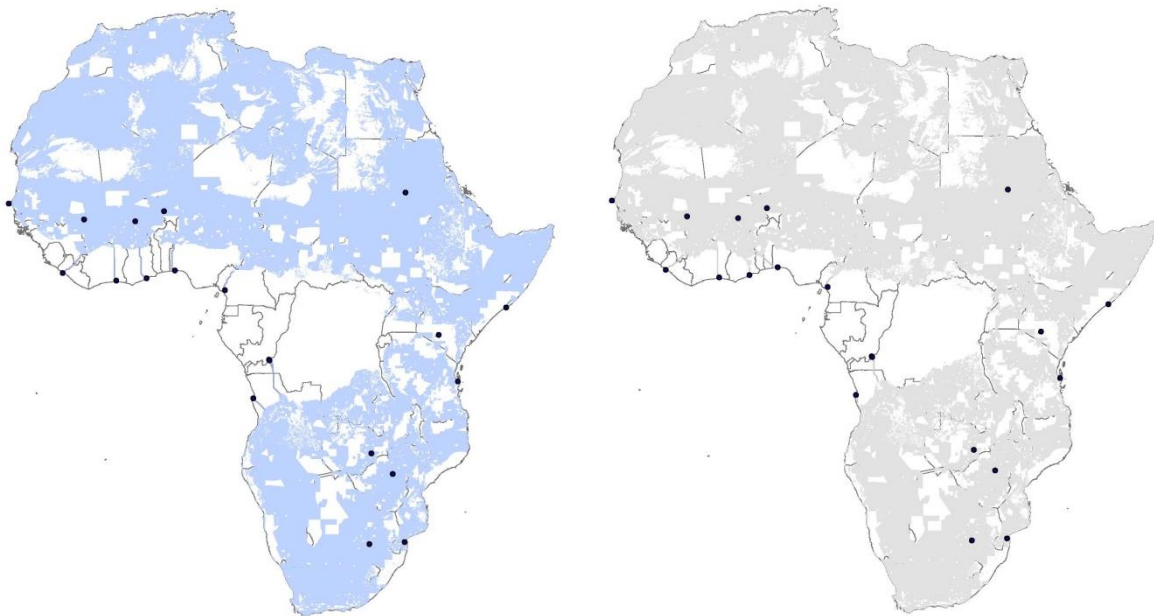


Figure 38 Suitable land for CSP plant location for $\text{DNI} \geq 1,200 \text{ KWh/m}^2/\text{year}$ and transmission corridors to demand centers.

Figure 39 Suitable land for CSP plant location for $\text{DNI} \geq 1,000 \text{ KWh/m}^2/\text{year}$ and transmission corridors to demand centers.

C

Appendix

Algorithm for Parabolic Trough and Solar Tower plant

```
import arcpy, os, sys
from arcpy import env
from arcpy.sa import *
env.workspace = "D:/path"
arcpy.CheckOutExtension("Spatial")
env.overwriteOutput = True

#Parabolic Trough 2050_change parameters depending on technology and year of implementation#
capacity = 100000
actualOutput = .94 * capacity
storageHours = 12
loadFactor = (.57 * storageHours + 7.1)/24
years = 30
eff = 0.17
power_block = 1007
mirror = 190
tower_receiver = 200 #in case of solar tower calculations#
storage = 25
om = 0.048
crf = 0.738327189 #csp plant from 2020 to 2050#
i = .1
ir = .01

productionPlant = loadFactor * actualOutput * 8760
DNI = "DNI.img" #input layer#
```

```
arcpy.env.cellSize = "MAXOF"
arcpy.env.extent = "MAXOF"

#Investment plant = Power block + storage + mirror Field + Power Tower#
size1 = (capacity * 8760 * loadFactor)
size2 = Times(DNI, eff)
size = Divide(size1, size2)

powerBlock = capacity * power_block * crf
storage = storageHours * capacity * storage * crf

mirror2050 = mirror * crf
mirrorField = Times(size, mirror2050)

powerTower = (capacity * tower_receiver) * crf #in case of solar tower calculations#

investment1 = powerBlock + storage
investment2 = Plus(mirrorField, powerTower) #in case of solar tower calculations#
investmentPlant = Plus(investment1, mirrorField)

omPlant = Times(investmentPlant, om)

FormulaPart1 = Times(investmentPlant,((i * ((1 + i)**years))/(((1 + i)**years))+ ir))
FormulaPart2 = FormulaPart1 + omPlant
FormulaPart3 = Divide(FormulaPart2, productionPlant)

lcoePlant = FormulaPart3 * 100 * 0.763
print "Total lcoePlant: ", lcoePlant

arcpy.CopyRaster_management(lcoePlant, os.path.join(env.workspace, "lcoePlant.img"))
lst_temp = [size2, size, mirrorField, investment2, omPlant, investmentPlant, FormulaPart1,
FormulaPart2, FormulaPart3]
for item in lst_temp:
    if arcpy.Exists(item):
        arcpy.Delete_management(item)
```


Algorithm for HVDC transmission lines

```
import arcpy, os, sys
from arcpy import env
from arcpy.sa import *
env.workspace = "D:/path"
arcpy.CheckOutExtension("Spatial")
env.overwriteOutput = True

#Transmission 2020 and 2050#
crf_trans = 0.843571
crf_trans2050 = 0.649222
i = .1
ir = .01
capacityTrans = 5000000
yearsTrans = 40
converter = 67.90957
converter_losses = 0.008
converter_om = 0.01
line_losses = 0.000026
HVDCcost = 0.06
line_om = 0.01

distance = "dist_N1.img" #input layer#
arcpy.env.cellSize = "MAXOF"
arcpy.env.extent = "MAXOF"

variable1_losses = Divide(distance, 1000)
losses = converter_losses * 2 * capacityTrans * 8760
variable2_losses = line_losses * capacityTrans * 8760
lossesAerialWires = Times (variable1_losses, variable2_losses)
total_losses = Plus(losses, lossesAerialWires)

variable1_Aerial_wires = Divide(distance, 1000)
variable2_Aerial_wires = HVDCcost * capacityTrans
Aerial_wires = Times (variable1_Aerial_wires, variable2_Aerial_wires)
```

```

variable1 = converter * capacityTrans * 2
variable2 = Plus(Aerial_wires, variable1)
investmentTrans = Times(variable2, crf_trans2050)

omTransWires = Times (Aerial_wires, line_om)
omTrans = converter * capacityTrans * 2 * converter_om
omTransTotal = Plus (omTrans, omTransWires)

tc1 = capacityTrans * 8760
transmissionCapacity = Minus (float(tc1), total_losses)

lcoeTrans1 = (i * (1 + i)**yearsTrans)/(((1 + i)**yearsTrans)+ ir)
lcoeTrans2 = Times(investmentTrans,lcoeTrans1)
lcoeTrans3 = Plus(lcoeTrans2, omTransTotal)
lcoeTrans4 = Divide (lcoeTrans3, transmissionCapacity)

lcoeTran = lcoeTrans4 * 100
print " lcoeTrans calculated:", lcoeTran

arcpy.CopyRaster_management(lcoeTran, os.path.join(env.workspace, "lcoeTran.img"))
lst_temp = [Aerial_wires, variable1_Aerial_wires, variable1_losses, total_losses, variable2,
omTransWires, lossesAerialWires, investmentTrans, omTrans, lcoeTrans2, lcoeTrans3, lcoeTrans4]
for item in lst_temp:
    if arcpy.Exists(item):
        arcpy.Delete_management(item)

LCOE Total = lcoePlant + lcoeTran

```

Equation 6 CSP site suitability model algorithm

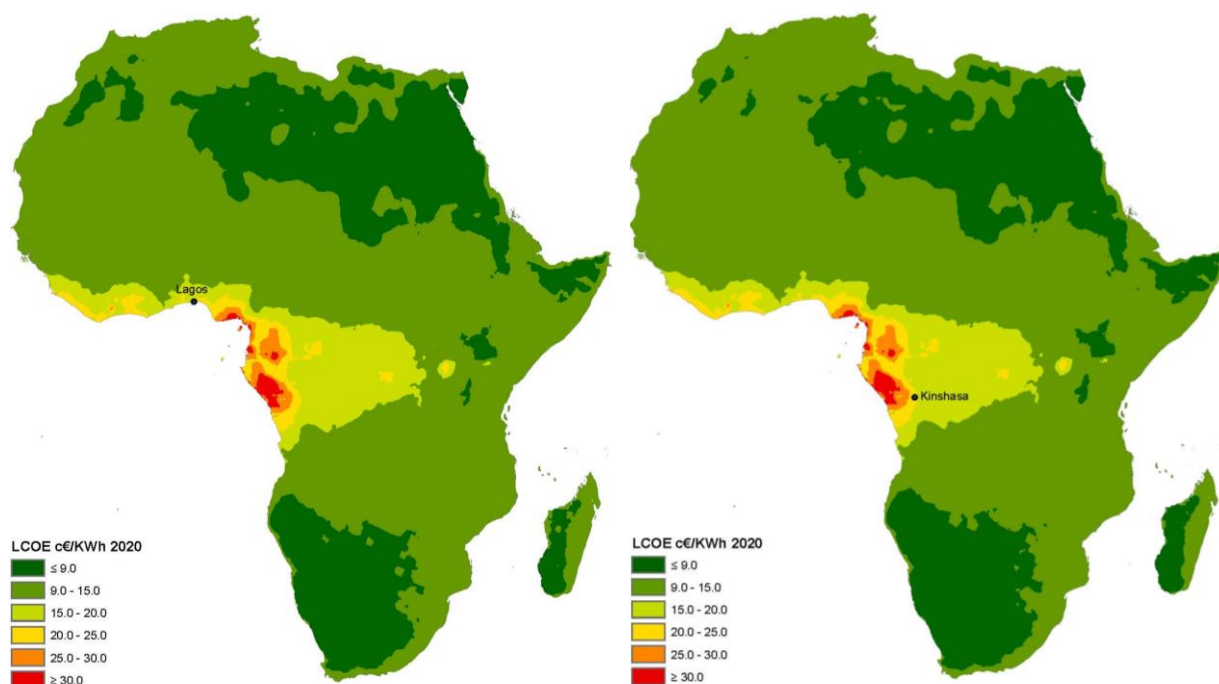


Figure 40 (Left) Spatial distribution of the levelized cost of the electricity (cEUR/kWh/year) in Africa by 2020 of a solar tower plant in relation with Lagos as center of demand (Western Africa Power Pool).

Figure 41 (Right) Spatial distribution of the levelized cost of the electricity (cEUR/kWh/year) in Africa by 2020 of a solar tower plant in relation with Kinshasa as center of demand (Southern Africa Power Pool).

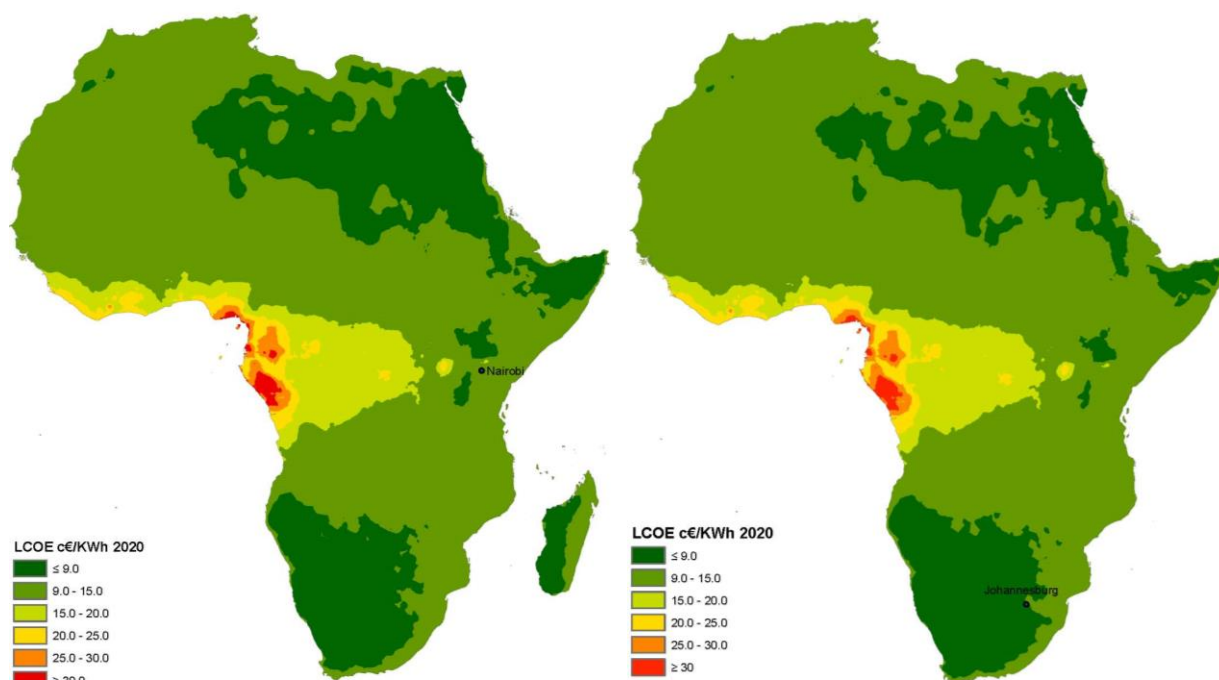


Figure 42 (Left) Spatial distribution of the levelized cost of the electricity (cEUR/kWh/year) in Africa by 2020 of a solar tower plant in relation with Nairobi as center of demand (Eastern Africa Power Pool).

Figure 43 (Right) Spatial distribution of the levelized cost of the electricity (cEUR/kWh/year) in Africa by 2020 of a solar tower plant in relation with Johannesburg as center of demand (Eastern Africa Power Pool).

Table 28 LCOE (cEUR/KWh) of a solar tower plant for 2020 in relation with the transmission distances to connect the SSA centers of demand with the nearest CSP suitable field for different DNI levels.

Demand centers	DNI (kWh/m ² /year)	Overhead line length (km)	LCOE Total cEUR/kWh
Lagos, Nigeria	1000	264,344	14,904
	1200	393,802	13,15
	1400	529,5	11,901
	1600	666,011	10,967
	1800	1034,456	10,262
	2000	1250,687	9,691
	2200	1541,84	9,234
	2400	2391,128	8,906
	2600	2604,205	8,585
	2800	4236,896	8,446
Kinshasa, Democratic Republic of the Congo	1000	348,628	14,91
	1200	510,836	13,159
	1400	652,828	11,911
	1600	780,218	10,976
	1800	917,472	10,252
	2000	1021,344	9,673
	2200	1207,887	9,207
	2400	1350,364	8,818
	2600	1513,626	8,492
	2800	1762,905	8,222
Khartoum, Sudan	1000	8,02	14,884
	1200	8,02	13,12
	1400	8,02	11,86
	1600	8,02	10,916
	1800	8,02	10,181
	2000	8,02	9,593
	2200	52,878	9,115
	2400	623,247	8,759
	2600	795,489	8,433
	2800	2129,162	8,253
Luanda, Angola	1000	121,015	14,893
	1200	216,376	13,136
	1400	250,682	11,879
	1600	317,853	10,94
	1800	465,535	10,216
	2000	573,69	9,637
	2200	675,362	9,164
	2400	806,151	8,773
	2600	971,94	8,448
	2800	1226,34	8,178
Abidjan, Côte d'Ivoire	1000	390,541	14,914
	1200	459,258	13,155

	1400	564,361	11,904
	1600	780,782	10,976
	1800	1479,508	10,298
	2000	1681,04	9,727
	2200	1975,775	9,271
	2400	3171,973	8,975
	2600	3384,756	8,656
	2800	5017,24	8,523
Johannesburg, South Africa	1000	10,5	14,884
	1200	10,5	13,12
	1400	10,5	11,861
	1600	10,5	10,916
	1800	10,5	10,181
	2000	10,5	9,593
	2200	20,747	9,113
	2400	208,905	8,727
	2600	385,298	8,401
	2800	569,511	8,125
Nairobi, Kenya	1000	52,298	14,887
	1200	52,298	13,124
	1400	52,298	11,864
	1600	71,279	10,92
	1800	75,593	10,186
	2000	137,532	9,603
	2200	182,583	9,125
	2400	184,394	8,725
	2600	220,103	8,388
	2800	3108,059	8,34
Dar es Salaam, United Republic of Tanzania	1000	8,015	14,884
	1200	10,889	13,12
	1400	216,057	11,876
	1600	294,189	10,938
	1800	308,218	10,204
	2000	360,047	9,62
	2200	543,101	9,153
	2400	876,477	8,779
	2600	922,001	8,444
	2800	2809,979	8,313
Dakar, Senegal	1000	14,661	14,884
	1200	14,661	13,121
	1400	55,61	11,864
	1600	350,603	10,942
	1800	2115,397	10,352
	2000	2526,086	9,799
	2200	2861,305	9,348
	2400	4173,64	9,069
	2600	4396,609	8,752

	2800	4364,654	8,458
Accra, Ghana	1000	109,127	14,892
	1200	444,768	13,154
	1400	554,8	11,903
	1600	840,997	10,981
	1800	1256,611	10,28
	2000	1470,402	9,709
	2200	1763,251	9,253
	2400	2755,46	8,938
	2600	2968,828	8,618
	2800	4519,308	8,473
Douala, Cameroon	1000	213,81	14,9
	1200	323,549	13,145
	1400	478,553	11,897
	1600	680,368	10,968
	1800	1103,15	10,267
	2000	1300,372	9,695
	2200	1696,115	9,247
	2400	1847,697	8,859
	2600	2060,888	8,538
	2800	2971,022	8,327
Ouagadougou, Burkina Faso	1000	3,5	14,884
	1200	3,5	13,12
	1400	31,31	11,862
	1600	78,571	10,921
	1800	629,202	10,229
	2000	977,581	9,669
	2200	1296,864	9,214
	2400	2391,128	8,906
	2600	2604,205	8,585
	2800	4601,02	8,481
Bamako, Mali	1000	2,915	14,884
	1200	2,915	13,12
	1400	8,105	11,86
	1600	131,044	10,925
	1800	1214,85	10,276
	2000	1591,418	9,719
	2200	1927,436	9,267
	2400	3221,535	8,98
	2600	3444,117	8,661
	2800	3396,221	8,366
Harare, Zimbabwe	1000	7,22	14,884
	1200	7,22	13,12
	1400	7,22	11,86
	1600	7,22	10,916
	1800	7,22	10,181
	2000	44,866	9,596
	2200	64,029	9,116

	2400	417,697	8,743
	2600	1043,413	8,453
	2800	1346,85	8,187
Maputo, Mozambique	1000	5,7	14,884
	1200	5,7	13,12
	1400	5,7	11,86
	1600	29,241	10,917
	1800	71,468	10,186
	2000	242,093	9,611
	2200	297,279	9,134
	2400	561,764	8,754
	2600	820,391	8,435
	2800	1055,957	8,164
Lusaka, Zambia	1000	5,5	14,884
	1200	5,5	13,12
	1400	5,5	11,86
	1600	5,5	10,915
	1800	10,609	10,181
	2000	88,328	9,599
	2200	387,326	9,141
	2400	493,605	8,749
	2600	1073,731	8,456
	2800	1269,129	8,181
Mogadishu, Somalia	1000	126,594	14,893
	1200	126,594	13,129
	1400	126,594	11,87
	1600	126,594	10,925
	1800	126,594	10,19
	2000	242,029	9,611
	2200	439,574	9,145
	2400	758,486	8,77
	2600	907,167	8,442
	2800	3907,197	8,414
Brazzaville, Congo	1000	370,078	14,912
	1200	526,63	13,161
	1400	668,279	11,912
	1600	775,546	10,976
	1800	924,721	10,253
	2000	1019,334	9,672
	2200	1198,03	9,206
	2400	1340,915	8,817
	2600	1505,797	8,491
	2800	1755,277	8,221
Niamey, Niger	1000	4,031	14,884
	1200	4,031	13,12
	1400	28,445	11,862
	1600	101,12	10,923
	1800	288,632	10,202

	2000	568,724	9,636
	2200	882,634	9,18
	2400	3017,871	8,961
	2600	2364,808	8,564
	2800	2332,913	8,271
Monrovia, Liberia	1000	322,218	14,908
	1200	382,359	13,149
	1400	574,481	11,905
	1600	941,98	10,989
	1800	1793,873	10,324
	2000	2215,181	9,772
	2200	2529,089	9,319
	2400	3797,719	9,034
	2600	4010,696	8,715
	2800	3963,14	8,419

Table 29 Electricity related impacts of drought in African countries.

Country	Period	Consequences of drought
Uganda	2004/2005	Reduction in water levels at Lake Victoria resulting in reduction in hydro-power generation by 50 MW
Kenya	1992	Failure of rains led to power rationing in April-May 1992
Kenya	1998 -2001	Massive drought decreased hydro generation (25 percent in 2000), which had to be replaced by more expensive fuel-based generation. Period of power rationing
Lesotho	1992	Hydro power operation limited to 6 months, leading to 20 percent reduction compared to 1991
Malawi	1997 - 1998	Engineering operations were affected by drought. The amount of hydro energy generated was 6 percent less than in years of normal rainfall
Mauritius	1999	Massive drought led to 70 percent drop in normal annual production of electricity
Tanzania	1997	The Mtera dam reached its lowest ever level resulting in a 17 percent drop in hydro generation, needed to use thermal generation to meet the shortfall. Period of power rationing
Zambia	1992	Poor rainfall resulted in a 35 percent reduction in hydro generation in relation to the previous year
Zimbabwe	1993	Drought led to a drop of over 9 percent in energy production compared to 1992

Source: Adapted from (UN, 2007)

Table 30 Cost difference when compared the total system levelized cost from fossil fuel technologies considering external costs with the levelized electricity cost in 2020 by means of CSP for selected centers of demand.

	Min. CSP cost 2020	CC	AC	AC with CCS	CGCC	Max. CSP cost 2020	CC	AC	AC with CCS	CGCC
Parabolic Trough										
Lagos, Nigeria	8.26	-37%	-45%	-48%	-3%	11.09	-16%	-26%	-30%	31%
Kinshasa, DRC	8.14	-38%	-45%	-49%	-4%	10.98	-17%	-26%	-31%	30%
Nairobi, Kenya	8.22	-38%	-45%	-48%	-3%	11.10	-16%	-26%	-30%	31%
Johannesburg, SA	8.03	-39%	-46%	-50%	-5%	11.26	-15%	-25%	-29%	33%
Solar Tower										
Lagos, Nigeria	7.65	-42%	-49%	-52%	-10%	9.71	-26%	-35%	-39%	15%
Kinshasa, DRC	7.53	-43%	-50%	-53%	-11%	9.60	-27%	-36%	-40%	13%
Nairobi, Kenya	7.61	-42%	-49%	-52%	-10%	9.71	-26%	-35%	-39%	15%
Johannesburg, SA	7.42	-44%	-50%	-53%	-12%	9.87	-25%	-34%	-38%	16%

CC: Conventional coal, AC: Advanced coal, AC with CCS, Advanced coal with Carbon Capture and Storage, CGCC: Conventional gas combined cycle.

Table 31 Cost difference when compared the total system levelized cost from fossil fuel technologies considering external costs with the levelized electricity cost in 2050 by means of CSP for selected centers of demand

	Min CSP cost 2050	CC	AC	AC with CCS	CGCC	Max CSP cost 2050	CC	AC	AC with CCS	CGCC
Parabolic Trough										
Lagos, Nigeria	6.12	-54%	-59%	-62%	-28%	8.22	-38%	-45%	-48%	-3%
Kinshasa, DRC	6.03	-54%	-60%	-62%	-29%	8.14	-38%	-45%	-49%	-4%
Nairobi, Kenya	6.09	-54%	-59%	-62%	-28%	8.23	-38%	-45%	-48%	-3%
Johannesburg, SA	5.94	-55%	-60%	-63%	-30%	8.35	-37%	-44%	-48%	-1%
Solar Tower										
Lagos, Nigeria	5.54	-58%	-63%	-65%	-35%	7.00	-47%	-53%	-56%	-17%
Kinshasa, DRC	5.45	-59%	-64%	-66%	-36%	6.92	-47%	-54%	-57%	-18%
Nairobi, Kenya	5.51	-58%	-63%	-65%	-35%	7.01	-47%	-53%	-56%	-17%
Johannesburg, SA	5.36	-59%	-64%	-66%	-37%	7.13	-46%	-52%	-55%	-16%

CC: Conventional coal, AC: Advanced coal, AC with CCS, Advanced coal with Carbon Capture and Storage, CGCC: Conventional gas combined cycle.

